Liu, Chapter 4: Some local properties

Solutions to the exercises

Conventions. Rings are commutative with 1. Results of the text are cited by their numbers in Liu ("Prop. 3.25", "Thm. 3.12", ...), and results of earlier chapters as "Prop. 2.3.12", "Ex. 3.2.4" etc. Exercise labels are those of the book, numbered within each section of the chapter: for instance Exercise 3.12 is the twelfth exercise at the end of Section 4.3. Each exercise is restated before its solution; sub-items that the book displays as nested lists are written as running prose.

Honest notes. Many printed statements need a correction or an interpretation, and each solution says what is proved: Ex. 1.5 (integrality over $A[T]$), Ex. 1.9(c) (the formula for the restriction to V), Ex. 1.15, 1.17, 1.18, 1.19(b) (empty open sets, an integrality convention, and counterexamples in characteristic p), Ex. 2.1 (the printed claim fails for $\mathbb{A}^1_k \to \operatorname{Spec} k$; replacement statements are proved), Ex. 3.1 (needs a one-dimensional local ring), Ex. 3.2, 3.8(b), 3.11 (counterexamples and corrected versions), Ex. 3.19(c) and 3.22(a). Some solutions cite standard facts which are named where used.

4.1 Normal schemes

Throughout, rings are commutative with 1, and (as in Section 4.1.1 of the book) a *normal ring* means a normal integral domain. Results of the text are cited by their numbers in Liu ("Prop. 1.5", "Thm. 1.14", ...), and results of earlier chapters as "Prop. 2.4.18" etc.

Honest notes. In Ex. 1.5 the book says "f integral over A"; this must be "integral over $A[T]$" (this is what (b) and (c) use), and that is what is proved. In Ex. 1.9(c) the printed formula for $\varphi|_V$ ("$A[T_1/T_0] \to \mathcal{O}_X(V)$, $T_1/T_0 \mapsto f$", although $A[T_1/T_0]$ is the coordinate ring of $D_+(T_0)$, not of $D_+(T_1)$) has to be read as $T_0/T_1 \mapsto f^{-1}$, the coordinate of $D_+(T_1)$; this is the only reading for which the two morphisms agree on $U \cap V$. In Ex. 1.4 the phrase "this property is verified at the closed points" is read as: the local rings at the closed points are reduced (resp. integral domains, resp. normal domains). In Ex. 1.10 I use, for the quotient X/G, only the description recalled at the beginning of that solution. Exercises 1.2, 1.5(e), 1.8 use earlier exercises, and the needed arguments are given.

Exercise 1.1

> **Problem.** Let A be a ring and B an A-algebra. We suppose that there exists a faithful B-module M (i.e., $bM = 0$ for $b \in B$ implies $b = 0$) which is finitely generated over A. Using the Cayley–Hamilton identity on the matrices $\operatorname{Hom}_A(M, M)$ (i.e., any matrix cancels its characteristic polynomial), show that B is integral over A (this generalizes Lemma 1.11).

Solution. Let $\varphi : A \to B$ be the structure map; M is an A-module through φ. "Integral" means that every $b \in B$ is a root of a monic polynomial whose coefficients lie in $\varphi(A)$. Let $m_1, \dots, m_n$ generate M over A. If $n = 0$ then $M = 0$, hence $B = 0$ since M is faithful, and there is nothing to prove.

Fix $b \in B$ and let $u : M \to M$ be multiplication by b; it is A-linear. Since $bm_i \in M$, we can write $bm_i = \sum_j \lambda_{ij} m_j$ with $\lambda_{ij} \in A$. Let $\varepsilon : A^n \to M$, $e_i \mapsto m_i$, which is surjective, and let $L : A^n \to A^n$ be the A-linear map $e_i \mapsto \sum_j \lambda_{ij} e_j$. Then $u \circ \varepsilon = \varepsilon \circ L$, hence $P(u) \circ \varepsilon = \varepsilon \circ P(L)$ for every $P \in A[X]$ (where $A[X]$ acts on M through φ and u). Let $\chi(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_0 \in A[X]$ be the characteristic polynomial of L. By Cayley–Hamilton $\chi(L) = 0$, so $\chi(u) \circ \varepsilon = 0$, and since ε is onto, $\chi(u) = 0$. This says

$$\big(b^n + \varphi(c_{n-1})b^{n-1} + \cdots + \varphi(c_0)\big)\, m = 0 \quad \text{for all } m \in M.$$

As M is faithful, $b^n + \varphi(c_{n-1})b^{n-1} + \cdots + \varphi(c_0) = 0$. So b is integral over A. □

Exercise 1.2

> **Problem.** Let A be an integral domain, and let I be a finitely generated ideal of A. Show that for any $\phi \in \mathrm{Hom}_A(I, I)$ and for any non-zero $a \in I$, $a^{-1}\phi(a) \in \mathrm{Frac}(A)$ is integral over A (use the Cayley–Hamilton identity).

Solution. Let $K = \mathrm{Frac}(A)$. Put $c = a^{-1}\phi(a) \in K$. For nonzero $x \in I$ we have, since $x, a \in A$ and ϕ is A-linear,

$$a\,\phi(x) = \phi(ax) = x\,\phi(a), \qquad \text{so} \qquad \phi(x) = cx.$$

This also holds for $x = 0$. Hence $cI = \phi(I) \subseteq I$. Let $B = A[c] \subseteq K$. Then I is a B-module (because $cI \subseteq I$), finitely generated over A by hypothesis, and it is faithful: if $b \in B \subseteq K$ and $bI = 0$, then $bx = 0$ for the nonzero $x = a \in I$, so $b = 0$ in the field K. By Exercise 1.1 (with $M = I$), B is integral over A. In particular c is integral over A. □

Exercise 1.3

> **Problem.** Let X be a normal Noetherian local scheme of dimension 2, with closed point s. Show that $X \setminus \{s\}$ is a non-affine Dedekind scheme.

Solution. Write $X = \operatorname{Spec} A$ with $(A, \mathfrak{m})$ a Noetherian local ring; $s = \mathfrak{m}$, $\dim A = 2$. Since X is normal, $A = \mathcal{O}_{X,s}$ is a normal domain, and X is irreducible. Put $U = X \setminus \{s\}$.

U is a Dedekind scheme. U is open in the Noetherian scheme X, hence Noetherian, and it is non-empty and irreducible (it contains the generic point). Its local rings are $A_{\mathfrak{p}}$ with $\mathfrak{p} \neq \mathfrak{m}$, which are localizations of the normal domain A, hence normal (Lemma 1.4(b)). So U is a normal locally Noetherian scheme. Let $\mathfrak{p} \in U$, i.e. $\mathfrak{p} \neq \mathfrak{m}$, and let $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_r = \mathfrak{p}$ be a chain of primes. Appending $\mathfrak{m}$ gives a chain of length $r + 1 \leq \dim A = 2$, so $\operatorname{ht} \mathfrak{p} \leq 1$. Thus every chain of irreducible closed subsets of U has length ≤ 1, i.e. $\dim U \leq 1$. Moreover, as $\dim A = 2$ there is a chain $0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{m}$, and $0 \subsetneq \mathfrak{p}_1$ is a chain in U; so $\dim U = 1$. By Definition 1.2, U is a Dedekind scheme.

U is not affine. Since $\operatorname{codim}(\{s\}, X) = \dim \mathcal{O}_{X,s} = \dim A = 2$, Thm. 1.14 shows that the restriction $\mathcal{O}_X(X) = A \to \mathcal{O}_X(U)$ is an isomorphism. Suppose $U = \operatorname{Spec} B$ were affine. The open immersion $j : U \to X$ corresponds to a ring homomorphism $A \to B$, and this is precisely the restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(U)$, i.e. an isomorphism. A morphism of affine schemes whose ring homomorphism is an isomorphism is an isomorphism; so j would be surjective. But $s \notin j(U)$, a contradiction. □

Exercise 1.4

> **Problem.** Let X be a connected algebraic variety. Show that X is reduced (resp. integral; resp. normal) if and only if this property is verified at the closed points of X.

Solution. An algebraic variety is of finite type over a field, hence Noetherian and quasi-compact. "At a point x" means: $\mathcal{O}_{X,x}$ is reduced (resp. an integral domain, resp. a normal domain). A closed point means a point closed in X.

Step 0. If $x \in X$ is any point, there is a closed point $y \in \overline{\{x\}}$ (Exercise 2.4.8, as used in the proof of Prop. 1.5). Every affine open neighbourhood $V = \operatorname{Spec} R$ of y contains x (an open set containing $y \in \overline{\{x\}}$ contains x), and x corresponds to a prime contained in the prime of y; hence $\mathcal{O}_{X,x}$ is a localization of $\mathcal{O}_{X,y}$. A localization of a reduced ring is reduced, of a domain is a domain, and of a normal domain is a normal domain (Lemma 1.4(b)). So if the property holds at all closed points, it holds at all points.

Reduced. By Step 0 and the definition of reducedness (all local rings reduced), X is reduced iff it is reduced at its closed points. (Connectedness is not needed.)

Integral. If X is integral, all local rings are domains. Conversely suppose $\mathcal{O}_{X,y}$ is a domain for every closed y. By Step 0 all local rings are domains, in particular reduced, so X is reduced; it remains to see that X is irreducible (reduced and irreducible means integral). Let $Z_1, \dots, Z_r$ be the irreducible components of X (finitely many since X is Noetherian). Suppose $Z_i \cap Z_j \neq \emptyset$ for some $i \neq j$. This is a nonempty closed subset, so it contains a closed point y of X (apply Exercise 2.4.8 to a point of it). Take an affine open $V = \operatorname{Spec} R \ni y$, $y \leftrightarrow \mathfrak{p}$. The irreducible components of V passing through y correspond to the minimal primes of R contained in $\mathfrak{p}$, i.e. to the minimal primes of $R_{\mathfrak{p}} = \mathcal{O}_{X,y}$, which is a domain; so exactly one irreducible component W of V passes through y. Now $Z_i \cap V$ and $Z_j \cap V$ are irreducible closed subsets of V containing y, so both lie in W. Hence $Z_i = \overline{Z_i \cap V} \subseteq \overline{W}$ and likewise $Z_j \subseteq \overline{W}$, with $\overline{W}$ irreducible; by maximality of components $Z_i = \overline{W} = Z_j$, a contradiction. Therefore the Z_i are pairwise disjoint closed sets covering X; each Z_i is the complement of the union of the others, hence also open. As X is connected, $r = 1$ and X is irreducible, hence integral.

Normal. If X is normal, it is normal at every point. Conversely, suppose $\mathcal{O}_{X,y}$ is a normal domain for all closed y. By the integral case (here connectedness is used), X is integral, in particular irreducible; by Step 0 it is normal at all points. So X is normal. □

Exercise 1.5

> **Problem.** Let A be a normal ring. We want to show that $A[T]$ is also normal. Let $f \in \operatorname{Frac}(A[T])$ be integral over $A[T]$ (the book writes "over A"; see the remark at the beginning).

(a) Show that $f \in \mathrm{Frac}(A)[T]$.

(b) Show that there exists a subring A_0 of A such that A_0 is finitely generated over $\mathbb{Z}$, $f \in \mathrm{Frac}(A_0)[T]$, and f is integral over $A_0[T]$.

(c) Show that there exists an $a \in A_0 \setminus \{0\}$ such that $af^n \in A_0[T]$ for every $n \geq 0$.

(d) Let us write $f = \sum_{i \leq d} \alpha_i T^i$. Show that $a\alpha_0^n \in A_0$ for every $n \geq 0$, and that $a^{i+1}\alpha_i^n \in A_0$ for every $i \leq d$ and for every $n \geq 0$.

(e) Show that $A_0[\alpha_i] \subseteq a^{-i-1}A_0$, and that $f \in A[T]$.

(f) (A second method when A is Noetherian.) Show that $A[T]$ is normal if A is a discrete valuation ring; then prove the general case using Lemma 1.13.

Solution. Let $K = \mathrm{Frac}(A)$, so $\mathrm{Frac}(A[T]) = K(T)$. Since $A[T]$ is a domain, showing that it is normal means showing that every $f \in K(T)$ integral over $A[T]$ lies in $A[T]$. We may assume $f \neq 0$. Fix an equation

$$f^N + g_{N-1}f^{N-1} + \cdots + g_0 = 0, \qquad g_j \in A[T].$$

(a) $K[T]$ is a principal ideal domain, hence factorial, hence normal by Lemma 1.4(a), and $\mathrm{Frac}(K[T]) = K(T)$. The element f is integral over $A[T] \subseteq K[T]$, hence over $K[T]$, so $f \in K[T]$.

(b) Write $f = \sum_{i \leq d} \alpha_i T^i$ with $\alpha_i \in K$, and write each $\alpha_i = b_i/c_i$ with $b_i, c_i \in A$, $c_i \neq 0$. Let A_0 be the subring of A generated over $\mathbb{Z}$ by all b_i, c_i and by all the (finitely many) coefficients of $g_0, \ldots, g_{N-1}$. Then A_0 is a domain finitely generated over $\mathbb{Z}$, $\alpha_i \in K_0 := \mathrm{Frac}(A_0) \subseteq K$, so $f \in K_0[T]$, and the displayed equation has $g_j \in A_0[T]$, so f is integral over $A_0[T]$. Note that A_0 is Noetherian (Hilbert's basis theorem).

(c) By the integral equation, $A_0[T][f] = \sum_{k<N} A_0[T]f^k$ (every f^n with $n \geq N$ is an $A_0[T]$-combination of $1, f, \ldots, f^{N-1}$, by induction on n). Each f^k ($k < N$) lies in $K_0[T]$, so it has finitely many coefficients in K_0. Let $a \in A_0 \setminus \{0\}$ be a common denominator of all these coefficients, so $af^k \in A_0[T]$ for $k < N$. For $n \geq N$, $f^n = \sum_{k<N} h_k f^k$ with $h_k \in A_0[T]$, so $af^n \in A_0[T]$ as well.

(d) Define $f_0 = f$ and recursively $\alpha_i := f_i(0) \in K_0$ and $f_{i+1} := (f_i - \alpha_i)/T \in K_0[T]$. By induction $f_i = \sum_{k \geq 0} \alpha_{i+k} T^k$ with α_j the coefficients of f (and $\alpha_j = 0$ for $j > d$); the constant term of f_i^n is α_i^n. Let $B_0 = A_0$ and $B_i = A_0[\alpha_0, \ldots, \alpha_{i-1}]$. We prove by induction on i that for all $n \geq 0$:

$$(1_i)\ \ af_i^n \in B_i[T]; \qquad (2_i)\ \ a\alpha_i^n \in B_i; \qquad (3_i)\ \ B_i \subseteq a^{-i}A_0.$$

For $i = 0$, (1_0) is (c), (2_0) is the constant term of (1_0), and (3_0) is trivial. Suppose $(1_i), (2_i), (3_i)$ hold. Note (2_i) follows from (1_i) by taking constant terms. Since $a\alpha_i^n \in B_i$ for all n, we get $B_{i+1} = B_i[\alpha_i] = \sum_n B_i\alpha_i^n \subseteq a^{-1}B_i \subseteq a^{-i-1}A_0$, which is (3_{i+1}). Next, $Tf_{i+1} = f_i - \alpha_i$, so

$$T^n f_{i+1}^n = (f_i - \alpha_i)^n = \sum_{k=0}^{n} \binom{n}{k} (-\alpha_i)^{n-k} f_i^k.$$

Multiply by a: by (1_i), $af_i^k \in B_i[T] \subseteq B_{i+1}[T]$, and $\alpha_i \in B_{i+1}$; hence $aT^n f_{i+1}^n \in B_{i+1}[T]$. The coefficients of af_{i+1}^n are those of $aT^n f_{i+1}^n$ shifted by n places, so $af_{i+1}^n \in B_{i+1}[T]$: this is (1_{i+1}), and (2_{i+1}) follows. This completes the induction.

Now (2_0) gives $a\alpha_0^n \in A_0$, and $(2_i), (3_i)$ give $a\alpha_i^n \in B_i \subseteq a^{-i}A_0$, i.e. $a^{i+1}\alpha_i^n \in A_0$ for all i, n.

(e) By (d), $\alpha_i^n \in a^{-i-1}A_0$ for all $n \geq 0$, so $A_0[\alpha_i] = \sum_n A_0\alpha_i^n \subseteq a^{-i-1}A_0$. Now A_0 is Noetherian and $a^{-i-1}A_0 \simeq A_0$ as A_0-modules, so $A_0[\alpha_i]$ is a finitely generated A_0-module. It is a faithful module over the ring $A_0[\alpha_i]$ (it contains 1), so by Exercise 1.1 (with $B = M = A_0[\alpha_i]$) the ring $A_0[\alpha_i]$ is integral over A_0; in particular α_i is integral over $A_0 \subseteq A$. As $\alpha_i \in K$ and A is normal, $\alpha_i \in A$. This holds for each i, so $f \in A[T]$. Hence $A[T]$ is normal.

(f) *Case A a discrete valuation ring*, with uniformizer t and fraction field K. Let $f \in K(T)$ be integral over $A[T]$; by (a) $f \in K[T]$. Let $r \geq 0$ be minimal with $g := t^r f \in A[T]$. Suppose $r > 0$; then $g \notin tA[T]$ (otherwise $t^{r-1}f \in A[T]$). Multiplying the integral equation by t^{rN} gives $g^N + \sum_{j<N} t^{r(N-j)} g_j g^j = 0$, so $g^N \in tA[T]$. But $A[T]/tA[T] \simeq (A/tA)[T]$ is a domain (as A/tA is a field), so $g \in tA[T]$, a contradiction. Hence $r = 0$ and $f \in A[T]$.

General Noetherian case. Let A be a normal Noetherian domain. If $\dim A = 0$, A is a field and $A[T]$ is a principal ideal domain, hence normal. Let $\dim A \geq 1$ and let $f \in K(T)$ be integral over $A[T]$. For each prime $\mathfrak{p}$ of height 1, $A_\mathfrak{p}$ is a Noetherian normal local domain of dimension 1 (Lemma 1.4(b)); it is a principal ideal domain by Prop. 1.12, and not a field, so it is a discrete valuation ring. Since $A[T] \subseteq A_\mathfrak{p}[T]$ and $\mathrm{Frac}(A_\mathfrak{p}[T]) = K(T)$, the first case shows $f \in A_\mathfrak{p}[T]$. Thus every coefficient of f lies in $\bigcap_{\mathrm{ht}\,\mathfrak{p}=1} A_\mathfrak{p} = A$ (Lemma 1.13), i.e. $f \in A[T]$. □

Exercise 1.6

> **Problem.** Let $A \to B$ be a faithfully flat ring homomorphism, and let us suppose B is normal.
>
> (a) Let I, J be ideals of A such that $IB \subseteq JB$. Show that $I \subseteq J$.
>
> (b) Let $f = ab^{-1} \in \mathrm{Frac}(A)$ be integral over A. Show that $aB \subseteq bB$. Deduce from this that $f \in A$ and that A is normal.

Solution. Recall that faithfully flat means: B is flat over A, and $M \otimes_A B = 0$ implies $M = 0$ for every A-module M.

(a) Let $M = (I + J)/J$. Tensoring the exact sequence $0 \to J \to I + J \to M \to 0$ with the flat module B, and using that for an ideal $\mathfrak{a}$ of A the map $\mathfrak{a} \otimes_A B \to B$ is injective with image $\mathfrak{a}B$, we get

$$M \otimes_A B \simeq \frac{(I+J)B}{JB} = \frac{IB + JB}{JB} = 0$$

because $IB \subseteq JB$. By faithful flatness $M = 0$, i.e. $I \subseteq J$.

(In particular, taking $I = \mathrm{Ker}(A \to B)$ and $J = 0$, we get $IB = 0$, so $I = 0$: the map $A \to B$ is injective. Thus $A \subseteq B$ is a domain.)

(b) Let $f = ab^{-1}$ with $a, b \in A$, $b \neq 0$, integral over A. Since $A \subseteq B$, $f \in \mathrm{Frac}(A) \subseteq \mathrm{Frac}(B)$ is integral over B, so $f \in B$ because B is normal. Hence $a = fb \in bB$, i.e. $aB \subseteq bB$. Applying (a) to the ideals aA and bA (note $(aA)B = aB$ and $(bA)B = bB$) gives $aA \subseteq bA$, so $a = bc$ with $c \in A$ and $f = c \in A$. Since A is a domain in which every element of $\mathrm{Frac}(A)$ integral over A lies in A, A is normal. □

Exercise 1.7

Problem. Let k be a field. Determine the normalization of $\operatorname{Proj} k[x,y,z]/(x^3-y^2z)$.

Solution. Let $C = \operatorname{Proj} k[x,y,z]/(x^3-y^2z)$ (any characteristic). **Answer:** the normalization is

$$\varphi : \mathbb{P}^1_k \to C, \qquad [s:u] \mapsto [s^2u : s^3 : u^3],$$

an isomorphism away from the cusp $[0:0:1]$, with $\varphi^{-1}([0:0:1]) = \{[0:1]\}$.

The charts of C. Let $U_y = D_+(y)$ and $U_z = D_+(z)$. If $y = z = 0$ at a point then $x^3 = 0$, so $U_y \cup U_z = C$. We have

$$\mathcal{O}(U_y) = k[X,Z]/(X^3 - Z) \simeq k[X] \ (X = x/y,\ Z = z/y), \qquad \mathcal{O}(U_z) = k[a,b]/(a^3 - b^2) \ (a = x/z,\ b = y/z).$$

The ring $k[a,b]/(a^3-b^2)$ is isomorphic to $k[v^2,v^3] \subset k[v]$ via $a \mapsto v^2$, $b \mapsto v^3$: this map is onto $k[v^2,v^3]$, and it is injective because every class has a representative $p_0(b) + p_1(b)a + p_2(b)a^2$ (reduce $a^3 = b^2$), whose image $p_0(v^3) + p_1(v^3)v^2 + p_2(v^3)v^4$ has the three summands supported on exponents $\equiv 0, 2, 1 \pmod 3$ respectively, so it vanishes only if all $p_i = 0$. Thus $U_y \simeq \mathbb{A}^1_k$ and $U_z \simeq \operatorname{Spec} k[v^2,v^3]$ are integral, and $U_y \cap U_z = D(X) \neq \emptyset$; two irreducible opens that meet generate an irreducible space, so C is irreducible and reduced, i.e. an integral scheme.

The morphism φ. Let $V_1 = \{s \neq 0\} = \operatorname{Spec} k[w]$ with $w = u/s$, and $V_2 = \{u \neq 0\} = \operatorname{Spec} k[v]$ with $v = s/u$. Define $\varphi_1 : V_1 \to U_y$ by $X \mapsto w$ (then $Z = X^3 \mapsto w^3$, i.e. $[x:y:z] = [w:1:w^3]$), and $\varphi_2 : V_2 \to U_z$ by $a \mapsto v^2$, $b \mapsto v^3$. On $V_1 \cap V_2 = \operatorname{Spec} k[w^{\pm 1}]$, φ_1 lands in $U_y \cap U_z = D(X)$, where $a = X/Z = X^{-2}$ and $b = 1/Z = X^{-3}$; so φ_1 sends $a \mapsto w^{-2} = v^2$ and $b \mapsto w^{-3} = v^3$, which is φ_2. So φ_1, φ_2 glue to a morphism $\varphi : \mathbb{P}^1_k \to C$, and it is given by the stated formula. Moreover $\varphi^{-1}(U_y) = V_1$ (a point of V_2 goes to U_y iff $b = v^3 \neq 0$, i.e. it lies in $V_1 \cap V_2$) and $\varphi^{-1}(U_z) = V_2$ (a point of V_1 goes to U_z iff $z/y = w^3 \neq 0$, i.e. it lies in V_2).

φ is the normalization. $\mathbb{P}^1_k$ is normal (Example 1.6) and φ is dominant (it is an isomorphism $V_1 \to U_y$, since $k[X] \to k[w]$, $X \mapsto w$, is an isomorphism). Let $\pi : C' \to C$ be the normalization (Prop. 1.22). By Def. 1.19, $\varphi = \pi \circ g$ for a morphism $g : \mathbb{P}^1_k \to C'$. Over U_y: $\mathcal{O}(U_y) = k[X]$ is normal, so by Lemma 1.21 the normalization of U_y is the identity, and $\varphi^{-1}(U_y) = V_1 \to U_y$ is an isomorphism. Over U_z: $A = k[v^2,v^3]$ has fraction field $k(v)$ (as $v = v^3/v^2$), and its integral closure in $k(v)$ is $k[v]$ (the ring $k[v]$ is factorial, hence normal, and it is integral over A because v is a root of the monic polynomial $T^2 - v^2 \in A[T]$). By Lemma 1.21, $\operatorname{Spec} k[v] = V_2 \to U_z$ is the normalization of U_z. Since the restriction of a normalization to an open subset of the target is a normalization (remark after Def. 1.19), $\pi^{-1}(U_y) \to U_y$ and $\pi^{-1}(U_z) \to U_z$ are the normalizations of U_y, U_z, and g restricts to morphisms $\varphi^{-1}(U_i) \to \pi^{-1}(U_i)$ between two normalizations of U_i over U_i, which are isomorphisms by uniqueness of normalization. As U_y, U_z cover C, g is an isomorphism, so φ is the normalization. □

Exercise 1.8

Problem. Let $\mathcal{O}_K$ be a Dedekind domain, and let $a \in \mathcal{O}_K$ be non-zero. Show that the scheme $\operatorname{Proj} \mathcal{O}_K[x,y,z]/(xy - az^2)$ is normal.

Solution. Write $A = \mathcal{O}_K$, $F = \operatorname{Frac}(A)$, and $X = \operatorname{Proj} A[x,y,z]/(xy - az^2)$. We use that $A[T]$ is normal for A a normal Noetherian domain: this was proved in Exercise 1.5(f) (for a DVR by

the t-adic argument, in general through $A = \bigcap_{\mathrm{ht}\,\mathfrak{p}=1} A_{\mathfrak{p}}$ of Lemma 1.13; if A is a field it is clear). Then $A[T]$ and its localizations (Lemma 1.4(b)) are normal.

The three charts. $X = D_+(x) \cup D_+(y) \cup D_+(z)$. We have

$$D_+(x) = \operatorname{Spec} A[Y,W]/(Y - aW^2) \simeq \operatorname{Spec} A[W], \quad D_+(y) = \operatorname{Spec} A[X',W']/(X' - aW'^2) \simeq \operatorname{Spec} A[W'],$$

with $Y = y/x$, $W = z/x$, $X' = x/y$, $W' = z/y$, and

$$D_+(z) = \operatorname{Spec} B, \qquad B = A[\xi,\eta]/(\xi\eta - a), \quad \xi = x/z,\ \eta = y/z.$$

Two of the charts are polynomial rings over A, hence normal domains. It remains to study B.

Structure of B. Since $\xi\eta = a \in A$, B is spanned over A by ξ^i $(i \geq 0)$ and η^j $(j \geq 1)$. Consider $\theta : B \to F[\xi,\xi^{-1}]$ (Laurent polynomial ring in a new variable ξ) with $\xi \mapsto \xi$, $\eta \mapsto a\xi^{-1}$; it is a well-defined ring homomorphism. The images ξ^i $(i \geq 0)$ and $a^j\xi^{-j}$ $(j \geq 1)$ of the spanning elements are linearly independent over A (as $a \neq 0$), so these elements of B form an A-basis, and θ is injective with image

$$B_0 = \Big\{ \sum_i c_i \xi^i \ :\ c_i \in A\ (i \geq 0),\ c_{-j} \in a^j A\ (j \geq 1) \Big\}.$$

Hence B is a domain with $\operatorname{Frac}(B) = F(\xi)$. Under θ:

- $B[1/\xi]$ becomes $A[\xi,\xi^{-1}] = \{\sum c_i\xi^i :\ c_i \in A\ \forall i\}$ (because $\eta = a\xi^{-1}$ lies in it);
- $B[1/\eta] = A[\eta,\eta^{-1}]$ (because $\xi = a\eta^{-1}$) becomes $\{\sum c_i\xi^i :\ c_{-j} \in a^jA\ (j \geq 0),\ c_i \in a^{-i}A\ (i > 0)\}$, since $\eta^j = a^j\xi^{-j}$ and, for $j > 0$, $\eta^{-j} = a^{-j}\xi^j$.

The intersection of these two rings inside $F[\xi,\xi^{-1}]$ is the set of $\sum c_i\xi^i$ with $c_i \in A \cap a^{-i}A = A$ for $i > 0$ and $c_{-j} \in A \cap a^jA = a^jA$ for $j \geq 0$, that is, exactly B_0. So

$$B = B[1/\xi] \cap B[1/\eta].$$

Both $B[1/\xi] \simeq A[\xi^{\pm 1}]$ and $B[1/\eta] \simeq A[\eta^{\pm 1}]$ are normal (localizations of polynomial rings over A). *B is normal*: let $g \in \operatorname{Frac}(B)$ be integral over B. It is integral over the normal domains $B[1/\xi]$ and $B[1/\eta]$ (both have fraction field $\operatorname{Frac}(B)$), hence lies in both, hence in B.

X is normal. Every point of X lies in one of the three affine charts, each the spectrum of a normal domain, so its local ring is a localization of a normal domain, hence normal. It remains to see that X is irreducible. The three charts are integral. The open set $D_+(z) \cap D_+(x) = \operatorname{Spec} B[1/\xi]$ is nonempty, and so is $D_+(z) \cap D_+(y)$. A nonempty open subset of an irreducible space contains its generic point, which is also the generic point of the open subset; so the generic points of $D_+(x)$, $D_+(z)$ coincide, and likewise for $D_+(y)$ and $D_+(z)$. Call it ζ. Then each chart is contained in $\overline{\{\zeta\}}$, so $X = \overline{\{\zeta\}}$ is irreducible. Thus X is normal. □

Exercise 1.9

Problem. Let $X = \operatorname{Spec} A$ be an affine Dedekind scheme. We are going to show that every open subscheme U of X is affine. We may assume that U is the complement of a closed point x_0.

(a) Let $\mathfrak{m}_0$ be the maximal ideal of A corresponding to x_0. Let $t \in A$ be such that $t\mathcal{O}_{X,x_0} = \mathfrak{m}_0\mathcal{O}_{X,x_0}$ and let $V(t) = \{x_0, x_1, \ldots, x_n\}$. Show that for every $1 \le i \le n$, there exists $t_i \in A$ such that $t_i \notin \mathfrak{m}_0$ and that t_i generates the maximal ideal of $\mathcal{O}_{X,x_i}$.

(b) Show that there exists an $f \in \mathcal{O}_X(U) \setminus A$ (take $f = t^{-1}\prod_{1\le i\le n} t_i^{d_i}$ with large d_i).

(c) Let $Z = V(f) \subset U$ and $V = X \setminus Z$. Show that there exists a (unique) morphism $\varphi : X \to \mathbb{P}^1_X = \operatorname{Proj} A[T_0, T_1]$ such that $\varphi|_U$ is the morphism of A-schemes $U \to D_+(T_0)$ corresponding to $A[T_1/T_0] \to \mathcal{O}_X(U)$, $T_1/T_0 \mapsto f$; and $\varphi|_V$ is the morphism of A-schemes $V \to D_+(T_1)$ corresponding to $A[T_0/T_1] \to \mathcal{O}_X(V)$, $T_0/T_1 \mapsto f^{-1}$ (the printed formula for V is corrected, see the beginning).

(d) Show that φ is a section of the canonical morphism $\mathbb{P}^1_X \to X$, and that $U = \varphi^{-1}(D_+(T_0))$. Deduce from this that U is affine, and that $\mathcal{O}_X(U) = A[f]$.

Solution. *Preliminaries.* A is a Noetherian normal domain of dimension ≤ 1; let $F = \operatorname{Frac}(A)$. If $\dim A = 0$, X is a point and every open subset ($\emptyset$ or X) is affine; so let $\dim A = 1$. Every nonzero prime is then maximal. For a maximal ideal $\mathfrak{m}$ the ring $A_\mathfrak{m}$ is a local principal ideal domain (Prop. 1.12) which is not a field, i.e. a discrete valuation ring; let $v_\mathfrak{m}$ be its valuation on F^*, so $A_\mathfrak{m} = \{g \in F : v_\mathfrak{m}(g) \ge 0\}$.

Reduction to the complement of one closed point. Let $U \subseteq X$ be open and nonempty; it contains the generic point. $X \setminus U = V(I)$ with $I \ne 0$, and A/I is a Noetherian ring of dimension 0, so $X \setminus U = \{x'_1, \ldots, x'_r\}$ is a finite set of closed points. Put $U_0 = X$ and $U_k = U_{k-1} \setminus \{x'_k\}$. If U_{k-1} is affine, it is an affine Dedekind scheme (an open subscheme of a Dedekind scheme is Dedekind, Rem. 1.3), and U_k is the complement of a closed point in it. So by induction it suffices to treat $U = X \setminus \{x_0\}$ with X an affine Dedekind scheme and x_0 a closed point, as in the statement. Also recall that for the integral scheme X one has $\mathcal{O}_X(W) = \bigcap_{x\in W} \mathcal{O}_{X,x} \subseteq F$ for every open W (compare Prop. 2.4.18 as used in the proof of Prop. 1.5).

(a) The existence of t: $\mathcal{O}_{X,x_0} = A_{\mathfrak{m}_0}$ is a discrete valuation ring, so $\mathfrak{m}_0 A_{\mathfrak{m}_0} = (\tau)$ is principal, with $\tau = t/s$, $t \in \mathfrak{m}_0$, $s \in A \setminus \mathfrak{m}_0$; as s is a unit in $A_{\mathfrak{m}_0}$, $tA_{\mathfrak{m}_0} = \mathfrak{m}_0 A_{\mathfrak{m}_0}$. Also $t \ne 0$, so $V(t)$ consists of closed points (it misses the generic point) and is finite (A/tA is Noetherian of dimension 0). Let $\mathfrak{m}_i$ correspond to x_i ($1 \le i \le n$); $\mathfrak{m}_i \ne \mathfrak{m}_0$.

Fix i. As $A_{\mathfrak{m}_i}$ is a discrete valuation ring, $\mathfrak{m}_i A_{\mathfrak{m}_i} = cA_{\mathfrak{m}_i}$ for some $c \in A$ (as above); then $c \in \mathfrak{m}_i A_{\mathfrak{m}_i} \cap A = \mathfrak{m}_i$. The ideals $\mathfrak{m}_0$ and $\mathfrak{m}_i^2$ are comaximal, so by the Chinese remainder theorem there is $t_i \in A$ with $t_i \equiv 1 \pmod{\mathfrak{m}_0}$ and $t_i \equiv c \pmod{\mathfrak{m}_i^2}$. Then $t_i \notin \mathfrak{m}_0$. Write $t_i = c + w$ with $w \in \mathfrak{m}_i^2$; in $A_{\mathfrak{m}_i}$ we have $w \in c^2 A_{\mathfrak{m}_i}$, so $t_i = c(1 + c\,w')$ with $w' \in A_{\mathfrak{m}_i}$, and $1 + cw'$ is a unit of the local ring $A_{\mathfrak{m}_i}$ (as $c \in \mathfrak{m}_i A_{\mathfrak{m}_i}$). Hence $t_i A_{\mathfrak{m}_i} = cA_{\mathfrak{m}_i} = \mathfrak{m}_i A_{\mathfrak{m}_i}$: t_i generates the maximal ideal of $\mathcal{O}_{X,x_i}$.

(b) Let $e_i = v_{\mathfrak{m}_i}(t) \ge 1$ (since $t \in \mathfrak{m}_i$), choose integers $d_i \ge e_i$, and set $f = t^{-1}\prod_{i=1}^n t_i^{d_i} \in F$. We check membership in each local ring $\mathcal{O}_{X,x}$, $x \in U$.

- x the generic point: $\mathcal{O}_{X,x} = F$. Fine.
- $x = \mathfrak{m}$ a maximal ideal with $t \notin \mathfrak{m}$, i.e. $x \notin V(t)$: t is a unit of $A_\mathfrak{m}$ and $t_i \in A$, so $f \in A_\mathfrak{m}$.
- $x = x_i$, $1 \le i \le n$: $v_{\mathfrak{m}_i}(t_i) = 1$ and $v_{\mathfrak{m}_i}(t_j) \ge 0$ for all j (as $t_j \in A$), so $v_{\mathfrak{m}_i}(f) \ge -e_i + d_i \ge 0$.

Hence $f \in \bigcap_{x\in U} \mathcal{O}_{X,x} = \mathcal{O}_X(U)$. But at x_0 we have $v_{\mathfrak{m}_0}(t) = 1$ and $v_{\mathfrak{m}_0}(t_i) = 0$ (as $t_i \notin \mathfrak{m}_0$), so $v_{\mathfrak{m}_0}(f) = -1 < 0$, i.e. $f \notin A_{\mathfrak{m}_0} \supseteq A$. So $f \in \mathcal{O}_X(U) \setminus A$. The same computation shows $f^{-1} \in \mathfrak{m}_0 A_{\mathfrak{m}_0}$, which we use next.

(c) $Z = V(f) = \{x \in U : f \in \mathfrak{m}_x A_x\}$ is closed in U and does not contain x_0; $V = X \setminus Z = \{x_0\} \cup \{x \in U : f \text{ is a unit in } \mathcal{O}_{X,x}\}$ is open in X, and $U \cup V = X$ since $x_0 \in V$. On $U \cap V = U \setminus Z$ the function f is invertible. Moreover $f^{-1} \in \mathcal{O}_X(V)$: for $x \in U \cap V$, $f^{-1} \in \mathcal{O}_{X,x}$; for $x = x_0$, $v_{\mathfrak{m}_0}(f^{-1}) = 1 \geq 0$. We have $D_+(T_0) = \operatorname{Spec} A[S]$ with $S = T_1/T_0$ and $D_+(T_1) = \operatorname{Spec} A[S']$ with $S' = T_0/T_1 = S^{-1}$, and a morphism from a scheme W to an affine scheme $\operatorname{Spec} R$ corresponds to a ring homomorphism $R \to \mathcal{O}(W)$. Let $\varphi_U : U \to D_+(T_0)$ correspond to $S \mapsto f \in \mathcal{O}_X(U)$ and $\varphi_V : V \to D_+(T_1)$ to $S' \mapsto f^{-1} \in \mathcal{O}_X(V)$. Both are morphisms of A-schemes. On $U \cap V$ both land in $D_+(T_0) \cap D_+(T_1) = \operatorname{Spec} A[S, S^{-1}]$ (since f and f^{-1} are units there) and both correspond to the ring homomorphism $A[S, S^{-1}] \to \mathcal{O}_X(U \cap V)$, $S \mapsto f$. Hence they agree on $U \cap V$ and glue to a morphism $\varphi : X \to \mathbb{P}^1_X$ with the required restrictions. It is unique because $U \cup V = X$ and its restrictions to U and V are prescribed.

(d) Let $\pi : \mathbb{P}^1_X \to X$ be the structure morphism. As φ_U, φ_V are A-morphisms, $\pi \circ \varphi$ restricts to the inclusions on U and on V; since $U \cup V = X$, $\pi \circ \varphi = \mathrm{id}_X$: φ is a section.

Clearly $U \subseteq \varphi^{-1}(D_+(T_0))$. The only point of X not in U is x_0, and $\varphi(x_0) \in D_+(T_1)$ is the point where the coordinate S' vanishes: indeed S' maps to $f^{-1} \in \mathfrak{m}_0 A_{\mathfrak{m}_0}$, whose value in $k(x_0)$ is 0. A point of $D_+(T_0) \cap D_+(T_1) = D(S') \subset D_+(T_1)$ has $S' \neq 0$ in its residue field, so $\varphi(x_0) \notin D_+(T_0)$. Hence $\varphi^{-1}(D_+(T_0)) = U$.

U is affine. A section of a separated morphism is a closed immersion (it is the base change of the diagonal $\mathbb{P}^1_X \to \mathbb{P}^1_X \times_X \mathbb{P}^1_X$ along $(\mathrm{id}, \varphi \circ \pi)$), and π is separated. So φ is a closed immersion, and therefore so is its restriction $\varphi_U : U = \varphi^{-1}(D_+(T_0)) \to D_+(T_0) = \operatorname{Spec} A[S]$. A closed subscheme of an affine scheme is affine, so U is affine; the closed immersion $\operatorname{Spec} \mathcal{O}_X(U) \to \operatorname{Spec} A[S]$ corresponds to a *surjective* homomorphism $A[S] \to \mathcal{O}_X(U)$, which is $S \mapsto f$. Hence $\mathcal{O}_X(U) = A[f]$. □

Exercise 1.10

> **Problem.** Let X be a quasi-projective scheme over a locally Noetherian scheme S. Let G be a finite group acting on X, so that the quotient scheme X/G exists (Exercise 3.3.23). Let us moreover suppose that X is normal.
>
> (a) Show that $Y := X/G$ is normal.
>
> (b) Show that the first projection $X \times_Y X \to X$ induces an isomorphism between any irreducible component Z of $X \times_Y X$, endowed with the reduced subscheme structure, and the scheme X. (Show that $Z \to X$ is finite and birational.)

Solution. *What is used about X/G.* Let $\pi : X \to Y = X/G$ be the quotient morphism. By Exercise 3.3.23 (and its proof, which goes through Exercise 2.3.21(c)), X is covered by G-stable affine open sets $W_i = \operatorname{Spec} B_i$; by Exercise 2.3.21(a),(b), Y is obtained by gluing the affine schemes $\operatorname{Spec}(B_i^G)$, which are open in Y and cover it, and on each of them $\mathcal{O}_Y = (\pi_* \mathcal{O}_X)^G$ (this is also the quotient as a ringed space, Exercise 2.2.14). Since $V \mapsto \mathcal{O}_X(\pi^{-1}(V))^G$ is a sheaf on Y, we get $\mathcal{O}_Y(V) = \mathcal{O}_X(\pi^{-1}(V))^G$ for every open $V \subseteq Y$. Moreover $\pi \circ \sigma = \pi$ for $\sigma \in G$, π is surjective (on each W_i it is $\operatorname{Spec} B_i \to \operatorname{Spec} B_i^G$, surjective because B_i is integral over B_i^G) and finite (Exercise 3.3.23, as S is locally Noetherian). Consequently, for every affine open $V \subseteq Y$ the open set $\pi^{-1}(V)$ is affine (finiteness) and G-stable, and with $C = \mathcal{O}_Y(V)$, $B = \mathcal{O}_X(\pi^{-1}(V))$ we have $\pi^{-1}(V) = \operatorname{Spec} B$, $V = \operatorname{Spec} C$ and $C = B^\Gamma$, where $\Gamma \subseteq \operatorname{Aut}(B)$ is the finite group of ring automorphisms $\sigma^\sharp$ induced by $\sigma \in G$.

(a) X is irreducible, and π is surjective and continuous, so Y is irreducible. Let $V = \operatorname{Spec} C \subseteq Y$ be an affine open, $\pi^{-1}(V) = \operatorname{Spec} B$, $C = B^\Gamma$. By Prop. 1.5 ((i)⇒(ii)), B is a normal domain, hence C, a subring of B, is a domain. Let $\alpha \in \operatorname{Frac}(C) \subseteq \operatorname{Frac}(B)$ be integral over C. Then α is integral over B, so $\alpha \in B$ since B is normal. The automorphisms in Γ extend to $\operatorname{Frac}(B)$ and fix C, hence fix $\operatorname{Frac}(C) \ni \alpha$; so $\alpha \in B^\Gamma = C$. Thus C is normal. Every local ring $\mathcal{O}_{Y,y}$ is a localization of such a C, hence normal (Lemma 1.4(b)). So Y is irreducible and normal at all its points: Y is normal.

(b) We first prove an algebraic lemma.

Lemma. Let B be a ring, $\Gamma \subseteq \operatorname{Aut}(B)$ a finite group, $C = B^\Gamma$, Ω a field and $j_1, j_2 : B \to \Omega$ ring homomorphisms with $j_1|_C = j_2|_C$. Then there is $\tau \in \Gamma$ with $j_2 = j_1 \circ \tau$.

Proof. Let Γ act on $B[X]$ coefficientwise, so $B[X]^\Gamma = C[X]$. Extend j_1, j_2 to $\tilde{j}_1, \tilde{j}_2 : B[X] \to \Omega[X]$ by $X \mapsto X$; they agree on $C[X]$. For $\beta \in B[X]$, the polynomial $P_\beta(T) = \prod_{\tau\in\Gamma}(T - \tau\beta) \in B[X][T]$ has Γ-invariant coefficients, hence coefficients in $C[X]$. Applying $\tilde{j}_1$ or $\tilde{j}_2$ to its coefficients gives the same polynomial $\prod_\tau(T - \tilde{j}_1(\tau\beta)) = \prod_\tau(T - \tilde{j}_2(\tau\beta))$ in $\Omega[X][T]$. Its right-hand form has the root $\tilde{j}_2(\beta)$ (the factor $\tau = 1$), so in the domain $\Omega[X]$ we get $\tilde{j}_2(\beta) = \tilde{j}_1(\tau\beta)$ for some $\tau = \tau(\beta) \in \Gamma$. Let $E_\tau = \{b \in B : j_2(b) = j_1(\tau b)\}$. Suppose no E_τ equals B: pick $b_\tau \in B \setminus E_\tau$ for each τ, pairwise distinct integers $n_\tau \geq 0$, and $\beta = \sum_\tau b_\tau X^{n_\tau}$. Then for some τ, $\tilde{j}_2(\beta) = \tilde{j}_1(\tau\beta)$; comparing the coefficients of X^{n_τ} gives $j_2(b_\tau) = j_1(\tau b_\tau)$, i.e. $b_\tau \in E_\tau$, a contradiction. Hence $E_\tau = B$ for some τ. □

Local description of $X \times_Y X$. Let $V = \operatorname{Spec} C$ be an affine open of Y, $\pi^{-1}(V) = \operatorname{Spec} B$ as above. Then $W_V := (\pi \circ p_1)^{-1}(V) = \pi^{-1}(V) \times_V \pi^{-1}(V) = \operatorname{Spec}(B \otimes_C B)$, and these W_V cover $X \times_Y X$. For $\sigma \in G$, let $\gamma_\sigma = (\mathrm{id}_X, \sigma) : X \to X \times_Y X$ (a morphism since $\pi\sigma = \pi$); then $p_1\gamma_\sigma = \mathrm{id}_X$ and $\gamma_\sigma^{-1}(W_V) = \pi^{-1}(V)$. Over V, γ_σ corresponds to $\psi_\sigma : B \otimes_C B \to B$, $b \otimes b' \mapsto b\,\sigma^\sharp(b')$, which is surjective ($b \otimes 1 \mapsto b$). So γ_σ is a closed immersion over each W_V, hence a closed immersion, and $\Gamma_\sigma := \gamma_\sigma(X)$ is a closed irreducible subset of $X \times_Y X$, with $\gamma_\sigma \colon X \to \Gamma_\sigma$ a homeomorphism.

Claim: $X \times_Y X = \bigcup_{\sigma\in G} \Gamma_\sigma$ as sets. It suffices to check this inside each $\operatorname{Spec}(B \otimes_C B)$. Let Q be a prime of $B \otimes_C B$, $\Omega = \operatorname{Frac}((B \otimes_C B)/Q)$, and let $j_1(b) = \overline{b \otimes 1}$, $j_2(b') = \overline{1 \otimes b'}$ in Ω. They agree on C since $c \otimes 1 = 1 \otimes c$ in $B \otimes_C B$. By the Lemma, $j_2 = j_1 \circ \tau$ with $\tau = \sigma^\sharp$ for some $\sigma \in G$, i.e. $1 \otimes b' \equiv \sigma^\sharp(b') \otimes 1 \pmod{Q}$ for all b'. Then for $x = \sum_i b_i \otimes b_i' \in \operatorname{Ker}\psi_\sigma$, $x \equiv (\sum_i b_i\sigma^\sharp(b_i')) \otimes 1 = \psi_\sigma(x) \otimes 1 = 0 \pmod{Q}$. So $Q \supseteq \operatorname{Ker}\psi_\sigma$, i.e. $Q \in \Gamma_\sigma \cap W_V$. This proves the claim.

Conclusion. The finitely many irreducible closed sets Γ_σ cover $X \times_Y X$. An irreducible closed subset lying in a finite union of closed sets lies in one of them, so the irreducible components of $X \times_Y X$ are the maximal ones among the Γ_σ. Let Z be such a component, $Z = \Gamma_\sigma$, with its reduced structure. Since X is reduced (it is integral, being normal) and $\gamma_\sigma : X \to X \times_Y X$ is a closed immersion with image Z, γ_σ factors through an isomorphism $X \xrightarrow{\sim} Z$ (the closed subscheme of $X \times_Y X$ defined by γ_σ is reduced with underlying set Z, hence equals Z with the reduced structure). Since $p_1 \circ \gamma_\sigma = \mathrm{id}_X$, the restriction $p_1|_Z : Z \to X$ is the inverse of this isomorphism, hence an isomorphism. In particular $Z \to X$ is finite and birational. □

Remark on (b): for the reduced structure only the reducedness of X was used; normality of X enters in (a) only. The route via "finite and birational onto a normal scheme" suggested in the book is replaced by the direct identification of the components with the graphs Γ_σ.

Exercise 1.11

Problem. Let X be an integral algebraic variety over a field k. Let $k' = K(X) \cap \bar{k}$.

(a) Let X' be the normalization of X. Show that X' is an algebraic variety over k'.

(b) Show that for any point $x \in X'$, k' is a subfield of $k(x)$. Deduce from this that k' is finite over k.

Solution. Reading of k'. Here $\bar{k}$ is an algebraic closure of k, and $K(X) \cap \bar{k}$ is the set of elements of $K(X)$ that are algebraic over k (any common overfield of $K(X)$ and $\bar{k}$ gives the same set). So k' is the algebraic closure of k in $K(X)$; it is a subfield of $K(X)$ containing k. An *algebraic variety* is a separated scheme of finite type over a field.

Step 1: k' lies in every local ring of X'. Let $\pi : X' \to X$ be the normalization. It is birational (Prop. 1.22), so $K(X') = K(X)$, and it is finite (Cor. 1.30). Let $x \in X'$. The ring $\mathcal{O}_{X',x}$ is a normal domain with fraction field $K(X)$ and contains k. An element $\alpha \in k'$ is algebraic over k, hence integral over k (divide its minimal polynomial by the leading coefficient; k is a field), hence integral over $\mathcal{O}_{X',x}$. It lies in the fraction field, so $\alpha \in \mathcal{O}_{X',x}$. Thus $k' \subseteq \mathcal{O}_{X',x}$ for all $x \in X'$. For a non-empty open $V \subseteq X'$ we have $\mathcal{O}_{X'}(V) = \bigcap_{x \in V} \mathcal{O}_{X',x}$ inside $K(X')$, so $k' \subseteq \mathcal{O}_{X'}(V)$ compatibly with restrictions.

(a). By Step 1, $\mathcal{O}_{X'}$ is a sheaf of k'-algebras, so X' is a k'-scheme (Cor. 2.3.26), and the k-structure is the one induced by $k \subseteq k'$. The scheme X' is of finite type over k (finite over X), so it is covered by finitely many affine opens $\operatorname{Spec} B_j$ with B_j finitely generated k-algebras. They are finitely generated k'-algebras (add k' to the generators, using $k \subseteq k'$). Hence $X' \to \operatorname{Spec} k'$ is of finite type. The morphism $X' \to X$ is finite, hence affine, hence separated, and X is separated over k, so X' is separated over k. If $g \circ f$ is separated then so is f; applying this to $X' \to \operatorname{Spec} k' \to \operatorname{Spec} k$ shows that $X' \to \operatorname{Spec} k'$ is separated. So X' is an algebraic variety over k' (it is even integral and normal).

(b). Let $x \in X'$. By Step 1 we have a ring homomorphism $k' \to \mathcal{O}_{X',x} \to k(x)$. It is injective because k' is a field, and it is k-linear. So k' is a subfield of $k(x)$, containing k.

Now choose a non-empty affine open $\operatorname{Spec} B \subseteq X'$ (B a finitely generated k-algebra, $B \neq 0$) and a maximal ideal $\mathfrak{m}$ of B, with corresponding point x. Then $k(x) = B/\mathfrak{m}$ is a finite extension of k (Cor. 2.1.12). Hence $k \subseteq k' \subseteq k(x)$ and $[k' : k] \leq [k(x) : k] < \infty$. □

Exercise 1.12

Problem. Let X be a normal and proper algebraic variety over a field k. Show that $\mathcal{O}_X(X) = K(X) \cap \bar{k}$.

Solution. Since X is normal, it is irreducible with normal (in particular reduced) local rings, hence integral. As in Ex. 1.11, $K(X) \cap \bar{k}$ means the set k' of elements of $K(X)$ algebraic over k.

$k' \subseteq \mathcal{O}_X(X)$. Let $\alpha \in k'$. It is integral over k. For each $x \in X$ the ring $\mathcal{O}_{X,x}$ is a normal domain with fraction field $K(X)$ containing k, so $\alpha \in \mathcal{O}_{X,x}$. As X is integral, $\mathcal{O}_X(X) = \bigcap_x \mathcal{O}_{X,x}$ in $K(X)$ (a family of germs in $K(X)$ that are all regular glues, since restriction maps to $K(X)$ are injective). Hence $\alpha \in \mathcal{O}_X(X)$.

$\mathcal{O}_X(X) \subseteq k'$. Since X is proper over k, $\mathcal{O}_X(X)$ is integral over k (Prop. 3.3.18). Thus every element of $\mathcal{O}_X(X) \subseteq K(X)$ is integral, hence algebraic, over k, so it lies in k'. □

Exercise 1.13

> **Problem.** Let X be a reduced algebraic variety. Show that X is normal at all of the generic points of X. Show that X contains an everywhere dense normal open subvariety.

Solution. Conventions. "Normal at x" means that $\mathcal{O}_{X,x}$ is a normal domain (Def. 1.1). Since X need not be irreducible, "normal open subvariety" is understood as: an open subscheme all of whose local rings are normal domains. It is a disjoint union of normal schemes in the sense of Def. 1.1, and it is normal in that sense if X is irreducible.

Generic points. The scheme X is Noetherian, so it has finitely many irreducible components $X_1, \ldots, X_n$ with generic points $\eta_1, \ldots, \eta_n$. The point η_i lies in no X_j with $j \neq i$ (otherwise $X_i = \overline{\{\eta_i\}} \subseteq X_j$, forcing $X_i = X_j$). The primes of $\mathcal{O}_{X,\eta_i}$ correspond to the points of X that generalize η_i, and η_i is the only one. So $\mathcal{O}_{X,\eta_i}$ is a local ring with a unique prime ideal, and it is reduced, so its maximal ideal (the nilradical) is 0. Hence $\mathcal{O}_{X,\eta_i}$ is a field, in particular a normal domain.

Dense normal open. Put $U_i = X_i \setminus \bigcup_{j \neq i} X_j$. It is open in X (a finite union of closed sets is closed) and contains η_i. It is irreducible (open in X_i) and reduced, hence integral, and it is an algebraic variety (open in X). The U_i are pairwise disjoint. By Cor. 1.30, the set N_i of normal points of U_i is open in U_i, hence in X, and it contains η_i by the first part. Put $N = \bigcup_i N_i$. It is open, the N_i are pairwise disjoint, and N contains all η_i, so $\overline{N} \supseteq \bigcup_i \overline{\{\eta_i\}} = X$, i.e. N is dense. For $x \in N_i$ we have $\mathcal{O}_{N,x} = \mathcal{O}_{X,x} = \mathcal{O}_{U_i,x}$, which is normal. Thus N is a dense open subvariety of X all of whose local rings are normal domains. □

Exercise 1.14

> **Problem.** Let X be an integral scheme of dimension 1, and let $f : X \to Y$ be a separated birational morphism of finite type from X to a Dedekind scheme. Show that f is an open immersion.

Solution. Conventions. Y is normal, hence integral, locally Noetherian of dimension ≤ 1. Let ξ, η be the generic points of X, Y. *Birational* (Ex. 3.2.6) means $f(\xi) = \eta$ and $f^{\#}_{\xi} : K(Y) \to K(X)$ is an isomorphism; we identify $K := K(Y) = K(X)$. The hypothesis $\dim X = 1$ will not be used.

Lemma A. Let S be a scheme, Z an integral S-scheme with function field $K(Z)$ and $j : \operatorname{Spec} K(Z) \to Z$ the canonical morphism, and T a separated S-scheme. If two S-morphisms $g_1, g_2 : Z \to T$ satisfy $g_1 \circ j = g_2 \circ j$, then $g_1 = g_2$.

Proof. Both send the generic point ζ of Z to the same point p. Take an affine open $W = \operatorname{Spec} R \ni p$ of T and a non-empty affine open $V_0 = \operatorname{Spec} C \subseteq g_1^{-1}(W) \cap g_2^{-1}(W)$. Then C is a domain with fraction field $K(Z)$. The restrictions $g_i|_{V_0} : V_0 \to W$ correspond to ring maps $\gamma_i : R \to C$. The compositions $R \to C \subseteq K(Z)$ correspond to $g_i \circ j$, so they agree, and since

$C \to K(Z)$ is injective, $\gamma_1 = \gamma_2$. So g_1, g_2 agree on the dense open V_0. As Z is reduced and T separated over S, Prop. 3.3.11 gives $g_1 = g_2$. $\square$

Step 1: local rings. Let $x \in X$, $y = f(x)$, $R = \mathcal{O}_{Y,y}$, $S = \mathcal{O}_{X,x}$. These are domains with fraction field K, and $R \subseteq K$, $S \subseteq K$ (localizations of domains). The local homomorphism $f_x^\# : R \to S$ is compatible with the embeddings into K (because $f_\xi^\#$ is the identity of K), so $R \subseteq S \subseteq K$. We claim $R = S$.

If $x = \xi$, then $y = \eta$ and $R = S = K$. Let $x \neq \xi$. Then S is not a field, since ξ is a proper generization of x and $\operatorname{Spec} S$ has at least two points. If $y = \eta$, then $K = R \subseteq S \subseteq K$ would make S a field; so $y \neq \eta$. Then R is a normal Noetherian local domain of dimension ≥ 1, hence of dimension 1 (as $\dim Y \leq 1$), hence a principal ideal domain (Prop. 1.12) and a discrete valuation ring; let π be a uniformizer. As $f_x^\#$ is local, $\pi \in \mathfrak{m}_x$, so π is not a unit of S. If $s \in S \setminus R$, write $s = u\pi^n$ with $u \in R^*$ and $n < 0$; then $\pi^{-1} = s\, u^{-1}\pi^{-n-1} \in S$, contradicting that π is not a unit of S. Hence $S = R$.

In particular we showed: $x \neq \xi \Rightarrow f(x) \neq \eta$.

Step 2: f is injective. Let $f(x_1) = f(x_2) = y$. If $y = \eta$, then $x_1 = x_2 = \xi$ by the remark at the end of Step 1. Otherwise $x_1, x_2 \neq \xi$ and $R := \mathcal{O}_{Y,y} = \mathcal{O}_{X,x_1} = \mathcal{O}_{X,x_2}$ as subrings of K. Let $g_i : \operatorname{Spec} R = \operatorname{Spec} \mathcal{O}_{X,x_i} \to X$ be the canonical morphisms. They are Y-morphisms (both composites with f are the canonical $\operatorname{Spec} \mathcal{O}_{Y,y} \to Y$), and both restrict on $\operatorname{Spec} K$ to the canonical morphism $\operatorname{Spec} K(X) \to X$. As X is separated over Y and $\operatorname{Spec} R$ is integral, Lemma A gives $g_1 = g_2$. Evaluating at the closed point of $\operatorname{Spec} R$ gives $x_1 = x_2$.

Step 3: f is a local isomorphism. Let $x \in X$, $y = f(x)$. Choose affine opens $W = \operatorname{Spec} A \ni y$ and $V = \operatorname{Spec} B \ni x$ with $f(V) \subseteq W$; $B = A[b_1, \ldots, b_n]$ is a finitely generated A-algebra since f is of finite type. Both are domains inside K, with $A \subseteq B$. Let $\mathfrak{p}, \mathfrak{q}$ be the primes of y, x. By Step 1, $B_\mathfrak{q} = A_\mathfrak{p}$. So each $b_i \in A_\mathfrak{p}$ and there is $s \in A \setminus \mathfrak{p}$ with all $b_i \in A_s$. Since $\mathfrak{q} \cap A = \mathfrak{p}$, $s \notin \mathfrak{q}$, so $x \in V' := D(s) \subseteq V$. Now $B_s = A_s[b_1, \ldots, b_n] = A_s$ (the inclusion $A_s \subseteq B_s \subseteq A_s$). So f maps V' isomorphically onto the open subset $D_A(s)$ of Y.

Conclusion. By Step 3, f is an open map and a local isomorphism; by Step 2 it is injective. Hence f is a homeomorphism onto the open subset $f(X)$ and induces isomorphisms on all stalks, i.e. an isomorphism $X \to f(X)$ onto an open subscheme of Y. So f is an open immersion. $\square$

Exercise 1.15

Problem. Let X be a locally Noetherian, separated, integral scheme, and U an affine open subset of X different from X. Let us suppose that the normalization X' of X is also locally Noetherian (e.g., if $X' \to X$ is finite). We want to show that the irreducible components of $X \setminus U$ are of codimension 1 in X. Let ξ be a generic point of $X \setminus U$.

(a) Show that there exists an affine open neighborhood V of ξ in X such that ξ is the unique generic point of $V \setminus (U \cap V)$.

(b) Show that $V \cap U$ is an affine open subset of V. Using Theorem 1.14, show that $\dim \mathcal{O}_{X,\xi} = 1$ if X is normal.

(c) Let $\pi : X' \to X$ be the normalization morphism. Show that any point of $\pi^{-1}(\xi)$ is a generic point of $X' \setminus U'$, where $U' = \pi^{-1}(U)$. Show that $\dim \mathcal{O}_{X,\xi} = \max_{\xi' \in \pi^{-1}(\xi)} \dim \mathcal{O}_{X',\xi'} = 1$.

(d) Show that the statement still holds if X is not necessarily integral, but is such that the normalization of its (reduced) irreducible components are locally Noetherian.

Solution. Hypothesis needed. We assume $U \neq \emptyset$. (If $U = \emptyset$, which is an affine open subset different from X, then $X \setminus U = X$ has the single component X, of codimension 0.) In (d) the analogous hypothesis is that U meets every irreducible component of X, and without it (d) is false (see (d)). Recall $\operatorname{codim}(\overline{\{\xi\}}, X) = \dim \mathcal{O}_{X,\xi}$. The statement to prove is therefore $\dim \mathcal{O}_{X,\xi} = 1$.

(a). Let V_0 be an affine open neighborhood of ξ. It is a Noetherian space, so $V_0 \cap (X \setminus U)$ has finitely many irreducible components $Z_0, Z_1, \ldots, Z_r$, where Z_0 is the closure of $\{\xi\}$ in V_0 (the closure of $\{\xi\}$ is an irreducible component of $X \setminus U$, so Z_0 is a component of the trace). We have $\xi \notin Z_j$ for $j \geq 1$ (otherwise $Z_0 \subseteq Z_j$, hence $Z_0 = Z_j$). Then $V_1 = V_0 \setminus \bigcup_{j\geq 1} Z_j$ is an open neighborhood of ξ and $V_1 \cap (X \setminus U) = Z_0 \setminus \bigcup_{j\geq 1} Z_j$ is a non-empty open subset of the irreducible Z_0, hence irreducible with generic point ξ. Finally choose a principal open $V = D(g) \subseteq V_1$ containing ξ (it is affine). Then $V \cap (X \setminus U)$ is a non-empty open subset of $V_1 \cap (X \setminus U)$, hence irreducible with generic point ξ.

(b). Since X is separated and U, V are affine, $U \cap V$ is affine (it is the preimage of the closed diagonal in $U \times V$, a closed subscheme of an affine scheme). Now assume X is normal. Write $V = \operatorname{Spec} A$. Then A is a normal Noetherian domain (Prop. 1.5, and X is locally Noetherian). Let $F = V \setminus (U \cap V) = V \cap (X \setminus U)$; by (a) it is irreducible with generic point ξ; let $\mathfrak{p}$ be the prime of ξ, so $F = V(\mathfrak{p})$ and $\operatorname{codim}(F, V) = \operatorname{ht} \mathfrak{p} = \dim \mathcal{O}_{X,\xi}$.

Suppose $\operatorname{ht} \mathfrak{p} \geq 2$. By Theorem 1.14 the restriction $A = \mathcal{O}(V) \to \mathcal{O}(V \setminus F) = \mathcal{O}(U \cap V)$ is an isomorphism. As $U \cap V$ is affine, the open immersion $U \cap V \to V$ is Spec of this restriction map, hence an isomorphism. So $F = \emptyset$, contradicting $\xi \in F$. Hence $\operatorname{ht} \mathfrak{p} \leq 1$. Also $\operatorname{ht} \mathfrak{p} \geq 1$: the generic point of X lies in $U \neq \emptyset$, so ξ is not the generic point of X and $\mathcal{O}_{X,\xi}$ is not a field. So $\dim \mathcal{O}_{X,\xi} = 1$.

(c). The morphism π is integral and birational (Prop. 1.22). By Def. 1.20, preimages of affine opens are affine, so U' is affine and π is affine, hence separated. So X' is separated (over $\mathbb{Z}$, as X is). The ring map of π over each affine open is injective and integral, so π is surjective (Prop. 2.5.10(b)). Thus $U' \neq X'$ (a point over ξ lies in $X' \setminus U'$) and $U' \neq \emptyset$ (it contains the generic point). The scheme X' is integral, normal, and locally Noetherian by hypothesis.

Points over ξ are generic points of $X' \setminus U'$. Take $V = \operatorname{Spec} A$ as in (a), so $\pi^{-1}(V) = \operatorname{Spec} A'$ with $A \subseteq A'$ integral (both inside $K(X)$). Let $\mathfrak{p} \subset A$ be the prime of ξ and $F = V(\mathfrak{p})$ as before. Then $\pi^{-1}(F) = V(\mathfrak{p}A') = \pi^{-1}(V) \cap (X' \setminus U')$. Let $\mathfrak{q} \subset A'$ be a prime with $\mathfrak{q} \cap A = \mathfrak{p}$ (these are the points of $\pi^{-1}(\xi)$). If $\mathfrak{q}'$ is a prime with $\mathfrak{p}A' \subseteq \mathfrak{q}' \subseteq \mathfrak{q}$, then $\mathfrak{p} \subseteq \mathfrak{q}' \cap A \subseteq \mathfrak{q} \cap A = \mathfrak{p}$, so $\mathfrak{q}'$ and $\mathfrak{q}$ have the same contraction, and by incomparability for integral extensions (proof of Prop. 2.5.10(a)) $\mathfrak{q}' = \mathfrak{q}$. So $\mathfrak{q}$ is a minimal prime over $\mathfrak{p}A'$, i.e. a generic point of an irreducible component of $\pi^{-1}(F)$. An irreducible component of the trace on an open subset of a closed set C has closure an irreducible component of C, so $\mathfrak{q}$ is a generic point of an irreducible component of $X' \setminus U'$.

Equality of dimensions. Applying (b) to X', U' and each $\xi' \in \pi^{-1}(\xi)$ gives $\dim \mathcal{O}_{X',\xi'} = 1$. It remains to show $\dim \mathcal{O}_{X,\xi} = \max_{\xi'} \dim \mathcal{O}_{X',\xi'}$. Let $T = A \setminus \mathfrak{p}$. Then $A_{\mathfrak{p}} \subseteq T^{-1}A' =: A'_{\mathfrak{p}}$ is injective and integral, so $\dim A_{\mathfrak{p}} = \dim A'_{\mathfrak{p}}$ (Prop. 2.5.10(b)). The maximal ideals of $A'_{\mathfrak{p}}$ are the $\mathfrak{q}A'_{\mathfrak{p}}$ with $\mathfrak{q} \cap A = \mathfrak{p}$ (a prime of an integral extension is maximal iff its contraction is, by

Prop. 2.5.10(b)), and $(A'_{\mathfrak{p}})_{\mathfrak{q}A'_{\mathfrak{p}}} = A'_{\mathfrak{q}} = \mathcal{O}_{X',\mathfrak{q}}$. By Prop. 2.5.8(c), $\dim A'_{\mathfrak{p}} = \max_{\mathfrak{q}} \dim A'_{\mathfrak{q}}$ (the set of such $\mathfrak{q}$ is non-empty by surjectivity). Therefore $\dim \mathcal{O}_{X,\xi} = \max_{\xi' \in \pi^{-1}(\xi)} \dim \mathcal{O}_{X',\xi'} = 1$.

(d). Let X be locally Noetherian and separated, with reduced irreducible components X_i (integral closed subschemes) whose normalizations are locally Noetherian, and let $U \neq \emptyset$ be an affine open subset with $X \setminus U \neq \emptyset$.

Hypothesis needed. We must assume that U meets every irreducible component of X. Otherwise the statement is false: for $X = \mathbb{A}^1_k \sqcup \mathbb{A}^1_k$ (locally Noetherian, separated, with normal Noetherian components) and U the first copy, $X \setminus U$ is the second copy, which is an irreducible component of X of codimension 0.

Let ξ be a generic point of an irreducible component D of $X \setminus U$ (so $\dim \mathcal{O}_{X,\xi}$ is what we must compute). The primes of $\mathcal{O}_{X,\xi}$ are the points generalizing ξ; since $\mathcal{O}_{X,\xi}$ is Noetherian it has finitely many minimal primes, corresponding to the (finitely many) irreducible components $X_1, \ldots, X_m$ of X through ξ, and $\mathcal{O}_{X,\xi}/\mathfrak{p}_i = \mathcal{O}_{X_i,\xi}$ for the reduced structure. Every chain of primes starts at a minimal prime, so $\dim \mathcal{O}_{X,\xi} = \max_{i \leq m} \dim \mathcal{O}_{X_i,\xi}$.

Fix $i \leq m$. The scheme X_i is integral, locally Noetherian, separated, with locally Noetherian normalization. The set $U_i = U \cap X_i$ is non-empty (by the hypothesis) and affine (a closed subscheme of the affine U). It differs from X_i because $\xi \in X_i \setminus U_i$. The point ξ is a generic point of an irreducible component of $X_i \setminus U_i = X_i \cap (X \setminus U)$: the closure D of $\{\xi\}$ is contained in $X_i \setminus U_i$ (as $\xi \in X_i$ and X_i is closed), and any irreducible closed subset of $X_i \setminus U_i \subseteq X \setminus U$ containing D equals D by maximality of D. By (c) applied to (X_i, U_i), $\dim \mathcal{O}_{X_i,\xi} = 1$. Hence $\dim \mathcal{O}_{X,\xi} = 1$, i.e. every irreducible component of $X \setminus U$ has codimension 1 in X. □

Exercise 1.16

Problem. Let C be a normal proper curve over a field k. We are going to show that C is projective over k.

(a) Let $\{U_i\}_i$ be a finite covering of C by affine open subsets. Each U_i admits an open immersion into a projective variety Y_i over k (Exercise 2.3.18). Let Y be the fibered product of the Y_i over k. Show that the canonical morphism $\bigcap_i U_i \to Y$ extends uniquely to a proper morphism $h : C \to Y$.

(b) Let $Z = h(C)$ be the closed subvariety of Y endowed with the reduced structure. Show that h induces a surjective morphism $f : C \to Z$ and that the projection $Y \to Y_i$ induces a dominant morphism $Z \to Y_i$.

(c) Show that for any open subset U_i, the open immersion $U_i \to Y_i$ factors into $U_i \to C \to Z \to Y_i$. Deduce from this that $f_x^{\#} : \mathcal{O}_{Z,f(x)} \to \mathcal{O}_{C,x}$ is an isomorphism for every $x \in U_i$. Show that f is an isomorphism and that C is projective over k.

Solution. A curve is an algebraic variety of dimension 1. As C is normal, it is irreducible with normal local rings, hence integral. We may discard empty U_i and take $r \geq 1$ pieces $U_1, \ldots, U_r$.

(a). *Construction of Y_i.* The affine variety U_i (integral) is a closed subscheme of some $\mathbb{A}^n_k \subseteq \mathbb{P}^n_k$. Let Y_i be the closure of U_i in $\mathbb{P}^n_k$ with its reduced structure. It is an integral projective k-scheme, and $U_i = Y_i \cap \mathbb{A}^n_k$ is a dense open subscheme of Y_i (because U_i is closed in $\mathbb{A}^n_k$). Let $j_i : U_i \to Y_i$ be the open immersion. The fibered product $Y = Y_1 \times_k \cdots \times_k Y_r$ is projective over k (Segre embedding); let $p_i : Y \to Y_i$ be the projections.

Let $W = \bigcap_i U_i$; it is non-empty and open since C is irreducible. Let $j : W \to Y$ be the morphism with $p_i \circ j = j_i|_W$. Apply Cor. 1.17 to $S = \operatorname{Spec} k$: C is normal of finite type over k of dimension 1, $Y \to S$ is proper, and j is defined on the non-empty open set W. It extends uniquely to a morphism $h : C \to Y$. Since $C \to \operatorname{Spec} k$ is proper and $Y \to \operatorname{Spec} k$ is separated, h is proper (Prop. 3.3.16(e)).

(b). A proper morphism is closed, so $Z = h(C)$ is closed in Y; it is irreducible (image of an irreducible space), so with the reduced structure it is an integral closed subscheme, proper over k. Since C is reduced, h factors through a morphism $f : C \to Z$, which is surjective by definition of Z. Let $q_i = p_i|_Z : Z \to Y_i$. It is proper (Prop. 3.3.16(e)), hence closed. Its image contains $p_i(h(W)) = j_i(W)$, which is dense in Y_i (W is dense in U_i, and U_i is dense in Y_i). So $q_i(Z) = Y_i$: the morphism q_i is surjective, in particular dominant.

(c). *Factorization.* The morphisms $p_i \circ h|_{U_i}$ and j_i from U_i to Y_i are k-morphisms that agree on the dense open W, U_i is reduced, and Y_i is separated over k. By Prop. 3.3.11 they coincide: $j_i = q_i \circ f|_{U_i}$. So $U_i \to C \to Z \to Y_i$ is the open immersion j_i.

Stalks. Let $x \in U_i$ and $z = f(x)$. Then $\mathcal{O}_{Y_i, j_i(x)} \xrightarrow{q_i^\#} \mathcal{O}_{Z,z} \xrightarrow{f_x^\#} \mathcal{O}_{C,x}$ has composite $j_{i,x}^\#$, an isomorphism. So $f_x^\#$ is surjective. It is also injective: f is surjective and C is irreducible, so f maps the generic point of C to the generic point of Z, and $f_x^\#$ is compatible with the embeddings $\mathcal{O}_{Z,z} \subseteq K(Z) \hookrightarrow K(C) \supseteq \mathcal{O}_{C,x}$ (all rings are domains, Z and C integral). So $f_x^\#$ is an isomorphism for every $x \in \bigcup U_i = C$. In particular $K(Z) \to K(C)$ is an isomorphism and we identify the two fields.

Injectivity of f. Let $f(x_1) = f(x_2) = z$. By the above, $\mathcal{O}_{C,x_1}$ and $\mathcal{O}_{C,x_2}$ both equal $R := \mathcal{O}_{Z,z}$ as subrings of $K(C)$. If $R = K(C)$ is a field, then $x_1 = x_2$ is the generic point of C (the local ring of an integral scheme at a point is a field only at the generic point). Otherwise R is a normal Noetherian local domain of dimension 1 (as $\dim C = 1$ and x_i is not generic), hence a discrete valuation ring. Let $g_i : \operatorname{Spec} R = \operatorname{Spec} \mathcal{O}_{C,x_i} \to C$ be the canonical morphisms. They are k-morphisms that agree on $\operatorname{Spec} K(C)$, and C is separated over k, so $g_1 = g_2$ by Lemma A of Ex. 1.14 (applied with $S = \operatorname{Spec} k$, $Z = \operatorname{Spec} R$, $T = C$). Hence $x_1 = g_1(\text{closed point}) = g_2(\text{closed point}) = x_2$.

Conclusion. The morphism f is bijective and proper (Prop. 3.3.16(e), Z being separated), hence closed, so it is a homeomorphism. For a homeomorphism the stalk of $f_*\mathcal{O}_C$ at z is $\mathcal{O}_{C,f^{-1}(z)}$, so $f^\# : \mathcal{O}_Z \to f_*\mathcal{O}_C$ is an isomorphism of sheaves (on stalks, by the above). Thus f is an isomorphism of schemes. Therefore $C \simeq Z$ is a closed subscheme of the projective k-scheme Y, so C is projective over k. □

Exercise 1.17

Problem. Let U be an integral affine algebraic curve over a field k.

(a) Show that there exists a proper curve $\hat{U}$ over k verifying the following properties: (1) there exists an open immersion $i : U \to \hat{U}$; (2) the points of $\hat{U} \setminus U$ are normal in $\hat{U}$.

(b) Show that $\hat{U}$ is the 'smallest proper completion' of U, in the sense that any morphism $U \to X$ into a proper variety over k factors uniquely through the open immersion $U \to \hat{U}$. Deduce from this that $\hat{U}$ is unique up to isomorphism.

(c) Show that there exists a finite morphism $U \to \mathbb{A}^1_k$, and that it extends to a finite morphism $f : \hat{U} \to \mathbb{P}^1_k$ such that $U = f^{-1}(\mathbb{A}^1_k)$.

Solution. Conventions. In (a) and (b) the curve $\hat{U}$ is required to be *integral* (equivalently $i(U)$ is dense in $\hat{U}$); without this the uniqueness in (b) fails (one could add a disjoint proper curve). Morphisms in (b) are k-morphisms. We call such a pair $(\hat{U}, i)$ a *completion* of U. We write $U = \operatorname{Spec} A$, $K = K(U) = \operatorname{Frac}(A)$. Lemma A of Ex. 1.14 is used again: two S-morphisms from an integral scheme to a separated S-scheme that agree on $\operatorname{Spec} K(Z)$ are equal.

Construction (giving (a) and the existence in (c)). A is a finitely generated k-domain of dimension 1. By Noether normalization (Prop. 2.1.9) there is $t \in A$, transcendental over k, with $k[t] \subseteq A$ finite. So $\psi : U \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$ is a finite morphism; this is the first claim of (c). Put $s = t^{-1} \in K$; the field K is finite over $k(t) = k(s)$. Let C_0 be the integral closure of $k[s]$ in K; it is a normal domain, finite over $k[s]$ (Prop. 1.27 applied to $\operatorname{Spec} k[s]$ and $L = K$), of dimension 1. Put

$$C' = C_0 \cap A_t, \quad \text{inside } K, \text{ where } A_t = A[s].$$

(i) C' is a finite $k[s]$-module (a submodule of the finite module C_0 over the Noetherian ring $k[s]$) and $k[s] \subseteq C' \subseteq C_0$, with C_0 integral over C'.

(ii) $C'[1/s] = A_t$. The inclusion $\subseteq$ is clear. Let $a \in A_t$. As A is finite over $k[t]$, A_t is finite over $k[t, t^{-1}] = k[s, s^{-1}]$, so a satisfies an equation $a^n + c_1 a^{n-1} + \cdots + c_n = 0$ with $c_i \in k[s, s^{-1}]$. Choose N with $s^N c_i \in k[s]$ for all i. Then $s^N a$ satisfies the monic equation with coefficients $s^{iN} c_i = s^{(i-1)N} \cdot s^N c_i \in k[s]$, so $s^N a \in C_0 \cap A_t = C'$ and $a = s^{-N}(s^N a) \in C'[1/s]$.

(iii) *For every prime $\mathfrak{q}'$ of C' with $s \in \mathfrak{q}'$, the ring $C'_{\mathfrak{q}'}$ is normal.* Let $J = \{c \in C_0 \mid cC_0 \subseteq C'\}$ (the conductor); it is an ideal of C_0, contained in C'. It is non-zero: C_0 is a finitely generated $k[s]$-module inside $K = \operatorname{Frac}(C')$ (by (ii), $\operatorname{Frac} C' = \operatorname{Frac} A_t = K$), so a common denominator $v \in C' \setminus \{0\}$ of generators satisfies $vC_0 \subseteq C'$. Moreover J is saturated with respect to s: if $x \in C_0$ and $sx \in J$, then for $y \in C_0$ we get $sxy \in C' \subseteq A_t$, hence $xy \in A_t \cap C_0 = C'$ (as s is a unit of A_t), so $x \in J$.

Suppose $J \subseteq \mathfrak{q}'$. By Prop. 2.5.10(b) there is a prime $\mathfrak{q}$ of C_0 with $\mathfrak{q} \cap C' = \mathfrak{q}'$; then $J \subseteq \mathfrak{q}$ and $s \in \mathfrak{q}$. As C_0 is a Noetherian domain of dimension 1 and $J \neq 0$, the prime $\mathfrak{q} \neq 0$ is maximal and minimal over J, hence an associated prime of C_0/J: $\mathfrak{q} = \operatorname{Ann}(\bar{z})$ for some $z \in C_0$ with $z \notin J$. Then $sz \in J$, so $z \in J$ by saturation, a contradiction. Hence $J \not\subseteq \mathfrak{q}'$. Pick $u \in J \setminus \mathfrak{q}'$. For every $x \in C_0$ we have $ux \in C'$, so $x = ux/u \in C'_{\mathfrak{q}'}$. Thus $C'_{\mathfrak{q}'} = (C' \setminus \mathfrak{q}')^{-1} C_0$, a localization of the normal domain C_0, hence normal (Lemma 1.4(b)).

Gluing. Let $E = \operatorname{Spec} C'$. By (ii), the open subschemes $D(t) = \operatorname{Spec} A_t \subseteq U$ and $D(s) = \operatorname{Spec} C'[1/s] \subseteq E$ are the same $\operatorname{Spec} A_t$. Let $\hat{U} = U \cup_{\operatorname{Spec} A_t} E$. Let $f : \hat{U} \to \mathbb{P}^1_k = \operatorname{Spec} k[t] \cup \operatorname{Spec} k[s]$ be given on U by ψ (from $k[t] \subseteq A$) and on E by $k[s] \subseteq C'$. They agree on the overlap since both come from $k[s, s^{-1}] \subseteq A_t$. We have $f^{-1}(\mathbb{A}^1_k) = U$ (the points of E outside $V(s)$ lie in $D(s) \subseteq U$) and $f^{-1}(\operatorname{Spec} k[s]) = E$. Both are finite over their targets, so f is finite.

Consequently $\hat{U}$ is separated (finite morphisms are affine, and $\mathbb{P}^1_k$ is separated), of finite type, and proper over k (a finite morphism is proper, and $\mathbb{P}^1_k$ is proper; Prop. 3.3.16(b)). It is integral, being glued from the integral schemes U, E along a non-empty open set, and has dimension 1. U is an open subscheme (so (1) holds), and $\hat{U} \setminus U = V(s) \subseteq E$. By (iii) its local rings are normal, so (2) holds. This proves (a) and the existence statement in (c).

(b). Let $(\hat{U}, i)$ be any completion; identify U with an open dense subscheme. The set $T = \hat{U} \setminus U$ is closed and proper in the integral curve $\hat{U}$, so it is a finite set of closed points. Let $\varphi : U \to X$ be a k-morphism to a proper variety.

Uniqueness of an extension $\Phi : \hat{U} \to X$ follows from Lemma A ($\hat{U}$ integral, X separated, and two extensions agree on $U \ni$ generic point).

Existence. Let $x \in T$ and $R = \mathcal{O}_{\hat{U},x}$. It is a normal Noetherian local domain of dimension 1 (as x is a closed point of a curve, normal by (2)), hence a discrete valuation ring (Prop. 1.12) with fraction field $K(\hat{U}) = K(U) = K$. The morphism $\operatorname{Spec} K \to U \xrightarrow{\varphi} X$ extends uniquely to a k-morphism $\operatorname{Spec} R \to X$ by the valuative criterion (Cor. 3.3.26, as $X \to \operatorname{Spec} k$ is proper), exactly as in the proof of Prop. 1.16. By Ex. 3.2.4 (as X is of finite type over $\operatorname{Spec} k$) it extends to a morphism $g_x : U_x \to X$ on an open neighborhood U_x of x in $\hat{U}$; shrinking, $U_x \cap T = \{x\}$. The restrictions of g_x and φ to the integral scheme $U \cap U_x$ agree on $\operatorname{Spec} K$, hence agree by Lemma A. Likewise g_x and $g_{x'}$ agree on $U_x \cap U_{x'}$. So φ and all g_x glue to a morphism Φ on $U \cup \bigcup_{x \in T} U_x = \hat{U}$, which extends φ.

Uniqueness of $\hat{U}$. Let $(\hat{U}, i)$ and $(\hat{U}', i')$ be two completions. Both $\hat{U}, \hat{U}'$ are proper varieties, so $i' : U \to \hat{U}'$ extends to $\alpha : \hat{U} \to \hat{U}'$ with $\alpha \circ i = i'$, and i extends to $\beta : \hat{U}' \to \hat{U}$ with $\beta \circ i' = i$. Then $\beta \circ \alpha$ and $\mathrm{id}_{\hat{U}}$ both extend $i : U \to \hat{U}$, so they are equal by uniqueness; similarly $\alpha \circ \beta = \mathrm{id}$. Hence $\hat{U} \simeq \hat{U}'$ compatibly with i, i'.

(c), conclusion. The finite morphism $\psi : U \to \mathbb{A}^1_k$ was constructed. For the completion built above, f is a finite morphism extending ψ (composed with $\mathbb{A}^1_k \subseteq \mathbb{P}^1_k$) with $f^{-1}(\mathbb{A}^1_k) = U$. By (b) any other completion is isomorphic to this one by an isomorphism compatible with i, and the extension of ψ to it is unique, so the same holds for any completion $\hat{U}$ of U. □

Exercise 1.18

Problem. Let k be a field, and let $P(x) \in k[x]$ be a monic polynomial.

(a) Show that $X = \operatorname{Spec} k[x, y]/(y^2 - P(x))$ is integral if and only if $P(x)$ is not a square.

(b) Let us suppose $\mathrm{char}(k) \neq 2$. Show that X is normal if and only if $P(x)$ is a polynomial without a square factor.

(c) Let us suppose $\mathrm{char}(k) = 2$ and k is perfect. Then $P(x)$ can be written in a unique way as $P(x) = xQ_1(x)^2 + Q_2(x)^2$. Show that X is normal if and only if $P(x) \in k^* x + k[x^2]$. If k is not perfect, show that this condition is sufficient.

(d) Supposing X integral, and $\mathrm{char}(k) \neq 2$ or k perfect, determine the normalization of X.

Solution. Let $B = k[x, y]/(y^2 - P)$, $K_0 = k(x)$. As a $k[x]$-module B is free with basis $1, y$ (division by the monic polynomial $y^2 - P$), so every element of B is uniquely $a + by$ with $a, b \in k[x]$, and B is integral over $k[x]$. By Prop. 1.5, $X = \operatorname{Spec} B$ is normal iff B is a normal domain.

(a). If $P = Q^2$ with $Q \in k[x]$, then $(y - Q)(y + Q) = y^2 - P = 0$ in B, while $y \pm Q \neq 0$ (the coefficient of y is 1). So B is not a domain. Conversely, suppose P is not a square in $k[x]$. If $y^2 - P$ were reducible in the UFD $k[x][y]$, being monic of degree 2 in y it would be a product $(y - a)(y - b)$ with $a, b \in k[x]$ (a factor of y-degree 0 would divide the monic polynomial, hence

be a unit); then $a+b=0$ and $ab=-P$, so $P=a^2$, a contradiction. So y^2-P is irreducible, hence prime in the UFD $k[x,y]$, and B is a domain. (We also need $B\neq 0$, which is clear.)

If B is a domain, then $y\notin K_0$ and $L:=\operatorname{Frac} B=K_0(y)=K_0\oplus K_0 y$ has degree 2 over K_0. Indeed $y\in K_0$ would make $P=y^2$ a square in K_0, hence in $k[x]$ ($k[x]$ is normal, Lemma 1.4(a)).

(b). Assume $\operatorname{char} k\neq 2$. We read "without a square factor" as: $P\neq 1$ and there is no non-constant Q with $Q^2\mid P$ (the constant $P=1$ is a square, and X is then not integral; so for $P=1$ the printed statement needs this reading).

Normal $\Rightarrow$ squarefree and $P\neq 1$. If X is normal then B is a domain, so P is not a square by (a); in particular $P\neq 1$. Suppose $P=Q^2R$ with Q non-constant and $R\in k[x]$. Let $z=y/Q\in L$. Then $z^2=R\in k[x]$, so z is integral over $k[x]$, hence over B. But $z=0+Q^{-1}y$ and $Q^{-1}\notin k[x]$, so $z\notin B$ by uniqueness of the form $a+by$ ($1,y$ is a basis of L over K_0). So B is not normal, a contradiction.

Squarefree and $P\neq 1\Rightarrow$ normal. Then P is not a square, so B is a domain and $L=K_0\oplus K_0y$. The automorphism σ of L over K_0 with $\sigma(y)=-y$ exists (the minimal polynomial T^2-P has the distinct roots $\pm y$, as $2\neq 0$). Let $z=a+by\in L$ ($a,b\in K_0$) be integral over B; it is integral over $k[x]$, and so is $\sigma(z)$ (apply σ to an integral equation with coefficients in $k[x]$). Hence $z+\sigma(z)=2a$ and $z\sigma(z)=a^2-Pb^2$ are elements of K_0 integral over the normal ring $k[x]$, so they lie in $k[x]$. So $a\in k[x]$ (as 2 is invertible) and $Pb^2\in k[x]$. Write $b=u/v$ with $u,v\in k[x]$ coprime, $v\neq 0$. Then $v^2\mid Pu^2$, so $v^2\mid P$, hence v is a constant since P has no square factor. Thus $b\in k[x]$ and $z\in B$. So B is integrally closed in L, i.e. normal.

(c). Let $\operatorname{char} k=2$.

Decomposition. If k is perfect, write $P=\sum a_ix^i$. The even part is $\sum a_{2j}x^{2j}=Q_2^2$ with $Q_2=\sum\sqrt{a_{2j}}x^j$, and the odd part is $x\cdot Q_1^2$ with $Q_1=\sum\sqrt{a_{2j+1}}x^j$. If $xQ_1^2+Q_2^2=xQ_1'^2+Q_2'^2$, then $x(Q_1-Q_1')^2=(Q_2-Q_2')^2$ (characteristic 2); the left side has odd degree if $Q_1\neq Q_1'$, the right side even degree, so $Q_1=Q_1'$ and then $Q_2=Q_2'$. Note $k[x^2]\supseteq Q_2^2$, and over a perfect field $k[x^2]$ consists of the squares of polynomials. Hence $P\in k^*x+k[x^2]$ iff $Q_1\in k^*$.

Sufficiency (any field k of characteristic 2). Suppose $P=cx+S(x^2)$ with $c\in k^*$ and $S\in k[T]$. Then $P'=c\neq 0$ (formal derivative in x), whereas a square Q^2 has derivative $2QQ'=0$; so P is not a square and B is a domain with $L=K_0\oplus K_0y$ by (a). Let $z=a+by\in L$ be integral over B (hence over $k[x]$). Then $z^2=a^2+b^2P$ (characteristic 2) is an element of K_0 integral over $k[x]$, so $a^2+b^2P\in k[x]$. Write $a=\alpha/v$, $b=\beta/v$ with $\alpha,\beta,v\in k[x]$, $v\neq 0$ and $\gcd(\alpha,\beta,v)=1$. Then $\alpha^2+\beta^2P=v^2g$ with $g\in k[x]$. Differentiating, $(\alpha^2)'=0$ and $(\beta^2P)'=\beta^2P'=c\beta^2$, $(v^2g)'=v^2g'$. So $c\beta^2=v^2g'$, hence $v^2\mid\beta^2$, so $v\mid\beta$. Then $v^2\mid\alpha^2=v^2g-\beta^2P$, so $v\mid\alpha$. Therefore $v\mid\gcd(\alpha,\beta,v)=1$, so $a,b\in k[x]$ and $z\in B$. Hence B is normal.

Necessity (k perfect). Suppose B is normal, so P is not a square and $Q_1\neq 0$ (otherwise $P=Q_2^2$). Suppose Q_1 is non-constant. Put $z=(y+Q_2)/Q_1\in L$. Then $z^2=(y^2+Q_2^2)/Q_1^2=(P+Q_2^2)/Q_1^2=x$ (characteristic 2), so z is integral over $k[x]\subseteq B$. But $z=Q_2/Q_1+Q_1^{-1}y\notin B$, as $Q_1^{-1}\notin k[x]$. This contradicts normality. Hence $Q_1\in k^*$, i.e. $P\in k^*x+k[x^2]$.

So for k perfect: X normal $\iff$ $P\in k^*x+k[x^2]$. For k not perfect the condition is still sufficient, as shown, but not necessary: for $k=\mathbb{F}_2(a)$ with $a\notin k^2$ and $P=x^2+a$ (not in $k^*x+k[x^2]$), put $w=x+y$; then $w^2=x^2+y^2=a$ in B, and $B=k[x,w]/(w^2-a)=k(\sqrt{a})[x]$ is a polynomial ring over a field, hence normal.

(d). Assume B is a domain (P not a square) and $\operatorname{char} k \neq 2$ or k perfect. In each case we exhibit a normal domain B' with $B \subseteq B' \subseteq L$, integral over B; then B' is the integral closure of B in L and $\operatorname{Spec} B' \to X$ is the normalization (Lemma 1.21).

Case $\operatorname{char} k \neq 2$. Factor $P = Q^2 R$ with $Q, R \in k[x]$ monic and R squarefree (take Q the product of $p_i^{\lfloor e_i/2 \rfloor}$ and R the product of $p_i^{e_i - 2\lfloor e_i/2 \rfloor}$ over the prime factorization $P = \prod p_i^{e_i}$). $R \neq 1$ since P is not a square. Put $w = y/Q \in L$, so $w^2 = R$, and $B' = k[x, w] = k[x][w] \subseteq L$; it is $\simeq k[x, w]/(w^2 - R)$ (the polynomial $T^2 - R$ is the minimal polynomial of w over K_0 and $k[x]$ is normal), which is normal by (b) applied to R. Moreover $B \subseteq B'$ ($y = Qw$), B' is integral over B ($w^2 = R \in k[x]$), and $\operatorname{Frac} B' = K_0(w) = K_0(y) = L$. So the normalization of X is

$$X' = \operatorname{Spec} k[x, w]/(w^2 - R) \longrightarrow X, \qquad (x, w) \mapsto (x, y = Q(x)w),$$

where $P = Q^2 R$ with R squarefree.

Case $\operatorname{char} k = 2$, *$k$ perfect.* Write $P = xQ_1^2 + Q_2^2$; $Q_1 \neq 0$ as P is not a square. Put $z = (y + Q_2)/Q_1 \in L$. Then $z^2 = x$ (as in (c)). The element z is transcendental over k (since $z^2 = x$ is), so $B' = k[z]$ is a polynomial ring, hence normal (Lemma 1.4(a)). We have $x = z^2 \in B'$ and $y = Q_1(z^2)z + Q_2(z^2) \in B'$, so $B \subseteq B'$; B' is integral over B since $z^2 = x \in B$; and $\operatorname{Frac} B' = k(z) \ni x, y$ equals L (as $z \in L$ and $L = k(x, y)$). So the normalization of X is

$$X' = \mathbb{A}^1_k = \operatorname{Spec} k[z] \longrightarrow X, \qquad z \mapsto \big(x = z^2,\ y = Q_1(z^2)\, z + Q_2(z^2)\big).$$

(If X is already normal, $Q_1 \in k^*$ and this map is an isomorphism.) □

Exercise 1.19

> **Problem.** Let k be a field of characteristic $p > 0$. Let $f : X \to Y$ be a finite surjective morphism of normal algebraic varieties over k. We identify $K(Y)$ with a subfield of $K(X)$ via f. Use Lemma 3.2.25 to show the following properties:
>
> (a) If $K(X)^p \subseteq K(Y)$, show that $F_{X/k}$ factors into $f : X \to Y$ followed by a morphism $Y \to X^{(p)}$.
>
> (b) If $K(Y) \subseteq kK(X)^p$, show that f factors into $F_{X/k} : X \to X^{(p)}$ followed by a morphism $X^{(p)} \to Y$.

Solution. Summary of what is proved. Part (a) is proved as stated. Part (b) is *false as stated when k is not perfect* (counterexample below). We prove it under the hypothesis $K(Y) \subseteq K(X)^p$, which is the same as the printed hypothesis when k is perfect (then $k = k^p \subseteq K(X)^p$, so $kK(X)^p = K(X)^p$).

Notation. X, Y are normal, hence integral. $X^{(p)} = X \times_k k$, where the second factor k is a k-algebra via $a \mapsto a^p$; $F_{X/k} : X \to X^{(p)}$ is the relative Frobenius, $\phi : X^{(p)} \to X$ the first projection, and $\phi \circ F_{X/k} = F_X$ (Defs. 3.2.21, 3.2.23). For an affine open $\operatorname{Spec} B \subseteq X$ we have $X^{(p)} \supseteq \operatorname{Spec}(B \otimes_k k)$, with $B \otimes_k k$ viewed via the second factor, and $F_{X/k}$ corresponds to $\Phi : B \otimes_k k \to B$, $b \otimes a \mapsto ab^p$ (the paragraph before Lemma 3.2.25; Lemma 3.2.25 is the coordinate version). The image of Φ is the subring $k[B^p]$ generated by k and B^p.

For an affine open $Y_0 = \operatorname{Spec} A \subseteq Y$, $f^{-1}(Y_0) = \operatorname{Spec} B$ is affine and $A \to B$ is finite (as f is finite). It is injective (f is surjective and A is a domain: the kernel lies in all primes, so it is nilpotent hence 0). So $A \subseteq B \subseteq K(X)$ and $\operatorname{Frac} A = K(Y)$.

(a). Assume $K(X)^p \subseteq K(Y)$. Let $Y_0 = \operatorname{Spec} A$, $f^{-1}(Y_0) = \operatorname{Spec} B$ as above. Each $b \in B$ is integral over A, hence so is b^p: if $b^n + a_{n-1}b^{n-1} + \cdots + a_0 = 0$ with $a_i \in A$, then $(b^p)^n + a_{n-1}^p(b^p)^{n-1} + \cdots + a_0^p = 0$ in characteristic p. Moreover $b^p \in K(X)^p \subseteq K(Y)$ and A is normal, so $b^p \in A$. Thus $B^p \subseteq A$, and since $k \subseteq A$, $k[B^p] \subseteq A$. The map

$$\rho : B \otimes_k k \to A, \qquad b \otimes a \mapsto ab^p$$

is a well-defined ring homomorphism ($b \mapsto b^p$ is additive and multiplicative, and $(\lambda b, a)$ and $(b, \lambda^p a)$ have the same image), compatible with the k-structure from the second factor, and $\Phi = \iota \circ \rho$, where $\iota : A \subseteq B$. This gives $g_{Y_0} : Y_0 \to X^{(p)}$ with $g_{Y_0} \circ f = F_{X/k}$ on $f^{-1}(Y_0)$.

Gluing. We use: **Lemma B.** If $V \subseteq Y$ is open and $g, g' : V \to T$ are morphisms of schemes with $g \circ f_V = g' \circ f_V$ ($f_V : f^{-1}V \to V$), then $g = g'$. Indeed, f is surjective so g, g' agree as maps of points. Fix an affine open $T_0 \subseteq T$ and an affine open $V_0 = \operatorname{Spec} A_0 \subseteq g^{-1}(T_0) = g'^{-1}(T_0)$; then $f^{-1}(V_0) = \operatorname{Spec} B_0$ and $A_0 \to B_0$ is injective as above. The ring maps $\mathcal{O}(T_0) \to A_0$ defined by g, g' become equal after composing with $A_0 \hookrightarrow B_0$, so they are equal. Hence $g = g'$ on the cover of V by such V_0.

For two affine opens Y_0, Y_1 the morphisms g_{Y_0}, g_{Y_1} restricted to $Y_0 \cap Y_1$ satisfy $g \circ f = F_{X/k} = g' \circ f$ there, so they agree by Lemma B. So the g_{Y_0} glue to $g : Y \to X^{(p)}$ with $g \circ f = F_{X/k}$. This proves (a).

(b), under the hypothesis $K(Y) \subseteq K(X)^p$. We construct $h' : X \to Y$ with $h' \circ F_X = f$. Then $h = h' \circ \phi : X^{(p)} \to Y$ satisfies $h \circ F_{X/k} = h' \circ \phi \circ F_{X/k} = h' \circ F_X = f$, as required.

Let $V \subseteq Y$ be a non-empty open set and $c \in \mathcal{O}_Y(V) \subseteq K(Y) \subseteq K(X)^p$. Let $u \in K(X)$ be the unique element with $u^p = c$ (the Frobenius of the field $K(X)$ is injective). We claim $u \in \mathcal{O}_X(f^{-1}V)$. Let $x \in f^{-1}(V)$, $y = f(x)$. Then $c \in \mathcal{O}_{Y,y} \subseteq \mathcal{O}_{X,x}$, so u is a root of the monic polynomial $T^p - c \in \mathcal{O}_{X,x}[T]$. As $\mathcal{O}_{X,x}$ is normal with fraction field $K(X)$, $u \in \mathcal{O}_{X,x}$. Hence $u \in \bigcap_{x \in f^{-1}V} \mathcal{O}_{X,x} = \mathcal{O}_X(f^{-1}V)$ (X integral). Set $\rho_V(c) = u = c^{1/p}$. The maps $\rho_V : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}V)$ are ring homomorphisms compatible with restriction, so they define $\rho : \mathcal{O}_Y \to f_*\mathcal{O}_X$. The stalk maps are local: if $c \in \mathfrak{m}_y$ and $x \in f^{-1}(y)$, then $\rho(c)^p = c \in \mathfrak{m}_x$ forces $\rho(c) \in \mathfrak{m}_x$ (prime ideal). So (f, ρ) is a morphism of schemes $h' : X \to Y$.

Finally $h' \circ F_X = f$: F_X is the identity on the underlying space (Lemma 3.2.22(b)), so both sides have the same underlying map; and for $c \in \mathcal{O}_Y(V)$, $F_X^{\#}(\rho(c)) = \rho(c)^p = c = f^{\#}(c)$. This proves (b) under $K(Y) \subseteq K(X)^p$, in particular for k perfect with the printed hypothesis.

(b) is false as printed for imperfect k. Let $p = 2$, $k = \mathbb{F}_2(t)$ and $B = k[x, y]/(y^2 - x^3 - t)$, $X = \operatorname{Spec} B$, $Y = \mathbb{A}^1_k = \operatorname{Spec} k[x]$, f induced by $k[x] \subseteq B$.

B is a normal domain. It is a domain as $x^3 + t$ is not a square (its derivative x^2 is non-zero, whereas squares have derivative 0). It is a 1-dimensional Noetherian domain, so it suffices to check that each localization $B_{\mathfrak{m}}$ at a maximal ideal is a discrete valuation ring, i.e. that $\mathfrak{m}B_{\mathfrak{m}}$ is principal. Let $G = y^2 + x^3 + t$ and $M \supseteq (G)$ be a maximal ideal of $k[x, y]$. If $x \notin M$, then $\partial G/\partial x = x^2 \notin M$, so $G \notin M^2$ (an element of M^2 has partial derivatives in M). If $x \in M$, then $y^2 + t \in M$ and $M = (x, y^2 + t)$ (the latter is maximal: $k[y]/(y^2 + t)$ is a field), and $G = (y^2 + t) + x^3 \notin M^2$ since $x, y^2 + t$ form a basis of M/M^2. In both cases $G \in M \setminus M^2$; as $\dim k[x, y]_M = 2$ and M/M^2 has dimension 2, $\bar{G}$ is part of a basis of M/M^2 and by Nakayama $\mathfrak{m}B_{\mathfrak{m}} = Mk[x, y]_M/(G)$ is generated by one element. So B is regular of dimension 1 at all maximal ideals, hence normal. $A = k[x]$ is normal, and f is finite (free of rank 2) and surjective,

so f is a finite surjective morphism of normal varieties.

The hypothesis holds: $K(X)^2 \ni x^2$ and $y^2 = x^3 + t$, so $kK(X)^2 \ni x^3 + t$, hence x^3 (as $t \in k$), hence $x = x^3/x^2$, so $K(Y) = k(x) \subseteq kK(X)^2$.

The conclusion fails: a morphism $h : X^{(2)} \to Y$ with $h \circ F_{X/k} = f$ corresponds to a ring homomorphism $\psi : k[x] \to B \otimes_k k$ with $\Phi \circ \psi = \iota$ ($\iota : k[x] \subseteq B$). Then $x = \Phi(\psi(x)) \in \operatorname{Im}\Phi = k[B^2]$. But every element of B^2 has the form $(\alpha+\beta y)^2 = \alpha^2+\beta^2(x^3+t) \in k[x^2, x^3]$ ($\alpha, \beta \in k[x]$), so $k[B^2] \subseteq k[x^2, x^3]$, which does not contain x. A contradiction. (Here the reason is that $X^{(2)}$ is the cuspidal curve $y'^2 = x^3$, which is not normal, so f cannot factor through it.) $\square$

4.2 Regular schemes

Conventions for this section. Rings are commutative with 1. Results of the text are cited as "Prop. 2.5", "Cor. 2.12", "Thm. 2.16" (Chapter 4, Section 2), and results of other chapters with the chapter number in front ("Cor. 2.5.24" is Chapter 2, Section 5). For a local ring $(\mathcal{O}, \mathfrak{m})$ with residue field k we write $\operatorname{emb}(\mathcal{O}) = \dim_k \mathfrak{m}/\mathfrak{m}^2$; recall emb $\geq$ dim (Cor. 2.5.14(b)), with equality exactly when $\mathcal{O}$ is regular. For a field k and a closed point x of $\mathbb{A}^n_k$ with maximal ideal $\mathfrak{m}$, the map D_x from $\mathfrak{m}$ to the dual of $k(x)^n$ is $D_xP : (t_i) \mapsto \sum_i \frac{\partial P}{\partial T_i}(x)t_i$, where $\frac{\partial P}{\partial T_i}(x) \in k(x)$ is the image of the partial derivative. From $D_x(PQ) = P(x)D_xQ + Q(x)D_xP$ one sees that D_x kills $\mathfrak{m}^2$ and is $k(x)$-linear on $\mathfrak{m}/\mathfrak{m}^2$.

Honest notes. In Ex. 2.1 the last assertion ("x is discrete in X_y") is false as printed for a point x that is not closed in X_y; it is proved for x closed in X_y (which is the case of interest), and what is true in general is also stated. In Ex. 2.4(a) the polynomials P_i are those of Ex. 2.1.5, i.e. with $k[T_1, \ldots, T_i] \cap \mathfrak{m} = (P_1, \ldots, P_i)$; this property is used, and its existence is recalled. In Ex. 2.10 the "x_0" of the book is read as x. Ex. 2.11: the curve is the closed subscheme (possibly non-reduced) defined by $x^n + y^n + z^n$. Ex. 2.13 is proved fully (with k infinite).

Exercise 2.1

> **Problem.** Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let $x \in X$, $y = f(x)$. Show that $T_{f,x}$ is injective if and only if the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_{X,x}$ is generated by $f_x^{\#}(\mathfrak{m}_y)$. Show that if f is moreover of finite type, then x is discrete in X_y.

Solution. *First assertion.* By Def. 2.1, $T_{f,x} : T_{X,x} \to T_{Y,y} \otimes_{k(y)} k(x)$ is the $k(x)$-dual of the $k(x)$-linear map

$$\psi : \ (\mathfrak{m}_y/\mathfrak{m}_y^2) \otimes_{k(y)} k(x) \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2$$

induced by $f_x^{\#}$ (which is a local homomorphism, so $f_x^{\#}(\mathfrak{m}_y) \subseteq \mathfrak{m}_x$). Both spaces are finite dimensional because $\mathfrak{m}_x, \mathfrak{m}_y$ are finitely generated (X, Y locally Noetherian), so $T_{Y,y} \otimes k(x) = (\mathfrak{m}_y/\mathfrak{m}_y^2)^\vee \otimes k(x) = ((\mathfrak{m}_y/\mathfrak{m}_y^2) \otimes k(x))^\vee$ and $T_{f,x} = \psi^\vee$. For linear maps of finite-dimensional vector spaces, $\psi^\vee$ is injective if and only if ψ is surjective. The image of ψ is the $k(x)$-span of the classes of the elements $f_x^{\#}(a)$, $a \in \mathfrak{m}_y$, that is $(f_x^{\#}(\mathfrak{m}_y)\mathcal{O}_{X,x} + \mathfrak{m}_x^2)/\mathfrak{m}_x^2$. Thus $T_{f,x}$ is injective if and only if $f_x^{\#}(\mathfrak{m}_y)\mathcal{O}_{X,x} + \mathfrak{m}_x^2 = \mathfrak{m}_x$. Since $\mathfrak{m}_x$ is finitely generated, Nakayama's lemma shows that this holds if and only if $f_x^{\#}(\mathfrak{m}_y)\mathcal{O}_{X,x} = \mathfrak{m}_x$ (if $N = \mathfrak{m}_x/\mathfrak{m}_y\mathcal{O}_{X,x}$ then $N = \mathfrak{m}_x N$ forces $N = 0$; the converse is trivial).

Second assertion. The statement "x is discrete in X_y" means that x is an isolated point of the fibre X_y. As printed it cannot hold for every x: for $f : \mathbb{A}^1_k \to \operatorname{Spec} k$ and x the generic point, $\mathfrak{m}_x = 0$ is generated by $f_x^{\#}(\mathfrak{m}_y) = 0$, but x is not isolated in $X_y = \mathbb{A}^1_k$. What is true, and what we prove, is:

(i) If $T_{f,x}$ is injective, then x is the generic point of an irreducible component of X_y. *(ii)* If moreover f is of finite type and x is closed in X_y (equivalently, $k(x)$ is finite over $k(y)$), then x is an isolated point of X_y.

The local ring of the fibre is $\mathcal{O}_{X_y,x} = \mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} k(y) = \mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$. By the first part, if $T_{f,x}$ is injective this equals $\mathcal{O}_{X,x}/\mathfrak{m}_x = k(x)$, a field. Take an affine open neighbourhood $\operatorname{Spec} C$ of x in X_y, with x corresponding to the prime $\mathfrak{p}$. Then $C_{\mathfrak{p}}$ is a field, so its only prime is 0, i.e. $\mathfrak{p}$ contains no smaller prime. So $\mathfrak{p}$ is a minimal prime of C, and x is the generic point of an irreducible component of $\operatorname{Spec} C$, whose closure Z in X_y is an irreducible component of X_y with generic point x. This proves (i).

For (ii), X_y is of finite type over $k(y)$, hence a Noetherian space with finitely many irreducible components. If x is closed in X_y then $Z = \overline{\{x\}} = \{x\}$. Let Z' be the union of the other irreducible components; it is closed. If $x \in Z'$, then x lies in another component Z_1, so $Z = \{x\} \subseteq Z_1$ contradicts the maximality of Z. Hence $x \notin Z'$ and $\{x\} = X_y \setminus Z'$ is open in X_y: x is isolated. □

Exercise 2.2

Problem. Let k be a field. Show that $\operatorname{Spec} k[x,y,z]/(x^2 - yz)$ is normal but singular.

Solution. Let $A = k[x,y,z]/(x^2 - yz)$ and $F = x^2 - yz$. The argument works in every characteristic.

Step 1: A is isomorphic to the subring $R = k[s^2, st, t^2]$ of $k[s,t]$. Let $\theta : k[x,y,z] \to k[s,t]$ be given by $x \mapsto st$, $y \mapsto s^2$, $z \mapsto t^2$. Then $\theta(F) = 0$, so θ induces $\bar{\theta} : A \to R$, which is surjective. Using $x^2 = yz$ in A, the monomials $x^\epsilon y^i z^j$ with $\epsilon \in \{0,1\}$, $i,j \geq 0$ span A over k. Their images are $s^{2i}t^{2j}$ ($\epsilon = 0$) and $s^{2i+1}t^{2j+1}$ ($\epsilon = 1$), which are pairwise distinct monomials of $k[s,t]$, hence linearly independent. Therefore these monomials form a basis of A and $\bar{\theta}$ is injective. So $A \simeq R$. In particular A is an integral domain. Moreover R is the set of polynomials all of whose homogeneous components have even degree: a monomial $s^i t^j$ with $i+j$ even has i,j of the same parity, so it is a product of s^2, st, t^2.

Step 2: A is normal. Let $\alpha \in \operatorname{Frac}(A) \subseteq k(s,t)$ be integral over A. Then α is integral over the UFD $k[s,t]$, hence $\alpha \in k[s,t]$. Write $\alpha = a/b$ with $a, b \in A$, $b \neq 0$, so $a = b\alpha$. Split $\alpha = \alpha' + \alpha''$ into its even-degree and odd-degree parts. Since a and b have only even-degree components, $b\alpha''$ has only odd-degree components while $a - b\alpha'$ has only even-degree components, and they are equal; so $b\alpha'' = 0$, whence $\alpha'' = 0$ because $k[s,t]$ is a domain. Thus $\alpha \in R = A$. So A is an integrally closed domain; all its localizations are integrally closed domains, and $X = \operatorname{Spec} A$ is normal.

Step 3: X is singular. The fraction field of R is $k(s^2, st, t^2)$; it is algebraic over $k(s^2, t^2)$ because $(st)^2 = s^2t^2$, and s^2, t^2 are algebraically independent. So $\operatorname{trdeg}_k = 2$ and $\dim X = 2$ (Prop. 2.5.19). Let x_0 be the rational point with maximal ideal $\mathfrak{m} = (x,y,z)A$; it is a closed

point, so $\dim \mathcal{O}_{X,x_0} = 2$ (Cor. 2.5.24). By Lemma 2.3,

$$\mathfrak{m}\mathcal{O}_{X,x_0}/\mathfrak{m}^2\mathcal{O}_{X,x_0} \simeq \mathfrak{m}/\mathfrak{m}^2 = (x,y,z)/\big((x,y,z)^2 + (F)\big) = (x,y,z)/(x,y,z)^2,$$

because $F \in (x,y,z)^2$; this has dimension $3 > 2 = \dim \mathcal{O}_{X,x_0}$. Hence $\mathcal{O}_{X,x_0}$ is not regular and X is singular (at x_0; in characteristic $\neq 2$ this is the only singular point, but we do not need it). □

Exercise 2.3

> **Problem.** Let $\mathcal{O}_K$ be a discrete valuation ring with uniformizing parameter t and field of fractions K. Let $n \geq 1$. We suppose that $\mathrm{char}(K) \neq 2, 3$. (a) Show that $K[x,y]/(y^2+x^3+t^n)$ is regular. (b) Let $A = \mathcal{O}_K[x,y]/(y^2+x^3+t^n)$. Show that the special fibre of $\operatorname{Spec} A \to \operatorname{Spec} \mathcal{O}_K$ is reduced. Deduce from this that A is normal (see Lemma 1.18). (c) Let $\mathfrak{m}$ be the maximal ideal of A generated by x, y and t. Show that $A_{\mathfrak{m}}$ is regular if and only if $n = 1$.

Solution. Put $F = y^2 + x^3 + t^n$ and let $k = \mathcal{O}_K/(t)$ be the residue field.

(a) The ring $K[x,y]/(F)$ is Noetherian, so by Cor. 2.17 it suffices to prove regularity at the closed points, i.e. at the images of the maximal ideals $\mathfrak{m}$ of $K[x,y]$ containing F. Fix such an $\mathfrak{m}$. The ring $S = K[x,y]_{\mathfrak{m}}$ is a regular local ring of dimension 2 (Example 2.18), hence a domain (Prop. 2.11), and $F \in \mathfrak{m}S$ is nonzero. The local ring of $\operatorname{Spec} K[x,y]/(F)$ at the point is S/FS, which by Cor. 2.12 is regular if and only if $F \notin \mathfrak{m}^2 S$. Since $\mathfrak{m} \subset K[x,y]$ is the preimage of $\mathfrak{m}S$ and $\mathfrak{m}S \cap K[x,y] = \mathfrak{m}$, $\mathfrak{m}^2 S \cap K[x,y] = \mathfrak{m}^2$, it suffices to show $F \notin \mathfrak{m}^2$.

Suppose $F \in \mathfrak{m}^2$. The partial derivatives ∂_x, ∂_y are K-linear derivations of $K[x,y]$, and a derivation D maps $\mathfrak{m}^2$ into $\mathfrak{m}$ (as $D(ab) = aDb + bDa$). Hence $\partial_x F = 3x^2 \in \mathfrak{m}$ and $\partial_y F = 2y \in \mathfrak{m}$. Since $2, 3 \in K^*$ (here $\mathrm{char}\, K \neq 2, 3$ is used), $x^2 \in \mathfrak{m}$ and $y \in \mathfrak{m}$, so $x \in \mathfrak{m}$ ($\mathfrak{m}$ is prime). Then $t^n = F - y^2 - x^3 \in \mathfrak{m}$. But t^n is a unit of K, so $\mathfrak{m}$ would be the unit ideal, a contradiction. Thus $F \notin \mathfrak{m}^2$, and $K[x,y]/(F)$ is regular.

(b) Since $t^n = 0$ in k, the special fibre is $\operatorname{Spec} k[x,y]/(y^2+x^3)$. The polynomial $y^2 + x^3 \in k[x][y]$ is monic of degree 2 in y, so it is reducible only if it has a root $g \in k[x]$, i.e. $g^2 = -x^3$. This is impossible by comparing degrees ($2 \deg g$ is even and 3 is odd; and $g = 0$ gives $x^3 = 0$). So $y^2 + x^3$ is irreducible, hence prime in the UFD $k[x,y]$, and $k[x,y]/(y^2+x^3)$ is a domain; in particular the special fibre is reduced. (This holds whatever the characteristic of k.)

To apply Lemma 1.18 we check the hypotheses. As an $\mathcal{O}_K[x]$-module, $A = \mathcal{O}_K[x][y]/(y^2+x^3+t^n)$ is free with basis $1, y$ (the relation is monic in y), and $\mathcal{O}_K[x]$ is free over $\mathcal{O}_K$; so A is free, hence flat, over $\mathcal{O}_K$. Flatness is local, so $\mathcal{O}_X(U)$ is flat over $\mathcal{O}_K$ for every affine open $U \subseteq X = \operatorname{Spec} A$ (all its localizations are localizations of A). The generic fibre is $X_K = \operatorname{Spec}(A \otimes_{\mathcal{O}_K} K) = \operatorname{Spec} K[x,y]/(F)$, which is regular by (a), hence normal (Thm. 2.16(b): regular local rings are factorial, hence normal). The special fibre X_k is reduced. By Lemma 1.18, X is normal, i.e. A is normal.

(c) Let $M = (t,x,y)$, a maximal ideal of $B = \mathcal{O}_K[x,y]$ with $B/M = k$, and $F \in M$. The ideal $\mathfrak{m}$ of A is $M/(F)$, and $A_{\mathfrak{m}} = S/FS$ with $S = B_M$. Since $M \cap \mathcal{O}_K = (t)$ has height 1, Lemma 2.5.16 gives $\mathrm{ht}\, M = 1 + 2 = 3$, so $\dim S = 3$; and MS is generated by the three elements t, x, y. Hence S is regular of dimension 3, a domain (Prop. 2.11), and t, x, y form a basis of the k-vector space

MS/M^2S (three generators of a 3-dimensional space). By Cor. 2.12, $A_{\mathfrak{m}} = S/FS$ is regular if and only if $F \notin M^2S$.

If $n \geq 2$, then t^n, x^3, $y^2 \in M^2S$, so $F \in M^2S$ and $A_{\mathfrak{m}}$ is not regular. If $n = 1$, then $F \equiv t \pmod{M^2S}$ because $y^2 + x^3 \in M^2S$; the class of t is a basis vector of MS/M^2S, hence nonzero, so $F \notin M^2S$ and $A_{\mathfrak{m}}$ is regular. Therefore $A_{\mathfrak{m}}$ is regular if and only if $n = 1$. □

Exercise 2.4

> **Problem.** Let k be a field and let $\mathfrak{m}$ be a maximal ideal of $k[T_1, \ldots, T_n]$ corresponding to a closed point x of $\mathbb{A}^n_k$. Let us consider the $k(x)$-linear map $\phi_x : \mathfrak{m}/\mathfrak{m}^2 \to E_x^\vee$ defined as before Lemma 2.4, with $E_x = k(x)^n$. (a) Let $P_1(T_1), P_2(T_1, T_2), \ldots, P_n(T_1, \ldots, T_n)$ be a system of generators of $\mathfrak{m}$ (see Exercise 2.1.5). Show that ϕ_x is an isomorphism if and only if $(\partial P_i/\partial T_i)(x) \neq 0$ for all $i \leq n$. (b) Deduce from (a) that ϕ_x is an isomorphism if and only if $k(x)$ is separable over k.

Solution. Let $x_i \in k(x) = k[T]/\mathfrak{m}$ be the image of T_i, $\mathfrak{m}_i = \mathfrak{m} \cap k[T_1, \ldots, T_i]$ and $k_i = k[x_1, \ldots, x_i] \subseteq k(x)$. Then $k[T_1, \ldots, T_i]/\mathfrak{m}_i \simeq k_i$ is a domain which is finite-dimensional over k (as $k(x)$ is finite over k by Cor. 2.1.12), hence a field: $k_i = k(x_1, \ldots, x_i)$, $k_0 = k$, $k_n = k(x)$.

In (a) the P_i are polynomials with $P_i \in k[T_1, \ldots, T_i]$ generating $\mathfrak{m}$; the proof of (a) uses only this. In (b) we choose them as in Exercise 2.1.5, i.e. with $\mathfrak{m}_i = (P_1, \ldots, P_i)k[T_1, \ldots, T_i]$ (this is harmless: by (a) the criterion holds for every such system, so (b) may use a convenient one). (Existence, for completeness: suppose $P_1, \ldots, P_{i-1}$ are found. Then $k[T_1, \ldots, T_i]/(P_1, \ldots, P_{i-1}) \simeq k_{i-1}[T_i]$, a principal ideal ring, and the image of $\mathfrak{m}_i$ is generated by one element; lifting it to $P_i \in k[T_1, \ldots, T_i]$ gives $\mathfrak{m}_i = (P_1, \ldots, P_i)$.)

(a) By Example 2.18 (regularity of $\mathbb{A}^n_k$ at closed points, Cor. 2.5.24 for the dimension n), $\dim_{k(x)} \mathfrak{m}/\mathfrak{m}^2 = n$. The classes of $P_1, \ldots, P_n$ span $\mathfrak{m}/\mathfrak{m}^2$, hence form a basis. The map ϕ_x sends the class of P_i to the i-th row of the $n \times n$ matrix $\left(\frac{\partial P_i}{\partial T_j}(x)\right)_{i,j}$ over $k(x)$. Since $P_i \in k[T_1, \ldots, T_i]$, $\frac{\partial P_i}{\partial T_j} = 0$ for $j > i$, so the matrix is lower triangular. Now ϕ_x is an isomorphism iff it maps a basis to a basis iff the matrix is invertible iff its determinant, the product of the diagonal entries $\frac{\partial P_i}{\partial T_i}(x)$, is nonzero iff all diagonal entries are nonzero.

(b) Let $\bar{P}_i \in k_{i-1}[T_i]$ be the image of P_i under reduction of the coefficients $k[T_1, \ldots, T_{i-1}] \to k_{i-1}$. The surjection $k[T_1, \ldots, T_i] \to k_{i-1}[T_i] \to k_i$ $(T_i \mapsto x_i)$ has kernel $\mathfrak{m}_i = (P_1, \ldots, P_i)$, and the first map has kernel $\mathfrak{m}_{i-1}k[T_1, \ldots, T_i] = (P_1, \ldots, P_{i-1})$. Hence the kernel of $k_{i-1}[T_i] \to k_i$ is generated by $\bar{P}_i$. This kernel is nonzero, so $\bar{P}_i = c\,\mu_i$ with $c \in k_{i-1}^*$ and μ_i the minimal polynomial of x_i over k_{i-1}. Since reducing coefficients commutes with $\partial/\partial T_i$, we get $\frac{\partial P_i}{\partial T_i}(x) = \bar{P}_i'(x_i) = c\,\mu_i'(x_i)$. As x_i is a root of μ_i, it is a simple root iff $\mu_i'(x_i) \neq 0$, i.e. iff x_i is separable over k_{i-1}. By (a), ϕ_x is an isomorphism iff x_i is separable over k_{i-1} for all i.

If every x_i is separable over k_{i-1}, then $k_n = k(x)$ is separable over k, since a tower of finite separable extensions is separable. Conversely if $k(x)/k$ is separable, each x_i is separable over k, and its minimal polynomial over k_{i-1} divides its (separable) minimal polynomial over k, so it is separable over k_{i-1}. Hence ϕ_x is an isomorphism iff $k(x)/k$ is separable. □

Exercise 2.5

Problem. Show that the Jacobian criterion (with J_x as a matrix with coefficients in $k(x)$) remains valid at the closed points $x \in X$ for which $k(x)$ is separable over k. See also Proposition 3.30. Give an example with X regular at x and $\dim \mathcal{O}_{X,x} \neq n - \operatorname{rank} J_x$.

Solution. *Statement.* Let $X = V(I) \subseteq \mathbb{A}^n_k$ be a closed subscheme, $F_1, \dots, F_r$ generators of I, $x \in X$ a closed point with $k(x)$ separable over k, and $J_x = \left(\frac{\partial F_i}{\partial T_j}(x)\right) \in M_{r\times n}(k(x))$. Then X is regular at x iff $\operatorname{rank} J_x = n - \dim \mathcal{O}_{X,x}$.

Proof. Let $\mathfrak{m} \subset k[T]$ be the maximal ideal of x and $\mathfrak{n} = \mathfrak{m}/I$ the maximal ideal of $A = k[T]/I$ at x. Then $\mathfrak{n}/\mathfrak{n}^2 = \mathfrak{m}/(I + \mathfrak{m}^2)$, and by Lemma 2.3 this is $\mathfrak{m}_x/\mathfrak{m}_x^2$ for $\mathfrak{m}_x$ the maximal ideal of $\mathcal{O}_{X,x}$. So X is regular at x iff $\dim_{k(x)} \mathfrak{m}/(I + \mathfrak{m}^2) = \dim \mathcal{O}_{X,x}$.

Since $k(x)/k$ is separable, $\phi_x : \mathfrak{m}/\mathfrak{m}^2 \to k(x)^{n\vee}$ is an isomorphism (Ex. 2.4(b)). The subspace $(I + \mathfrak{m}^2)/\mathfrak{m}^2 \subseteq \mathfrak{m}/\mathfrak{m}^2$ is the $k(x)$-span of the classes of the elements of I. If $G = \sum_i A_i F_i \in I$ ($A_i \in k[T]$), then the class of G in $\mathfrak{m}/\mathfrak{m}^2$ equals $\sum_i A_i(x)\,[F_i]$ because $F_i \in \mathfrak{m}$ (the class of aF is $a(x)[F]$ for $F \in \mathfrak{m}$). So $(I + \mathfrak{m}^2)/\mathfrak{m}^2$ is the span of $[F_1], \dots, [F_r]$, and ϕ_x maps it onto the span of the rows of J_x, which has dimension $\operatorname{rank} J_x$. As ϕ_x is bijective,

$$\dim_{k(x)} \mathfrak{m}_x/\mathfrak{m}_x^2 = n - \dim_{k(x)} (I + \mathfrak{m}^2)/\mathfrak{m}^2 = n - \operatorname{rank} J_x.$$

Thus X is regular at x iff $n - \operatorname{rank} J_x = \dim \mathcal{O}_{X,x}$. (When x is rational this is Thm. 2.19.) $\square$

Example where separability fails. Let p be a prime, $k = \mathbb{F}_p(s)$, $n = 1$ and $X = \operatorname{Spec} k[T]/(T^p - s)$ (the polynomial $T^p - s$ is irreducible, by Eisenstein at the prime s of $\mathbb{F}_p[s]$). Then $k[T]/(T^p - s) = k(s^{1/p})$ is a field, so X is a single point x with $\mathcal{O}_{X,x}$ a field: X is regular and $\dim \mathcal{O}_{X,x} = 0$. But $J_x = (pT^{p-1}) = (0)$ has rank 0, so $n - \operatorname{rank} J_x = 1 \neq 0$. (Here $k(x) = k(s^{1/p})$ is purely inseparable over k.)

Exercise 2.6

Problem. Let k be a field. Let $F_1, \dots, F_n \in k[T_1, \dots, T_n]$. Let us consider the homomorphism $\phi : k[T_1, \dots, T_n] \to k[T_1, \dots, T_n]$ defined by $\phi(T_i) = F_i(T)$ and let us suppose that the associated morphism $\mathbb{A}^n_k \to \mathbb{A}^n_k$ is an isomorphism. Show that $\det\left(\frac{\partial F_i}{\partial T_j}\right)_{1\le i,j\le n}$ belongs to k^*.

Solution. A morphism of affine schemes is an isomorphism iff the ring homomorphism is (Prop. 2.3.25), so ϕ is a k-algebra automorphism of $k[T] = k[T_1, \dots, T_n]$. Let $\psi = \phi^{-1}$ and $G_i = \psi(T_i) \in k[T]$. Then $T_i = \phi(\psi(T_i)) = \phi(G_i) = G_i(F_1, \dots, F_n)$.

The chain rule holds for composition of polynomials over any ring: for $G \in k[T]$, $\frac{\partial}{\partial T_k}\big(G(F_1, \dots, F_n)\big) = \sum_j \frac{\partial G}{\partial T_j}(F_1, \dots, F_n)\,\frac{\partial F_j}{\partial T_k}$. (Both sides are k-linear in G, and for a monomial $G = T_1^{e_1} \cdots T_n^{e_n}$ this is the Leibniz rule.) Applying it to $G = G_i$ and using $G_i(F) = T_i$,

$$\delta_{ik} = \frac{\partial T_i}{\partial T_k} = \sum_j \frac{\partial G_i}{\partial T_j}(F)\,\frac{\partial F_j}{\partial T_k}.$$

In matrix form, with entries in $k[T]$: $\left(\frac{\partial G_i}{\partial T_j}(F)\right)_{ij} \cdot \left(\frac{\partial F_j}{\partial T_k}\right)_{jk} = \mathrm{Id}$. Taking determinants, $\det\left(\frac{\partial G_i}{\partial T_j}(F)\right) \cdot \det\left(\frac{\partial F_i}{\partial T_j}\right) = 1$ in $k[T]$. So $\det(\partial F_i/\partial T_j)$ is a unit of the polynomial ring $k[T]$, i.e. an element of k^*. $\square$

Exercise 2.7

> **Problem.** Let X be a scheme over a field k, and let $x \in X(k)$. Show that there exists a canonical bijection between $T_{X,x}$ and the set of k-morphisms $\operatorname{Spec}(k[\epsilon]/\epsilon^2) \to X$ with image equal to x.

Solution. Let $D = k[\epsilon]/(\epsilon^2)$; $\operatorname{Spec} D$ has one point, corresponding to the maximal ideal ϵD. Let $\mathcal{O} = \mathcal{O}_{X,x}$, with maximal ideal $\mathfrak{m}$. Since $x \in X(k)$, $k(x) = k$, and the structure $k \to \mathcal{O} \to \mathcal{O}/\mathfrak{m} = k$ is the identity; so $\mathcal{O} = k \oplus \mathfrak{m}$ and we write $\pi : \mathcal{O} \to k$ for the residue map.

Step 1: morphisms with image x are local k-algebra homomorphisms $\mathcal{O} \to D$. Let $g : \operatorname{Spec} D \to X$ be a k-morphism with image $\{x\}$. Take an affine open neighbourhood $U = \operatorname{Spec} C$ of x; g factors through U, so by Prop. 2.3.25 it corresponds to a k-algebra homomorphism $h : C \to D$, and "image $= x$" means $h^{-1}(\epsilon D) = \mathfrak{p}_x$. Then h maps $C \setminus \mathfrak{p}_x$ into D^* (the elements outside the maximal ideal), so it extends uniquely to $\bar{h} : C_{\mathfrak{p}_x} = \mathcal{O} \to D$, and $\bar{h}(\mathfrak{m}) \subseteq \epsilon D$ (local). Conversely a local k-homomorphism $\mathcal{O} \to D$ composed with $C \to \mathcal{O}$ gives such an h, and the two constructions are inverse to each other. They do not depend on U.

Step 2: local k-algebra homomorphisms $\mathcal{O} \to D$ correspond to $(\mathfrak{m}/\mathfrak{m}^2)^\vee$. Let $h : \mathcal{O} \to D$ be k-linear, write $h(a) = \lambda(a) + \delta(a)\epsilon$ with $\lambda, \delta : \mathcal{O} \to k$ k-linear. Then h is a local k-algebra homomorphism iff λ is a k-algebra homomorphism with $\lambda(\mathfrak{m}) = 0$, and $h(ab) = h(a)h(b)$ holds in the ϵ-part. The first condition forces $\lambda = \pi$ (a k-algebra homomorphism $\mathcal{O} \to k$ killing $\mathfrak{m}$ factors through $\mathcal{O}/\mathfrak{m} = k$ and is the identity there). The multiplicativity of h then reads

$$\delta(ab) = \pi(a)\delta(b) + \pi(b)\delta(a) \qquad (a, b \in \mathcal{O}),$$

i.e. δ is a k-derivation $\mathcal{O} \to k$ (with k an $\mathcal{O}$-module via π); in particular $\delta(1) = 0$. So local k-homomorphisms $\mathcal{O} \to D$ correspond bijectively to $\operatorname{Der}_k(\mathcal{O}, k)$.

Now $\operatorname{Der}_k(\mathcal{O}, k) \simeq (\mathfrak{m}/\mathfrak{m}^2)^\vee$ via $\delta \mapsto \delta|_{\mathfrak{m}}$: if $a, b \in \mathfrak{m}$ then $\delta(ab) = 0$, so $\delta|_{\mathfrak{m}}$ factors through $\mathfrak{m}/\mathfrak{m}^2$. Conversely, given a k-linear $\ell : \mathfrak{m}/\mathfrak{m}^2 \to k$ define $\delta_\ell(a) = \ell\big(\overline{a - \pi(a)}\big)$ (note $a - \pi(a) \in \mathfrak{m}$). Writing $a = \lambda + u$, $b = \mu + v$ with $\lambda, \mu \in k$, $u, v \in \mathfrak{m}$, we get $ab = \lambda\mu + (\lambda v + \mu u) + uv$ with $uv \in \mathfrak{m}^2$, so $\delta_\ell(ab) = \lambda\ell(v) + \mu\ell(u) = \pi(a)\delta_\ell(b) + \pi(b)\delta_\ell(a)$: δ_ℓ is a derivation. The two maps are mutually inverse, because a derivation vanishes on $k \cdot 1$ and so is determined by its restriction to $\mathfrak{m}$.

Composing Steps 1 and 2 we get a bijection between the set of k-morphisms $\operatorname{Spec} D \to X$ with image x and $(\mathfrak{m}/\mathfrak{m}^2)^\vee = T_{X,x}$ (Def. 2.1); it involves no choices, so it is canonical. Explicitly, ℓ corresponds to the morphism given by $a \mapsto \pi(a) + \ell(\overline{a - \pi(a)})\,\epsilon$. □

Exercise 2.8

> **Problem.** Let us once again consider the scheme $\operatorname{Proj} B$ of Exercise 2.3.13 (so $B = A[X, Y, Z]$ with B_d the sub-A-module generated by the monomials $X^a Y^b Z^c$ with $a + 2b + 3c = d$). Show that the fibres of $\operatorname{Proj} B \to \operatorname{Spec} A$ are all singular.

Solution. Here X, Y, Z have weights $1, 2, 3$, and $Z \in B_3 \subseteq B_+$ is homogeneous, so $D_+(Z) = \operatorname{Spec} B_{(Z)}$ is an affine open subscheme of $\operatorname{Proj} B$ (Prop. 2.3.38). The ring $B_{(Z)}$ is the degree-0

part of $B_Z = A[X, Y, Z, Z^{-1}]$, i.e. the free A-module with basis the Laurent monomials $X^aY^bZ^c$ ($a, b \geq 0$, $c \in \mathbb{Z}$) with $a + 2b + 3c = 0$.

Claim: $B_{(Z)} \simeq A[U, V, W]/(UW - V^3)$ *via* $U \mapsto X^3/Z$, $V \mapsto XY/Z$, $W \mapsto Y^3/Z^2$. These three elements lie in $B_{(Z)}$ (their weights are 0), and $UW - V^3 \mapsto X^3Y^3/Z^3 - X^3Y^3/Z^3 = 0$, so we get an A-algebra homomorphism $\bar{\theta} : Q := A[U, V, W]/(UW - V^3) \to B_{(Z)}$. We have $U^iV^jW^l \mapsto X^{3i+j}Y^{j+3l}Z^{-i-j-2l}$.

Surjective. Let $X^aY^bZ^c$ be a basis monomial, so $a + 2b = -3c \equiv 0 \pmod 3$, whence $a \equiv b \pmod 3$. If $a \geq b$, write $a - b = 3i$; then $c = -(a + 2b)/3 = -i - b$ and the monomial is the image of U^iV^b. If $b > a$, write $b - a = 3l$; then $c = -a - 2l$ and the monomial is the image of V^aW^l.

Injective. In Q the relation $UW = V^3$ lets one reduce every monomial, so Q is spanned over A by $\mathcal{S} = \{U^iV^j\}_{i,j\geq 0} \cup \{V^jW^l\}_{j\geq 0, l\geq 1}$. The images of the elements of $\mathcal{S}$ are $X^aY^bZ^c$ with $(a, b) = (3i + j, j)$ (then $a \geq b$) or $(a, b) = (j, j + 3l)$ (then $b > a$); the pair (a, b) determines the element of $\mathcal{S}$ and the monomial, so distinct elements of $\mathcal{S}$ go to distinct monomials, which are A-linearly independent in B_Z. Hence $\mathcal{S}$ is an A-basis of Q and $\bar{\theta}$ is an isomorphism.

The fibres. Let $\mathfrak{p} \in \operatorname{Spec} A$, $\kappa = k(\mathfrak{p})$. The fibre $X_\mathfrak{p} = \operatorname{Proj} B \times_{\operatorname{Spec} A} \operatorname{Spec} \kappa$ contains the open subscheme $D_+(Z) \times_A \kappa = \operatorname{Spec} R$ with $R = B_{(Z)} \otimes_A \kappa = \kappa[U, V, W]/(UW - V^3)$. (Applying the claim with $A = \kappa$ shows that R is isomorphic to a subring of the domain $\kappa[X, Y, Z, Z^{-1}]$, so R is a domain.) Let o be the rational point of $\operatorname{Spec} R$ with maximal ideal $M = (U, V, W)$.

By Lemma 2.3, $\mathfrak{m}_o/\mathfrak{m}_o^2 = M/M^2$, and $M/M^2 = (U, V, W)/\big((U, V, W)^2 + (UW - V^3)\big) = (U, V, W)/(U, V, W)^2$, since $UW - V^3 \in (U, V, W)^2$; so $\dim_\kappa \mathfrak{m}_o/\mathfrak{m}_o^2 = 3$. On the other hand the domain R is generated over κ by u, v, w (the images of U, V, W) with $v^3 = uw$, and $u \neq 0$, so $\operatorname{Frac}(R) = \kappa(u, v)$ has transcendence degree ≤ 2, whence $\dim \mathcal{O}_{X_\mathfrak{p},o} \leq \dim R \leq 2$ (Prop. 2.5.19). Thus $\dim \mathcal{O}_{X_\mathfrak{p},o} \leq 2 < 3 = \dim \mathfrak{m}_o/\mathfrak{m}_o^2$, and $\mathcal{O}_{X_\mathfrak{p},o}$ is not regular. So every fibre $X_\mathfrak{p}$ is singular (at the point $[0 : 0 : 1]$). □

Exercise 2.9

Problem. Let $X = V_+(I)$ be a closed subvariety of $\mathbb{P}^n_k$ over a field k. Let $x \in X(k)$, and let us consider the linear subvariety of $\mathbb{P}(k^{n+1})$

$$PT_{X,x} = \Big\{(t_0, \ldots, t_n) \in \mathbb{P}(k^{n+1}) \;\Big|\; \sum_{0\leq i\leq n} \frac{\partial F}{\partial T_i}(x)\, t_i = 0,\ \forall F(T_0, \ldots, T_n) \in I\Big\}.$$

We will abusively call this linear subvariety the tangent space to X at x. Show that $\dim_k T_{X,x} = \dim PT_{X,x}$.

Solution. Here $I \subseteq k[T_0, \ldots, T_n]$ is a homogeneous ideal and $X = \operatorname{Proj} k[T]/I$, and in the formula x stands for a fixed representative $(x_0, \ldots, x_n) \in k^{n+1} \setminus \{0\}$ of the point (Lemma 2.3.43). Rescaling x by $\lambda \in k^*$ multiplies the partial derivatives of a homogeneous F of degree d by λ^{d-1}, so the set $PT_{X,x}$ does not depend on the representative. For $F \in I$ put $L_F(t) = \sum_j \frac{\partial F}{\partial T_j}(x) t_j$, a linear form on k^{n+1}, and let

$$W = \{t \in k^{n+1} :\ L_F(t) = 0\ \ \forall F \in I\}.$$

Then $PT_{X,x} = \mathbb{P}(W)$ has dimension $\dim_k W - 1$. As I is homogeneous and $F \mapsto L_F$ is linear, W is cut out by the L_F with $F \in I$ homogeneous.

Choose i with $x_i \neq 0$; scaling, we may assume $x_i = 1$. Put $S_j = T_j/T_i$ $(j \neq i)$; then $D_+(T_i) \simeq \mathbb{A}^n_k$ with coordinates S_j, and x is the rational point $x^a = (x_j)_{j\neq i}$. By Prop. 2.3.38 and Lemma 2.3.36(d),(e), $X \cap D_+(T_i)$ is the closed subscheme of $\mathbb{A}^n_k$ defined by the ideal I^a generated by the dehomogenizations $F^a(S) = F(S_0, \ldots, 1, \ldots, S_n)$ (1 in the i-th place) of homogeneous $F \in I$; and $T_{X,x}$ is the tangent space of this affine scheme at x^a. By Prop. 2.5,

$$T_{X,x} \simeq T' := \Big\{ s \in k^n \;\Big|\; \sum_{j\neq i} \frac{\partial F^a}{\partial S_j}(x^a)\, s_j = 0 \;\; \forall F^a \in I^a \Big\}.$$

For $G = \sum_m A_m F^a_m \in I^a$ ($F_m \in I$ homogeneous), $D_{x^a} G = \sum_m A_m(x^a) D_{x^a} F^a_m$ since $F^a_m(x^a) = F_m(x) = 0$ $(x \in X)$. So it is enough to impose the conditions for F^a with $F \in I$ homogeneous. Moreover $\frac{\partial F^a}{\partial S_j}(x^a) = \frac{\partial F}{\partial T_j}(x)$ for $j \neq i$, because setting $T_i = 1$ commutes with $\partial/\partial T_j$, $j \neq i$.

For homogeneous $F \in I$ of degree d, Euler's identity $\sum_j T_j \frac{\partial F}{\partial T_j} = dF$ (valid over any ring) at the point x gives $\sum_j x_j \frac{\partial F}{\partial T_j}(x) = dF(x) = 0$, i.e. $\frac{\partial F}{\partial T_i}(x) = -\sum_{j\neq i} x_j \frac{\partial F}{\partial T_j}(x)$ (using $x_i = 1$). Hence for all $t \in k^{n+1}$

$$L_F(t) = \sum_{j\neq i} \frac{\partial F}{\partial T_j}(x)\,(t_j - x_j t_i). \qquad (*)$$

Define $\rho : k^{n+1} \to k^n$, $\rho(t) = (t_j - x_j t_i)_{j\neq i}$. By $(*)$ and the description of T', for homogeneous $F \in I$: $L_F(t) = 0$ iff $\rho(t)$ satisfies the condition of F in T'. So $W = \rho^{-1}(T')$. The map ρ is surjective ($\rho(t) = s$ for $t_i = 0$, $t_j = s_j$), with kernel $\{t : \; t_j = x_j t_i \;\forall j \neq i\} = k \cdot x$ (recall $x_i = 1$), a line. Hence $\dim_k W = \dim_k T' + 1$ and

$$\dim PT_{X,x} = \dim_k W - 1 = \dim_k T' = \dim_k T_{X,x}.$$

This proves the assertion. □

Exercise 2.10

> **Problem.** Let X be a closed subvariety of $\mathbb{P}^n_k = \operatorname{Proj} k[T_0, \ldots, T_n]$ defined by some homogeneous polynomials $F_1, \ldots, F_r$. Let $x \in X(k)$. Let J_x denote the matrix $J_x = \big(\frac{\partial F_i}{\partial T_j}(x)\big)_{1\leq i\leq r,\; 0\leq j\leq n}$. Show that x is regular in X if and only if $\operatorname{rank} J_x = n - \dim \mathcal{O}_{X,x}$. This is the Jacobian criterion for projective varieties.

Solution. (The point called x_0 in the statement is x.) Here $I = (F_1, \ldots, F_r)$, and x is represented by a fixed vector $x \in k^{n+1} \setminus \{0\}$; rescaling it multiplies the i-th row of J_x by $\lambda^{\deg F_i - 1} \neq 0$, so the rank of J_x does not depend on the representative.

We use the notation L_F, W of the proof of Ex. 2.9. If $F = \sum_i A_i F_i \in I$, then $\frac{\partial F}{\partial T_j}(x) = \sum_i A_i(x) \frac{\partial F_i}{\partial T_j}(x) + \sum_i F_i(x) \frac{\partial A_i}{\partial T_j}(x) = \sum_i A_i(x) \frac{\partial F_i}{\partial T_j}(x)$, since $F_i(x) = 0$. So L_F is a combination of $L_{F_1}, \ldots, L_{F_r}$, and $W = \{t \in k^{n+1} : \; J_x t = 0\}$ has dimension $n + 1 - \operatorname{rank} J_x$. By the proof of Ex. 2.9, $\dim_k T_{X,x} = \dim_k W - 1 = n - \operatorname{rank} J_x$.

By Def. 2.8, X is regular at x iff $\dim_k T_{X,x} = \dim \mathcal{O}_{X,x}$ (here $k(x) = k$), i.e. iff $n - \operatorname{rank} J_x = \dim \mathcal{O}_{X,x}$. □

Exercise 2.11

> **Problem.** Let k be an algebraically closed field, and let $n \geq 1$. Show that the curve $\operatorname{Proj} k[x,y,z]/(x^n+y^n+z^n)$ is regular if and only if n is prime to $\operatorname{char}(k)$.

Solution. Let $F = x^n + y^n + z^n$ and $X = \operatorname{Proj} k[x,y,z]/(F) \subseteq \mathbb{P}^2_k$ (as a scheme, possibly non-reduced). The scheme X is Noetherian, so it is regular iff it is regular at its closed points (Cor. 2.17). Since k is algebraically closed, every closed point x of X is rational (its residue field is finite over k, Cor. 2.1.12). Conversely every rational point of X is closed.

Local dimension. A closed point x lies in a chart $D_+(x_i) \cap X = \operatorname{Spec} k[u,v]/(f)$, where f is the dehomogenization of F, a non-constant polynomial ($n \geq 1$). Then $\mathcal{O}_{X,x} = S/fS$ with $S = k[u,v]_M$ for a maximal ideal $M \ni f$. The ring S is a regular local ring of dimension 2 (Example 2.18, Cor. 2.5.24), hence a domain (Prop. 2.11), and $f \neq 0$ in $\mathfrak{m}_S$, so $\dim \mathcal{O}_{X,x} = \dim S - 1 = 1$ (Thm. 2.5.15, as in the proof of Cor. 2.12). By Ex. 2.10 (with $n=2$, $r=1$), X is regular at x iff $\operatorname{rank} J_x = 2-1 = 1$, i.e. iff $J_x \neq 0$, where

$$J_x = \big(n\,a^{n-1},\ n\,b^{n-1},\ n\,c^{n-1}\big), \qquad x = (a:b:c).$$

Case $\operatorname{char}(k) \nmid n$ *(including* $\operatorname{char} k = 0$*).* Then $n \in k^*$. If $n = 1$, $J_x = (1,1,1) \neq 0$. If $n \geq 2$, $J_x = 0$ would force $a^{n-1} = b^{n-1} = c^{n-1} = 0$, i.e. $a = b = c = 0$, impossible for a projective point. So $J_x \neq 0$ at every closed point and X is regular.

Case $p = \operatorname{char}(k) > 0$ *divides* n. Then $n = 0$ in k, so $J_x = 0$ at every point. The scheme X has a rational point: since k is algebraically closed, there is $a \in k$ with $a^n = -1$, and $(a:0:1) \in X$; it is a closed point. At this point $\operatorname{rank} J_x = 0 \neq 1$, so X is not regular there. Hence X is regular if and only if n is prime to $\operatorname{char}(k)$. □

Exercise 2.12

> **Problem.** Let X, Y be algebraic varieties over an algebraically closed field k. If X, Y are regular, show that $X \times_k Y$ is also regular. Show that this is false in general if k is not algebraically closed.

Solution. *Regularity of the product.* Let $Z = X \times_k Y$, with projections $p : Z \to X$, $q : Z \to Y$. It is of finite type over k, hence Noetherian, so by Cor. 2.17 it suffices to show that Z is regular at its closed points. Let $z \in Z$ be a closed point; $k(z)$ is finite over k, hence equal to k (k algebraically closed). Put $x = p(z)$, $y = q(z)$; then $k \subseteq k(x) \subseteq k(z) = k$, so $k(x) = k = k(y)$: x, y are rational points. Let $A = \mathcal{O}_{X,x}$, $B = \mathcal{O}_{Y,y}$, $d = \dim A$, $e = \dim B$ and $C = \mathcal{O}_{Z,z}$. By hypothesis A, B are regular, so there are $f_1, \ldots, f_d$ generating $\mathfrak{m}_x = \mathfrak{m}_A$ and $g_1, \ldots, g_e$ generating $\mathfrak{m}_B$.

(1) $\dim C = d + e$. The morphism $p : Z \to X$ is the base change of $Y \to \operatorname{Spec} k$, which is flat ($k$ is a field), so p is flat (Prop. 3.3(b)). By Thm. 3.12, $\dim \mathcal{O}_{Z_x,z} = \dim C - \dim A$. The fibre is $Z_x = Z \times_X \operatorname{Spec} k(x) = Y \times_k \operatorname{Spec} k = Y$ (as $k(x) = k$), and z corresponds to y, so $\mathcal{O}_{Z_x,z} = B$. Hence $\dim C = d + e$.

(2) $\mathfrak{m}_C$ *is generated by* $d + e$ *elements.* Take affine open neighbourhoods $\operatorname{Spec} A' \ni x$, $\operatorname{Spec} B' \ni y$; then $\operatorname{Spec}(A' \otimes_k B') \ni z$, and z corresponds to the kernel of the surjection $A' \otimes B' \to$

$A'/\mathfrak{m}_x \otimes B'/\mathfrak{m}_y = k \otimes_k k = k$. This kernel is generated by $\mathfrak{m}_x \otimes 1$ and $1 \otimes \mathfrak{m}_y$. Therefore $\mathfrak{m}_C = \mathfrak{m}_x C + \mathfrak{m}_y C$, where $\mathfrak{m}_x C$ (resp. $\mathfrak{m}_y C$) is the ideal generated by the image of $\mathfrak{m}_A$ (resp. $\mathfrak{m}_B$) under $A \to C$ (resp. $B \to C$). Since $\mathfrak{m}_A = (f_1, \dots, f_d)$ and $\mathfrak{m}_B = (g_1, \dots, g_e)$, $\mathfrak{m}_C = (f_1 \otimes 1, \dots, f_d \otimes 1, 1 \otimes g_1, \dots, 1 \otimes g_e)$. By Nakayama, $\dim_k \mathfrak{m}_C/\mathfrak{m}_C^2 \le d + e$.

By (1), (2) and Cor. 2.5.14(b), $d + e = \dim C \le \dim_k \mathfrak{m}_C/\mathfrak{m}_C^2 \le d + e$. So C is regular. Hence Z is regular.

Counterexample over a non-closed field. Let p be a prime, $k = \mathbb{F}_p(s)$, $L = k[\tau]/(\tau^p - s) = k(s^{1/p})$ (a field, since $\tau^p - s$ is irreducible by Eisenstein). Let $X = Y = \operatorname{Spec} L$, a variety over k, regular because it is a point whose local ring is a field. Let $\sigma_0 \in L$ be the class of τ. Then

$$L \otimes_k L = L[\tau]/(\tau^p - s) = L[\tau]/\big((\tau - \sigma_0)^p\big) = L[\eta]/(\eta^p), \qquad \eta = \tau - \sigma_0,$$

because $\tau^p - s = \tau^p - \sigma_0^p = (\tau - \sigma_0)^p$ in characteristic p. This is a local Artinian ring with maximal ideal $(\eta) \neq 0$ (for $p \ge 2$, $\eta \neq 0$), so $X \times_k Y$ is a single point with local ring of dimension 0 and $\dim \mathfrak{m}/\mathfrak{m}^2 = 1$. So $X \times_k Y$ is not regular (it is not even reduced), although X and Y are regular. □

Exercise 2.13

> **Problem.** Let $(A, \mathfrak{m})$ be a Noetherian local ring with residue field k. We set $\operatorname{gr}_{\mathfrak{m}}(A) = \bigoplus_{n \ge 0} \mathfrak{m}^n/\mathfrak{m}^{n+1}$. This is the graded ring associated to the ideal $\mathfrak{m}$. It naturally has the structure of a k-algebra. Let us suppose that A is regular of dimension d. We are going to show that $\operatorname{gr}_{\mathfrak{m}}(A)$ is a polynomial ring in d variables over k. To simplify, we suppose that k is infinite. (a) Let $f_1, \dots, f_d$ be a coordinate system for $\mathfrak{m}$. Let $\varphi : k[T_1, \dots, T_d] \to \operatorname{gr}_{\mathfrak{m}}(A)$ be the k-algebra homomorphism defined by $\varphi(T_i) =$ the image of f_i in $\mathfrak{m}/\mathfrak{m}^2$. Show that φ is surjective, and that φ is an isomorphism if $d = 0$. (b) Let $\tilde{\mathfrak{m}}$ be the maximal ideal of $\operatorname{gr}_{\mathfrak{m}}(A)$ generated by the $\varphi(T_i)$. Show that if $\dim \operatorname{gr}_{\mathfrak{m}}(A)_{\tilde{\mathfrak{m}}} = 0$, then $\dim A = 0$. (c) Let us suppose $d \ge 1$. Show that $\sum_i \varphi(T_i)k$ is not contained in the union of the minimal prime ideals of $\operatorname{gr}_{\mathfrak{m}}(A)$ (here we use the k infinite hypothesis). (d) Let $f \in \mathfrak{m}$. Show that we have a canonical surjective homomorphism $\operatorname{gr}_{\mathfrak{m}}(A) \to \operatorname{gr}_{\mathfrak{m}/fA}(A/fA)$. Show by induction on d that $\dim \operatorname{gr}_{\mathfrak{m}}(A) = d$ and that φ is an isomorphism.

Solution. Write $G = \operatorname{gr}_{\mathfrak{m}}(A) = \bigoplus_{n \ge 0} G_n$, $G_n = \mathfrak{m}^n/\mathfrak{m}^{n+1}$, so $G_0 = k$, $G_1 = \mathfrak{m}/\mathfrak{m}^2$, and $\bar{f}_i = \varphi(T_i) \in G_1$ the class of f_i.

(a) Since $\mathfrak{m} = (f_1, \dots, f_d)$, $\mathfrak{m}^n$ is generated by the monomials of degree n in the f_i; so $G_n = \mathfrak{m}^n/\mathfrak{m}^{n+1}$ is spanned over $k = A/\mathfrak{m}$ by the classes of these monomials, which are the monomials of degree n in $\bar{f}_1, \dots, \bar{f}_d$. Thus $G_n \subseteq \operatorname{Im} \varphi$ for all n, and φ is surjective. In particular G is a Noetherian ring and $G_+ := \bigoplus_{n \ge 1} G_n = (\bar{f}_1, \dots, \bar{f}_d)$. If $d = 0$, then $\mathfrak{m} = 0$ (Def. 2.7), A is a field, $G = G_0 = A = k$, and $\varphi : k \to k$ (the polynomial ring in 0 variables) is the identity, an isomorphism.

(b) The ideal $\tilde{\mathfrak{m}} = G_+$ is the kernel of $G \to G_0 = k$, hence a maximal ideal. Suppose $\dim G_{\tilde{\mathfrak{m}}} = 0$. Then the maximal ideal of the Noetherian local ring $G_{\tilde{\mathfrak{m}}}$ is nilpotent: there is N with $\bar{f}_i^N = 0$ in $G_{\tilde{\mathfrak{m}}}$ for all i, i.e. there are $s_i \in G \setminus \tilde{\mathfrak{m}}$ with $s_i \bar{f}_i^N = 0$ in G. Put $s = \prod s_i \notin \tilde{\mathfrak{m}}$ (as $\tilde{\mathfrak{m}}$ is prime). Write $s = c + h$ with $c \in G_0 = k$, $h \in G_+$; since $s \notin G_+$, $c \neq 0$. Then $0 = s\bar{f}_i^N$ has degree-N component $c\bar{f}_i^N$, so $\bar{f}_i^N = 0$ in G. Hence $G_+ = (\bar{f}_1, \dots, \bar{f}_d)$ is nilpotent: $G_+^m = 0$ for

$m = d(N-1)+1$, so $G_n = 0$ for $n \geq m$. This means $\mathfrak{m}^n = \mathfrak{m}^{n+1} = \mathfrak{m} \cdot \mathfrak{m}^n$ for $n \geq m$, so $\mathfrak{m}^n = 0$ by Nakayama. Then every prime ideal of A contains $\mathfrak{m}$, so $\operatorname{Spec} A$ is a point and $\dim A = 0$.

(c) Let $d \geq 1$. G is Noetherian, so it has finitely many minimal primes $P_1, \ldots, P_s$. The k-vector space $G_1 = \sum_i k\bar{f}_i$ is nonzero (its dimension is d). Suppose $G_1 \subseteq P_1 \cup \cdots \cup P_s$. Then $G_1 = \bigcup_j (G_1 \cap P_j)$ is a finite union of k-subspaces. Since k is infinite, a vector space is never a finite union of proper subspaces (take such a union $V = V_1 \cup \cdots \cup V_s$ with s minimal, $s \geq 2$; pick $u \in V_1 \setminus \bigcup_{j \geq 2} V_j$ and $w \notin V_1$; the infinitely many vectors $w + \lambda u$ ($\lambda \in k$) are not in V_1, so two of them lie in the same V_j, $j \geq 2$, so their difference $(\lambda - \lambda')u$ is in V_j, contradicting the choice of u). Hence $G_1 \subseteq P_j$ for some j, so $G_+ = G_1 G \subseteq P_j$. As G_+ is a prime ideal contained in the minimal prime P_j, $G_+ = P_j$ is a minimal prime of G, so $\dim G_{\tilde{\mathfrak{m}}} = 0$. By (b), $\dim A = 0$, contradicting $\dim A = d \geq 1$.

(d) Let $f \in \mathfrak{m}$, $\bar{A} = A/fA$ with maximal ideal $\bar{\mathfrak{m}} = \mathfrak{m}/fA = \mathfrak{m}\bar{A}$. Then $\bar{\mathfrak{m}}^n = (\mathfrak{m}^n + fA)/fA$, and the composite $\mathfrak{m}^n \to \bar{\mathfrak{m}}^n \to \bar{\mathfrak{m}}^n/\bar{\mathfrak{m}}^{n+1}$ kills $\mathfrak{m}^{n+1}$. So it induces surjections $\mathfrak{m}^n/\mathfrak{m}^{n+1} \to \bar{\mathfrak{m}}^n/\bar{\mathfrak{m}}^{n+1}$ for all n, compatible with multiplication (it is induced by the ring homomorphism $A \to \bar{A}$ mapping $\mathfrak{m}^n$ into $\bar{\mathfrak{m}}^n$). Their sum is a surjective homomorphism of graded k-algebras $\pi : G \to \bar{G} := \operatorname{gr}_{\bar{\mathfrak{m}}}(\bar{A})$. If $f \in \mathfrak{m} \setminus \mathfrak{m}^2$, the class of f in $\mathfrak{m}/\mathfrak{m}^2 = G_1$ lies in $\ker \pi$, since $f \mapsto 0 \in \bar{A}$.

Induction on d. We prove: for every regular local ring $(A, \mathfrak{m})$ of dimension d with infinite residue field k, $\dim G = d$ and φ is an isomorphism (for any coordinate system). For $d = 0$ this is (a) ($G = k$ has dimension 0). Let $d \geq 1$ and assume the result in dimension $d-1$.

By (c) there is $\ell \in G_1$ lying in no minimal prime of G. Then $\ell \neq 0$ (0 lies in every prime; $G \neq 0$). Write $\ell = \sum c_i \bar{f}_i$ ($c_i \in k$) and let $f = \sum \tilde{c}_i f_i \in \mathfrak{m}$ with lifts $\tilde{c}_i \in A$ of c_i; its class in $\mathfrak{m}/\mathfrak{m}^2$ is $\ell \neq 0$, so $f \in \mathfrak{m} \setminus \mathfrak{m}^2$. Since A is regular, hence a domain (Prop. 2.11), $f \neq 0$ and Cor. 2.12 shows that $\bar{A} = A/fA$ is regular, of dimension $d-1$ (Cor. 2.12), with residue field k, still infinite. By the induction hypothesis $\dim \bar{G} = d-1$.

Take the surjection $\pi : G \to \bar{G}$ of the first part; $\pi(\ell) = 0$. A chain of primes $Q_0 \subsetneq \cdots \subsetneq Q_{d-1}$ of $\bar{G}$ pulls back under π to a chain $P_0 \subsetneq \cdots \subsetneq P_{d-1}$ of primes of G, all containing $\ker \pi \ni \ell$. Since ℓ lies in no minimal prime of G, P_0 is not minimal, so there is a prime $P_{-1} \subsetneq P_0$. This gives a chain of length d, so $\dim G \geq d$.

On the other hand φ is surjective by (a), so $G \simeq k[T_1, \ldots, T_d]/\ker \varphi$ and $\dim G \leq \dim k[T_1, \ldots, T_d] = d$ (Cor. 2.5.17). Hence $\dim G = d$. If $\ker \varphi$ were nonzero, then every prime of $k[T_1, \ldots, T_d]$ containing $\ker \varphi$ would be nonzero, so any chain of such primes could be extended below by the prime (0) of the domain $k[T_1, \ldots, T_d]$, giving $\dim G \leq d-1$, a contradiction. So $\ker \varphi = 0$ and φ is an isomorphism: $\operatorname{gr}_{\mathfrak{m}}(A) \simeq k[T_1, \ldots, T_d]$. □

4.3 Flat morphisms and smooth morphisms

Results of the text are cited by their numbers in Liu ("Prop. 3.8", "Thm. 3.12", ...), meaning Chapter 4, Section 4.3, unless a Chapter is indicated. Honest notes. Ex. 3.1 needs $\dim \mathcal{O}_{S,s} = 1$ (otherwise $t = 0$). Ex. 3.2 is false as printed for a non-Jacobson base; it is proved under the hypothesis that closed points of X map to closed points of Y (automatic for varieties), and a counterexample is given. In Ex. 3.8(b) the assertion "there is an open $V \ni y$ such that $Z \cap f^{-1}(V) \to f^{-1}(V)$ is an open immersion" is false for non-proper f of finite type; the correct statement (an open neighbourhood of the fibre X_y in X) is proved, and the proper case is proved

as stated. Ex. 3.11 is false as printed (take $X_1 = Y$), and the correct statements are proved. Ex. 3.9 uses Chevalley's theorem in place of the suggested Ex. 3.2.17. Ex. 3.12 uses Ex. 3.9, and Ex. 3.13(c) uses Ex. 3.2.

Exercise 3.1

> **Problem.** Let X be a flat scheme of finite type over a Dedekind scheme S. Let $s \in S$ be a closed point and $x \in X_s$. Let t denote a generator of the maximal ideal of $\mathcal{O}_{S,s}$. We suppose that X is regular at x. Show that X_s is regular at x if and only if $t \in \mathfrak{m}_x \setminus \mathfrak{m}_x^2$, where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$.

Solution. We assume $\dim \mathcal{O}_{S,s} = 1$, i.e. $A := \mathcal{O}_{S,s}$ is a discrete valuation ring with uniformizing element t (if $\dim \mathcal{O}_{S,s} = 0$ then $t = 0$, the statement is empty or false: X_s is regular at x but $t = 0 \notin \mathfrak{m}_x \setminus \mathfrak{m}_x^2$). Put $B = \mathcal{O}_{X,x}$, $\mathfrak{m} = \mathfrak{m}_x$, $d = \dim B$, $k = B/\mathfrak{m}$. The point x lies over s, so $t \in \mathfrak{m}$. The local ring of the fibre is

$$\mathcal{O}_{X_s,x} = B/\mathfrak{m}_s B = B/tB.$$

As B is regular, it is a domain. Since B is flat over A and multiplication by t is injective on A, multiplication by t is injective on $B = B \otimes_A A$. Thus $t \neq 0$ in B, and B is not a field, so $d \geq 1$. We have $\dim B/tB = d - 1$: the inequality $\geq d - 1$ is Krull's principal ideal theorem (Thm. 2.5.15), and $\leq d-1$ holds because B is a domain and $t \neq 0$ (a chain of primes containing t can be extended by (0) at the bottom). The maximal ideal of $\bar{B} := B/tB$ is $\bar{\mathfrak{m}} = \mathfrak{m}/tB$ and

$$\bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2 = \mathfrak{m}/(\mathfrak{m}^2 + tB),$$

a quotient of $\mathfrak{m}/\mathfrak{m}^2$ by the line spanned by the class of t. Since B is regular, $\dim_k \mathfrak{m}/\mathfrak{m}^2 = d$.

If $t \notin \mathfrak{m}^2$, the class of t in $\mathfrak{m}/\mathfrak{m}^2$ is non-zero, so $\dim_k \bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2 = d - 1 = \dim \bar{B}$, and $\bar{B}$ is regular.

If $t \in \mathfrak{m}^2$, then $\mathfrak{m}^2 + tB = \mathfrak{m}^2$, so $\dim_k \bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2 = d > d - 1 = \dim \bar{B}$, and $\bar{B}$ is not regular.

Hence X_s is regular at x if and only if $t \in \mathfrak{m} \setminus \mathfrak{m}^2$. □

Exercise 3.2

> **Problem.** Let $f : X \to Y$ be a morphism of finite type. Show that f is flat if and only if for every closed point $y \in Y$ and for every closed point x of X_y, f is flat at x.

Solution. *Correction of the statement.* As printed the statement needs a hypothesis. We assume that Y is quasi-compact and that f maps closed points of X to closed points of Y; the latter is automatic if Y is a scheme of finite type over a field or over $\mathbb{Z}$ (Nullstellensatz), in particular for morphisms of algebraic varieties, which is the situation where the statement is used. Without it the statement is false (counterexample at the end). Note that if y is closed in Y, then X_y is closed in X, so the closed points of X_y are exactly the closed points x of X with $f(x) = y$ closed. Hence, under our hypothesis, the closed points x occurring in the statement are exactly the closed points of X. The implication "⇒" is trivial.

Step 1: flatness at x implies flatness at every generization x' of x. Let $y = f(x)$, $y' = f(x')$, $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$. Then $\mathcal{O}_{X,x'} = B_{\mathfrak{q}}$ for a prime $\mathfrak{q}$ of B, and $\mathcal{O}_{Y,y'} = A_{\mathfrak{p}}$ where $\mathfrak{p}$ is the inverse image of $\mathfrak{q}$ in A. The module $B_{\mathfrak{q}}$ is flat over B, hence over A. Every element of $A \setminus \mathfrak{p}$ is mapped outside $\mathfrak{q}$, so it is a unit in $B_{\mathfrak{q}}$, and for an $A_{\mathfrak{p}}$-module N we have $B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} N = B_{\mathfrak{q}} \otimes_A N$. The right-hand side is exact in N, so $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$.

Step 2. Suppose f is flat at all closed points of X. Since f is of finite type and Y is quasi-compact, X is quasi-compact. Let $x \in X$ be arbitrary. The closure of $\{x\}$ is a non-empty closed subset of a quasi-compact scheme, hence contains a closed point x_0 of X. Then f is flat at x_0 by hypothesis, and x is a generization of x_0, so f is flat at x by Step 1. Thus f is flat. □

Counterexample without the hypothesis on closed points. Let k be a field, $A = k[s,t]_{(s,t)}$, $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$ with $B = A[T]/(s,\, tT - 1)$. Then $B \simeq (A/sA)[1/t] = k(t)$, so $X \to Y$ is of finite type. The only closed point of Y is $\mathfrak{m} = (s,t)$ and $\mathfrak{m}B = B$ because t is invertible in B; so $X_{\mathfrak{m}} = \emptyset$ and the condition of the exercise holds vacuously. But $B \neq 0$ is killed by $s \neq 0$, so B is not torsion free over the domain A, hence not flat.

Exercise 3.3

> **Problem.** Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Show that $\dim X \leq \dim Y$ if f is injective, and $\dim X \geq \dim Y$ if f is flat and surjective.

Solution. *Injective case.* Let $Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n$ be irreducible closed subsets of X, with generic points $z_0, \ldots, z_n$. Then $z_i \in \overline{\{z_{i+1}\}}$ and $z_i \neq z_{i+1}$. By continuity, $f(z_i) \in \overline{\{f(z_{i+1})\}}$, so $\overline{\{f(z_i)\}} \subseteq \overline{\{f(z_{i+1})\}}$. Since f is injective, $f(z_i) \neq f(z_{i+1})$. A scheme is a T_0 space (an irreducible closed subset has a unique generic point), so the two closures are distinct. Thus $\overline{\{f(z_0)\}} \subsetneq \cdots \subsetneq \overline{\{f(z_n)\}}$ is a chain of length n of irreducible closed subsets of Y. Hence $\dim X \leq \dim Y$ (trivially true if $\dim Y = \infty$).

Flat surjective case. Recall that $\dim Y = \sup_{y \in Y} \dim \mathcal{O}_{Y,y}$ and $\dim \mathcal{O}_{X,x} \leq \dim X$ for all x. Let $y \in Y$. Since f is surjective, there is $x \in X$ with $f(x) = y$. By Thm. 3.12 (equality for flat f),

$$\dim \mathcal{O}_{X_y,x} = \dim \mathcal{O}_{X,x} - \dim \mathcal{O}_{Y,y}.$$

The left-hand side is ≥ 0, so $\dim X \geq \dim \mathcal{O}_{X,x} \geq \dim \mathcal{O}_{Y,y}$. Taking the supremum over y gives $\dim X \geq \dim Y$. □

Exercise 3.4

> **Problem.** Let k be a field. Let A be the polynomial ring $k[x,y]$. Show, by computing the dimension of the fibers, that the canonical morphism $\operatorname{Proj} A[T_0,T_1]/(yT_0 - xT_1) \to \operatorname{Spec} A$ is not flat.

Solution. Let $X = \operatorname{Proj} A[T_0,T_1]/(yT_0 - xT_1)$ and $f : X \to \operatorname{Spec} A = \mathbb{A}^2_k$. The scheme X is covered by the two affine open sets

$$D_+(T_0) = \operatorname{Spec} k[x,y,u]/(y - xu) \simeq \operatorname{Spec} k[x,u], \qquad D_+(T_1) = \operatorname{Spec} k[x,y,v]/(yv - x) \simeq \operatorname{Spec} k[y,v]$$

(here $u = T_1/T_0$, $v = T_0/T_1$; the relation allows us to eliminate y, respectively x). So X is covered by two copies of the affine plane, and for every $x' \in X$ the local ring $\mathcal{O}_{X,x'}$ is a localization of a polynomial ring in two variables over k; hence $\dim \mathcal{O}_{X,x'} \leq 2$.

Now consider the origin $o = (x, y) \in \mathbb{A}^2_k$, with $k(o) = k$. The fibre is $X_o = \operatorname{Proj} k[T_0, T_1] = \mathbb{P}^1_k$, since the relation $yT_0 - xT_1$ vanishes in $k = A/(x, y)$. Take a closed point $x' \in X_o$ (for instance $[1 : 0]$); then $\dim \mathcal{O}_{X_o,x'} = 1$.

Suppose f were flat. Then Thm. 3.12 gives

$$1 = \dim \mathcal{O}_{X_o,x'} = \dim \mathcal{O}_{X,x'} - \dim \mathcal{O}_{\mathbb{A}^2,o} = \dim \mathcal{O}_{X,x'} - 2,$$

so $\dim \mathcal{O}_{X,x'} = 3$, contradicting $\dim \mathcal{O}_{X,x'} \leq 2$. Hence f is not flat. (Alternatively, via Cor. 3.14: f is surjective and X is irreducible of dimension 2, so if f were flat all fibres would have dimension $\dim X - \dim Y = 0$, but $\dim X_o = 1$.) □

Exercise 3.5

> **Problem.** Let A be a reduced ring, and let $a \in A$ be a non-zero divisor. Let us consider a ring B containing $A[T]$ and an element $s \in B$ such that $s^n = aT$ for some $n \geq 1$. Show that the subalgebra $A[T][s]$ of B is free (and hence flat) over $A[T]$ (consider a linear relation between $1, \ldots, s^{n-1}$ and show that the coefficients are zero in $A_{\mathfrak{p}}[T]$ for every minimal prime ideal $\mathfrak{p}$ of A).

Solution. We show that $1, s, \ldots, s^{n-1}$ is a basis of $A[T][s]$ over $A[T]$.

Generation. For $j \geq 0$ write $j = qn + r$ with $0 \leq r < n$. Then $s^j = (s^n)^q s^r = (aT)^q s^r \in A[T]s^r$. So $A[T][s] = \sum_{r<n} A[T]s^r$.

Independence. Let $c_0, \ldots, c_{n-1} \in A[T]$ with $\sum_{i<n} c_i s^i = 0$ in B. Let $\mathfrak{p}$ be a minimal prime of A, and $S = A \setminus \mathfrak{p}$. Localizing the inclusion $A[T] \subseteq B$ at S (localization is exact) gives an inclusion $K[T] \subseteq S^{-1}B$, where $K = A_{\mathfrak{p}}$. Since A is reduced, $\mathfrak{p}A_{\mathfrak{p}}$ is the unique prime and equals the nilradical, which is 0; so K is a field. Since A is reduced, every element $c \in \mathfrak{p}$ is a zero divisor (there is $u \notin \mathfrak{p}$ with uc nilpotent, so $uc = 0$ and $u \neq 0$); hence the non-zero divisor a is not in $\mathfrak{p}$ and is a unit in K. In $S^{-1}B$ we have $s^n = aT$ with $a \in K^*$.

Let $K[S]$ be the polynomial ring in one variable and define the K-algebra homomorphism $\psi : K[S] \to S^{-1}B$, $S \mapsto s$. Its image is $K[s] \ni aT$, hence contains T. Let $I = \ker \psi$. We claim that $I = 0$. The subring $K[S^n] \subseteq K[S]$ maps onto $K[s^n] = K[aT] = K[T]$, and ψ restricted to $K[S^n]$ is injective because $K[T] \to S^{-1}B$ is injective and $S^n \mapsto aT$ (the elements aT and T generate the same polynomial ring). So $I \cap K[S^n] = 0$. If $I \neq 0$, choose $0 \neq g \in I$. As $K[S]$ is integral over $K[S^n]$, there is an equation $g^m + e_{m-1}g^{m-1} + \cdots + e_0 = 0$ with $e_i \in K[S^n]$ and m minimal. Then $e_0 \neq 0$ (otherwise, $K[S]$ being a domain, we could cancel a factor g and lower m), and $e_0 = -g(g^{m-1} + \cdots + e_1) \in I \cap K[S^n]$, a contradiction. So ψ is injective, and $1, s, \ldots, s^{n-1}$ are linearly independent over $K[T]$: indeed $K[S]$ is free over $K[S^n] = K[aT] = K[T]$ with basis $1, S, \ldots, S^{n-1}$.

Consequently the image of each c_i in $K[T] = A_{\mathfrak{p}}[T]$ is zero, for every minimal prime $\mathfrak{p}$. Since A is reduced, the intersection of its minimal primes is 0, so $A \to \prod_{\mathfrak{p}} A/\mathfrak{p}$ is injective. The field $A_{\mathfrak{p}}$ is $\operatorname{Frac}(A/\mathfrak{p})$ and $A \to A_{\mathfrak{p}}$ factors through $A/\mathfrak{p} \subseteq \operatorname{Frac}(A/\mathfrak{p})$. Hence $A \to \prod_{\mathfrak{p}} A_{\mathfrak{p}}$ is injective, and so is $A[T] \to \prod_{\mathfrak{p}} A_{\mathfrak{p}}[T]$ (coefficientwise). Therefore all $c_i = 0$.

So $A[T][s]$ is a free $A[T]$-module with basis $1, \dots, s^{n-1}$, and in particular flat. $\square$

Exercise 3.6

> **Problem.** Let $f : X \to Y$ be a morphism of integral schemes. We suppose that the generic fiber of f is geometrically integral (the definition is the same as for algebraic varieties 3.2.8). Show that for any flat morphism $Z \to Y$ with Z integral, we have $X \times_Y Z$ integral. This partially generalizes Exercise 3.2.14.

Solution. Let η be the generic point of Y, $K = K(Y) = k(\eta)$, ξ the generic point of X, ζ the generic point of Z, $g : Z \to Y$ the given flat morphism, and $W = X \times_Y Z$.

Preliminaries.

(i) The generic fibre X_η is geometrically integral, hence non-empty. Since ξ is a generization of every point of X, $f(\xi)$ is a generization of η, so $f(\xi) = \eta$; thus $\xi \in X_\eta$. Similarly $g(\zeta) = \eta$: by Lemma 3.7, $g(Z)$ is dense in Y, and $g(Z) \subseteq \overline{\{g(\zeta)\}}$.

(ii) *If X_η is geometrically integral over K, then $X_\eta \otimes_K L$ is integral for every field extension L/K.* First a lemma: *if Ω is an algebraically closed field, T an integral Ω-scheme (not necessarily of finite type) and L/Ω a field extension, then T_L is integral.* Let $\operatorname{Spec} B \subseteq T$ be a non-empty affine open, so B is a domain. Write B and L as directed unions of finitely generated Ω-subalgebras B_0, L_0 (domains). By Ex. 3.2.15 (the case where $\operatorname{Spec} B_0$ is integral over the algebraically closed field Ω, hence geometrically integral, and $\operatorname{Spec} L_0$ is integral) each $B_0 \otimes_\Omega L_0$ is a domain. These rings are subrings of $B \otimes_\Omega L$ (tensor products over a field preserve injections), and $B \otimes_\Omega L$ is their directed union; so it is a domain. Thus T_L is covered by the non-empty integral affine opens U_L, $U \subseteq T$ non-empty affine open. Two of them meet: $U \cap U'$ is a non-empty open subset of T, it contains a non-empty affine open W, and $W_L = \operatorname{Spec}(\mathcal{O}(W) \otimes_\Omega L)$ is non-empty since $\mathcal{O}(W) \otimes_\Omega L \neq 0$. A space covered by irreducible opens which pairwise meet is irreducible, and T_L is reduced, so T_L is integral.

Now choose an algebraic closure $\bar{L}$ of L and let $\bar{K}$ be the algebraic closure of K inside $\bar{L}$. The scheme $(X_\eta)_{\bar{K}}$ is integral by hypothesis, so by the lemma $(X_\eta)_{\bar{L}} = (X_\eta)_{\bar{K}} \times_{\bar{K}} \bar{L}$ is integral. Finally $(X_\eta)_{\bar{L}} \to (X_\eta)_L$ is faithfully flat (base change of $L \to \bar{L}$), hence surjective, so $(X_\eta)_L$ is irreducible, and it is reduced because its local rings embed in those of $(X_\eta)_{\bar{L}}$ (flat local homomorphisms are faithfully flat, hence injective).

(iii) The same holds for every non-empty open subscheme $U \subseteq X_\eta$, since U_L is a non-empty open subscheme of the integral scheme $(X_\eta)_L$.

The affine case. Let $V = \operatorname{Spec} A \subseteq Y$, $\operatorname{Spec} B \subseteq f^{-1}(V)$, $\operatorname{Spec} C \subseteq g^{-1}(V)$ be non-empty affine open subsets, where A, B, C are domains. The generic points ξ, ζ lie in them, so $A \to B$ and $A \to C$ are injective (the primes $\ker(A \to B)$ and $\ker(A \to C)$ are the images of the generic points, i.e. 0). The A-module C is flat. Let $K = \operatorname{Frac}(A)$, $B_K = B \otimes_A K$, $C_K = C \otimes_A K$ and $L = \operatorname{Frac}(C) \supseteq K$. We have

$$B \otimes_A C \;\hookrightarrow\; B_K \otimes_A C = B_K \otimes_K C_K \;\hookrightarrow\; B_K \otimes_K L.$$

The first map is injective because $B \to B_K$ is injective (B is a domain containing A) and C is A-flat. The equality holds because $B_K \otimes_A C = B_K \otimes_K (K \otimes_A C)$. The last map is injective

because $C_K \subseteq L$ and B_K is free over the field K. Now $\operatorname{Spec} B_K$ is a non-empty open subscheme of X_η (it is the part of X_η inside $\operatorname{Spec} B$), so $B_K \otimes_K L$ is a domain by (ii) and (iii). Hence $B \otimes_A C$ is a domain: every such affine open piece $U \times_V Z'$ of W is integral.

Globalization. The schemes $U \times_V Z'$, where $V \subseteq Y$, $U \subseteq f^{-1}(V)$, $Z' \subseteq g^{-1}(V)$ are non-empty affine opens, form an open cover of W by integral schemes; in particular W is reduced. They are non-empty and pairwise intersect: given two of them, $W_1 = U_1 \times_{V_1} Z_1$ and $W_2 = U_2 \times_{V_2} Z_2$, the open subscheme $(U_1 \cap U_2) \times_Y (Z_1 \cap Z_2)$ of W is contained in both, and it is non-empty: $\xi \in U_1 \cap U_2$ and $\zeta \in Z_1 \cap Z_2$ both lie over η, and the ring $k(\xi) \otimes_K k(\zeta)$ of two field extensions of K is non-zero, so there is a point of W over (ξ, ζ). A topological space covered by non-empty irreducible open subsets that pairwise intersect is irreducible. So W is irreducible and reduced, i.e. integral. □

Exercise 3.7

> **Problem.** Let $f : X \to Y$ be a morphism of schemes. Let X_1, X_2 be two closed subschemes of X, flat over Y. We suppose that there exists a dense open subscheme V of Y such that $X_1 \cap f^{-1}(V)$ and $X_2 \cap f^{-1}(V)$ are equal as closed subschemes of $f^{-1}(V)$. Let us suppose that Y is integral. We want to show that $X_1 = X_2$ as schemes.
>
> (a) Show that one can suppose that $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and V are affine.
>
> (b) Let I, J be the ideals of B defining X_1 and X_2. Tensoring the exact sequence $0 \to A \to \mathcal{O}_Y(V)$ by B/I, show that $I = \operatorname{Ker}(B \to \mathcal{O}_X(f^{-1}(V))/(I))$. Deduce from this that $I = J$.

Solution. (a) Equality of two closed subschemes of X can be checked on an open cover of X. Cover Y by non-empty affine open sets Y' and each $f^{-1}(Y')$ by affine open sets X'. Then $V \cap Y'$ is a non-empty open subset of Y' (as V is dense and Y' is open and non-empty), so it contains a non-empty principal open subset $D(a) = \operatorname{Spec} A_a$ of $Y' = \operatorname{Spec} A$; this $D(a)$ is dense in the integral scheme Y', and A is a domain with $a \neq 0$ (as $D(a) \neq \emptyset$). The hypotheses are inherited: $X_i \cap X'$ are closed subschemes of X', flat over Y' (flatness is local), and they coincide over $D(a)$. So we may assume that $Y = \operatorname{Spec} A$ with A a domain, $X = \operatorname{Spec} B$, and $V = D(a) = \operatorname{Spec} A_a$ with $a \neq 0$.

(b) The closed subschemes X_1, X_2 are $\operatorname{Spec} B/I$ and $\operatorname{Spec} B/J$, and B/I, B/J are flat over A. Since A is a domain and $a \neq 0$, the map $A \to A_a$ is injective. Tensoring with the flat A-module B/I gives an injection

$$B/I \;\hookrightarrow\; (B/I) \otimes_A A_a = (B/I)_a = B_a/IB_a.$$

Note $B_a = \mathcal{O}_X(f^{-1}(V))$, so the right-hand side is $\mathcal{O}_X(f^{-1}(V))/(I)$. The injectivity of the displayed map says: the kernel of the composition $B \to B/I \to B_a/IB_a$ is exactly I, that is,

$$I = \operatorname{Ker}\big(B \to B_a/IB_a\big).$$

(Concretely: $b \mapsto 0$ iff $a^n b \in I$ for some n, and flatness gives $b \in I$.) The same holds for J. By hypothesis the closed subschemes of $\operatorname{Spec} B_a$ defined by IB_a and JB_a coincide, i.e. $IB_a = JB_a$. Hence

$$I = \operatorname{Ker}(B \to B_a/IB_a) = \operatorname{Ker}(B \to B_a/JB_a) = J.$$

So $X_1 = X_2$ on each piece, hence globally. □

Exercise 3.8

Problem. Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let Z be a closed subscheme of X. We suppose that there exists a point $y \in Y$ such that $Z_y = X_y$ as schemes.

(a) Show that if Z is flat over Y at $z \in X_y$, then Z equals X in a neighborhood of z (use Exercise 1.2.14 and Nakayama's lemma).

(b) Show that if Z is flat over Y and if f is moreover of finite type (resp. proper) over Y, then there exists an open neighborhood $V \ni y$ such that $Z \cap f^{-1}(V) \to f^{-1}(V)$ is an open immersion (resp. an isomorphism) (use Exercise 3.2.5).

Solution. (a) Let $A = \mathcal{O}_{Y,y}$ with maximal ideal $\mathfrak{m}_y$ and residue field $k(y)$, $B = \mathcal{O}_{X,z}$, and let $I \subseteq B$ be the ideal of Z at z (so $\mathcal{O}_{Z,z} = B/I$). The ring B is Noetherian, so I is a finitely generated B-module. The local ring at z of the fibre $Z_y \subseteq X_y$ is $B/(I + \mathfrak{m}_y B)$, while that of X_y is $B/\mathfrak{m}_y B$. The hypothesis $Z_y = X_y$ means $I \subseteq \mathfrak{m}_y B$.

Since B/I is flat over A, $\operatorname{Tor}_1^A(B/I, k(y)) = 0$, so tensoring $0 \to I \to B \to B/I \to 0$ with $k(y) = A/\mathfrak{m}_y$ gives an exact sequence

$$0 \to I/\mathfrak{m}_y I \to B/\mathfrak{m}_y B \to (B/I)/\mathfrak{m}_y(B/I) \to 0.$$

The last map is the quotient $B/\mathfrak{m}_y B \to B/(I+\mathfrak{m}_y B)$, an isomorphism because $I \subseteq \mathfrak{m}_y B$. So the map $I/\mathfrak{m}_y I \to B/\mathfrak{m}_y B$ is both zero (its image is the kernel of the iso, which is 0) and injective; hence $I = \mathfrak{m}_y I$. As $\mathfrak{m}_y B \subseteq \mathfrak{m}_z$, we get $I = \mathfrak{m}_z I$, and Nakayama's lemma gives $I = 0$. Thus the coherent ideal sheaf $\mathcal{I}_Z$ of Z has zero stalk at z; being of finite type, it vanishes on an open neighbourhood U_z of z in X. So $Z \cap U_z = U_z$ as closed subschemes.

(b) *What is true.* Since $Z_y = X_y$, every point of X_y lies in Z, and Z is flat over Y at it. By (a) there is for each $z \in X_y$ an open set $U_z \ni z$ of X with $Z \cap U_z = U_z$. Let $U = \bigcup_{z \in X_y} U_z$. It is an open subset of X containing X_y and $Z \cap U = U$; in other words $Z \cap U \to U$ is an isomorphism (in particular an open immersion). This holds for any locally Noetherian f.

The proper case. Suppose f is proper. The set $C = X \setminus U$ is closed, so $f(C)$ is closed in Y, and $y \notin f(C)$ because $X_y \subseteq U$. Put $V = Y \setminus f(C)$, an open neighbourhood of y. Then $f^{-1}(V) \subseteq U$, so $Z \cap f^{-1}(V) = f^{-1}(V)$, i.e. $Z \cap f^{-1}(V) \to f^{-1}(V)$ is an isomorphism.

The finite type case: correction. For f merely of finite type, the neighbourhood V of y in Y cannot be found in general, so the statement must be read as: there is an open neighbourhood U of X_y in X such that $Z \cap U \to U$ is an open immersion (even an isomorphism); this was proved above. Here is a counterexample to the version with V. Let $\mathcal{V}$ be a discrete valuation ring with fraction field K, closed point y and residue field k, and $Y = \operatorname{Spec} \mathcal{V}$. Let $X = \operatorname{Spec} \mathcal{V}[x] \sqcup \operatorname{Spec} K[x]$ (finite type over Y), and let $Z = \operatorname{Spec} \mathcal{V}[x] \sqcup V(x)$, where $V(x) = \operatorname{Spec} K \subset \operatorname{Spec} K[x]$. Then Z is a closed subscheme of X, flat over Y (a disjoint union of the flat schemes $\operatorname{Spec} \mathcal{V}[x]$ and $\operatorname{Spec} K$), and $Z_y = \operatorname{Spec} k[x] = X_y$. The only open neighbourhood of y in Y is Y itself, and $Z \to X$ is not an open immersion, since $\{x = 0\}$ is a closed non-open point of the irreducible curve $\operatorname{Spec} K[x]$. □

Exercise 3.9

Problem. Let $f : X \to Y$ be a flat morphism of finite type with X, Y Noetherian. We will show that f is open.

(a) Use Exercise 3.2.17 to show that $f(X)$ contains a non-empty open subset V of Y.

(b) By considering the morphism $X \times_Y (Y \setminus V) \to Y \setminus V$, show that $f(X) \setminus V$ contains a non-empty open subset of $Y \setminus V$.

(c) Show that $Y \setminus f(X)$ is closed in Y by using the fact that the topological space Y is Noetherian; that is, every descending chain of closed subsets in Y is stationary.

Solution. We prove the statement by a direct argument which contains the content of the steps (a)–(c) (the finiteness of the situation being used through Chevalley's theorem, of which Ex. 3.2.17 is the special case needed for (a), and through the Noetherian property of Y, as suggested in (c)). It suffices to prove that $f(X)$ is open for every flat morphism $f : X \to Y$ of finite type of Noetherian schemes: if $W \subseteq X$ is open, then W is Noetherian and $W \to Y$ is flat of finite type, and then $f(W)$ is open.

Step 1: $f(X)$ is stable under generization. Let $x \in X$, $y = f(x)$, and let y' be a generization of y. The local homomorphism $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ is flat, hence faithfully flat, hence $\operatorname{Spec} \mathcal{O}_{X,x} \to \operatorname{Spec} \mathcal{O}_{Y,y}$ is surjective. The points of $\operatorname{Spec} \mathcal{O}_{Y,y}$ (resp. $\operatorname{Spec} \mathcal{O}_{X,x}$) are the generizations of y in Y (resp. of x in X), so $y' = f(x')$ for some generization x' of x. In particular $y' \in f(X)$.

Step 2: $f(X)$ is constructible by Chevalley's theorem ([EGA IV, 1.8.4]; for f of finite type between Noetherian schemes it is the content of Ex. 3.2.17 together with a Noetherian induction as in (b), (c)).

Step 3: a constructible subset E of a Noetherian scheme Y stable under generization is open. Let $C = Y \setminus E$; it is constructible and stable under specialization (if $c \in C$ and $c' \in \overline{\{c\}}$, then $c' \in C$, else $c' \in E$ would give $c \in E$). Write $C = \bigcup_{i=1}^{r} U_i \cap F_i$ with U_i open, F_i closed irreducible and $U_i \cap F_i \neq \emptyset$ (decompose each closed part into its finitely many irreducible components and discard empty pieces). The non-empty open subset $U_i \cap F_i$ of the irreducible space F_i contains the generic point ξ_i of F_i. As C is stable under specialization, $F_i = \overline{\{\xi_i\}} \subseteq C$. Therefore $C = \bigcup F_i$ is closed, i.e. E is open.

By Steps 1–3 with $E = f(X)$, $f(X)$ is open, hence f is open. (Step 3 is the Noetherian induction of (c), and the existence of a non-empty open subset in a dense constructible set is (a), (b).) □

Exercise 3.10

Problem. Let k be a field of characteristic $\operatorname{char}(k) \neq 2$. Let $Y = \operatorname{Spec} k[u, v]/(v^2 - u^2(u+1))$, and let $f : X \to Y$ be the normalization morphism. Show that f is unramified, surjective, but not etale.

Solution. Let $R = k[u, v]/(v^2 - u^2(u + 1))$.

R is a domain. If $F = v^2 - u^2(u+1)$ were reducible in $k[u][v]$ (monic in v), it would have a root $g \in k[u]$, so $g^2 = u^2(u + 1)$; then $u \mid g$ and $u + 1 = (g/u)^2$ would be a square in $k[u]$, impossible by degree. So F is irreducible, and R is a domain of dimension 1.

The normalization. Put $t = v/u \in \operatorname{Frac}(R)$. Then $t^2 = v^2/u^2 = u + 1$. Define $\varphi : R \to k[t]$ by $u \mapsto t^2 - 1$, $v \mapsto t(t^2 - 1)$; this is well defined since $v^2 = t^2(t^2 - 1)^2 = u^2(u + 1)$. It is injective:

its kernel is a prime ideal of the one-dimensional domain R; if non-zero it would be maximal, and the image would be a field, but the image contains $k[t^2-1]$, a polynomial ring, which is not a field. Moreover $\varphi(v)/\varphi(u) = t$, so $\operatorname{Frac}(R) = k(t)$ and $k[t]$ is integral over R ($t^2 = u+1$). Since $k[t]$ is integrally closed, it is the integral closure of R. Therefore $X = \operatorname{Spec} k[t] = \mathbb{A}^1_k$ and f is given by φ. It is a finite morphism.

Structure of f. Localizing at u: the map $R_u \to k[t]_{t^2-1}$ is injective, and surjective since $t = \varphi(v)/\varphi(u)$ and $1/(t^2-1) = 1/\varphi(u)$ lie in the image. So f restricts to an isomorphism $f^{-1}(D(u)) = D(t^2-1) \to D(u)$. The complement of $D(u)$ in Y is $V(u) = V(u, v^2) = \{o\}$, the point $o = (u, v)$ with $k(o) = k$ (note $R/uR = k[v]/(v^2)$). Its fibre is

$$X_o = \operatorname{Spec} k[t]/(t^2-1,\, t^3-t) = \operatorname{Spec} k[t]/(t^2-1).$$

As $\operatorname{char} k \neq 2$, $t^2-1 = (t-1)(t+1)$ with distinct roots in k, so $X_o = \operatorname{Spec} k \sqcup \operatorname{Spec} k$.

Surjective. f maps $f^{-1}(D(u))$ isomorphically onto $D(u)$, and the fibre over o is non-empty. So f is surjective.

Unramified. By Lemma 3.20 it suffices to check that every fibre is finite, reduced, with separable residue fields. Over $y \in D(u)$ the fibre is $\operatorname{Spec} k(y)$ (as f is an isomorphism there). Over o the fibre is $\operatorname{Spec} k \sqcup \operatorname{Spec} k$, reduced with residue fields equal to $k(o) = k$. So f is unramified.

Not etale. Let x be the point $t = 1$ of X; $f(x) = o$. The ring $\mathcal{O}_{Y,o} = R_{(u,v)}$ has dimension 1. Its maximal ideal $\mathfrak{m}_o = (u, v)$ satisfies $\mathfrak{m}_o/\mathfrak{m}_o^2 = \mathfrak{m}/(\mathfrak{m}^2 + (F))$ computed in $k[u, v]_{(u,v)}$, and $F = v^2 - u^3 - u^2 \in (u, v)^2$, so $\dim_k \mathfrak{m}_o/\mathfrak{m}_o^2 = 2 > 1$. Hence Y is not regular at o, whereas $X = \mathbb{A}^1_k$ is regular at x. If f were etale at x, then by Cor. 3.24 X regular at x would be equivalent to Y regular at $f(x) = o$, a contradiction. So f is not etale at x (equivalently, as f is unramified, f is not flat at x). □

Exercise 3.11

> **Problem.** Let $f_1 : X_1 \to Y$, $f_2 : X_2 \to Y$ be morphisms of locally Noetherian schemes of finite type. Let us suppose that for every $y \in Y$, there exists $i = 1$ or 2 such that f_i is unramified (resp. etale; resp. smooth) at all points of $f_i^{-1}(y)$. Show that $X_1 \times_Y X_2 \to Y$ is unramified (resp. etale; resp. smooth).

Solution. *The statement is false as printed.* Take $X_1 = Y$, $f_1 = \mathrm{id}$ (etale, smooth and unramified at all points), and $f_2 : X_2 \to Y$ any morphism of finite type, e.g. $\mathbb{A}^1_Y \to Y$ with Y the spectrum of a field. Then $X_1 \times_Y X_2 = X_2$, and the claim would say that $\mathbb{A}^1_k \to \operatorname{Spec} k$ is unramified, which is false (the fibre is not finite) and, for "etale", too. The correct statements are the two following ones, which I prove: with $W = X_1 \times_Y X_2$, $h : W \to Y$ the structure morphism and $p_i : W \to X_i$ the projections, and where "P" stands for unramified, etale, or smooth and a morphism has P over y if it has P at all points of the fibre over y:

(A) if for a given $y \in Y$ *both* f_1 and f_2 have P over y, then h has P over y (so if the hypothesis holds with "both", h has P; this is the pointwise version of Prop. 3.22(c) and Prop. 3.38);

(B) if for a given $y \in Y$, f_i has P over y for one index i, then the projection $p_j : W \to X_j$ ($j \neq i$) has P at all points of $h^{-1}(y)$.

For all of them, W is locally Noetherian and h, p_j are of finite type.

Fibres. For $y \in Y$ the fibre of h is $W_y = X_{1,y} \times_{k(y)} X_{2,y}$. Moreover, if $x_2 \in X_2$ lies over y, the fibre of p_2 over x_2 is $X_{1,y} \times_{k(y)} \operatorname{Spec} k(x_2) = (X_{1,y})_{k(x_2)}$, and similarly for p_1.

Unramified. By Lemma 3.20 and its proof, f_i is unramified over y if and only if $X_{i,y}$ is finite, reduced, with separable residue field extensions $k(x)/k(y)$; this means $X_{i,y} = \bigsqcup_m \operatorname{Spec} L_m$ with $L_m/k(y)$ finite separable. We use the following fact. If L/κ is finite separable and M is a field extension of κ, then $L \otimes_\kappa M$ is a finite product of finite separable field extensions of M: write $L = \kappa[T]/(P)$ with P separable; over M the polynomial P is a product of distinct irreducible factors with distinct roots, so $L \otimes_\kappa M = M[T]/(P)$ is the product of the corresponding finite separable extensions of M.

(A), unramified: $X_{1,y} = \bigsqcup \operatorname{Spec} L_m$ and $X_{2,y} = \bigsqcup \operatorname{Spec} L'_n$ with L_m, L'_n finite separable over $k(y)$. So $W_y = \bigsqcup \operatorname{Spec}(L_m \otimes_{k(y)} L'_n)$, and each $L_m \otimes L'_n$ is a finite product of fields, finite and separable over L'_n and hence over $k(y)$ (a tower of finite separable extensions is separable). Thus W_y is finite, reduced and has separable residue fields, i.e. h is unramified over y.

(B), unramified: the fibre of p_2 over x_2 is $(X_{1,y})_{k(x_2)} = \bigsqcup_m \operatorname{Spec}(L_m \otimes_{k(y)} k(x_2))$, finite, reduced with separable residue fields over $k(x_2)$ by the same computation. Hence p_2 is unramified at all points over x_2, by Lemma 3.20 applied to p_2 (and its fibre over x_2; the local-ring condition $\mathfrak{m}_{x_2}\mathcal{O}_{W,w} = \mathfrak{m}_w$ is a property of the fibre).

Flatness. Let $w \in W$ with images x_1, x_2, y, and put $A = \mathcal{O}_{Y,y}$, $B_i = \mathcal{O}_{X_i,x_i}$. Then $\mathcal{O}_{W,w}$ is a localization of $B_1 \otimes_A B_2$. If $A \to B_1$ and $A \to B_2$ are flat, then $B_1 \otimes_A B_2$ is flat over B_2 (base change of $A \to B_1$) and B_2 is flat over A, so $B_1 \otimes_A B_2$ is flat over A; localizing, $\mathcal{O}_{W,w}$ is flat over A. Similarly, if $A \to B_1$ is flat then $B_2 \to B_1 \otimes_A B_2$ is flat, so p_2 is flat at w. Hence: if f_1, f_2 are flat over y, then h is flat over y (needed in (A)); if f_1 is flat over y, p_2 is flat at all points over y (needed in (B)).

Etale. Etale means unramified and flat, so (A) and (B) for etale follow from the unramified and flat cases.

Smooth. Smooth at x means flat at x and $X_y \to \operatorname{Spec} k(y)$ smooth at x. For (A): h is flat over y by the above, and $W_y = X_{1,y} \times_{k(y)} X_{2,y}$ is smooth over $k(y)$ at all its points by Prop. 3.38 (stability under fibered products), since $X_{i,y}$ is smooth over $k(y)$ (a smooth-at-x statement at all points of the fibre is the same as smoothness of $X_{i,y} \to \operatorname{Spec} k(y)$, and over a field flatness is automatic). For (B): p_2 is flat at points over y by the above, and the fibre of p_2 over x_2 is $(X_{1,y})_{k(x_2)}$, smooth over $k(x_2)$ by stability of smoothness under base change (Prop. 3.38). □

Exercise 3.12

> **Problem.** Let $f : X \to Y$ be a flat morphism of algebraic varieties. We suppose that X_y is irreducible for every closed point $y \in Y$ and that Y is irreducible. Show that X is irreducible.

Solution. Algebraic varieties are of finite type over a field k, so X and Y are Noetherian and the closed points of Y are dense in every non-empty open subset of Y (Nullstellensatz). Irreducible includes non-empty: $Y \neq \emptyset$ has a closed point y, and $X_y \neq \emptyset$, so $X \neq \emptyset$.

Let $X_1, \ldots, X_r$ be the irreducible components of X (X is Noetherian). Suppose $r \geq 2$. For $i = 1, 2$ let

$$U_i = X \setminus \bigcup_{j \neq i} X_j.$$

This is a non-empty open subset of X (it is non-empty since no X_i is contained in the union of the others), and $U_1 \cap U_2 = \emptyset$ because a point of U_1 lies in no X_j with $j \neq 1$, in particular not in X_2, whereas $U_2 \subseteq X_2$. The morphisms $f|_{U_i} : U_i \to Y$ are flat and of finite type, so $f(U_i)$ is open in Y by Ex. 3.9. Moreover, $f(U_i) \neq \emptyset$. Since Y is irreducible, $f(U_1) \cap f(U_2)$ is a non-empty open subset of Y (two non-empty opens of an irreducible space meet), and so it contains a closed point y of Y. Then X_y meets both U_1 and U_2, i.e. $U_1 \cap X_y$ and $U_2 \cap X_y$ are non-empty open subsets of X_y. As X_y is irreducible, they meet, which contradicts $U_1 \cap U_2 = \emptyset$. Hence $r = 1$ and X is irreducible. □

Exercise 3.13

Problem. Let X be a smooth morphism over a scheme S of characteristic $p > 0$.

(a) Show that $X^{(p)} \to S$ is smooth (use Proposition 3.38).

(b) Let us suppose that S is the spectrum of a field k. Let $F_{X/k} : X \to X^{(p)}$ be the relative Frobenius. Show that $F_{X/k}$ is flat at the rational points of X (use Proposition 2.27).

(c) Show that $F_{X/k}$ is faithfully flat, that is flat and surjective.

(d) Let us suppose $p = 2$. Let $Y = \operatorname{Spec} k[x, y]/(y^2 + x^3)$. Show that $F_{Y/k}$ is not flat.

Solution. Recall the definitions: $X^{(p)} = X \times_{S,F_S} S$, where $F_S : S \to S$ is the absolute Frobenius (S has characteristic p, and F_S is the identity on the underlying space and $a \mapsto a^p$ on $\mathcal{O}_S$), viewed as an S-scheme by the second projection; and $F_{X/S} : X \to X^{(p)}$ is the morphism induced by the absolute Frobenius F_X and the structure map $X \to S$. We assume S locally Noetherian, so that smoothness is defined.

(a) The morphism $X^{(p)} \to S$ is the base change of the smooth morphism $X \to S$ by $F_S : S \to S$. Smooth morphisms are stable under base change (Prop. 3.38), so $X^{(p)} \to S$ is smooth.

For (b)–(d) let $S = \operatorname{Spec} k$, $\operatorname{char} k = p$, and k' denote the ring k viewed as a k-algebra via $F : \lambda \mapsto \lambda^p$. Then for an affine open $\operatorname{Spec} P$ of X, the corresponding open subscheme of $X^{(p)}$ is $\operatorname{Spec}(P \otimes_k k')$, a k-algebra via the second factor, and $F_{X/k}$ corresponds to the k-algebra map $\Phi : P \otimes_k k' \to P$, $a \otimes \mu \mapsto \mu a^p$.

(b) Let $x \in X(k)$ be a rational point, $A = \mathcal{O}_{X,x}$ with maximal ideal $\mathfrak{m}$, $A/\mathfrak{m} = k$, $d = \dim A$. Since X is smooth over k, X is regular at x (Prop. 3.30). Let $f_1, \ldots, f_d$ be a coordinate system of A. (Since x is a rational point, x is closed, and we may replace X by an affine open neighbourhood of x, which is an algebraic variety, so that Prop. 3.30 and Prop. 2.27 apply.) The k-algebra map $k[[T_1, \ldots, T_d]] \to \widehat{A}$, $T_i \mapsto f_i$, is an isomorphism (this is Prop. 2.27, with the given coordinate system): it exists as $\widehat{A}$ is complete and contains k, it is surjective because $\widehat{\mathfrak{m}} = (f_1, \ldots, f_d)$ and $\widehat{A}/\widehat{\mathfrak{m}} = k$, and its kernel is a prime ideal of the d-dimensional domain $k[[T]]$ with quotient of dimension $\dim \widehat{A} = \dim A = d$ (Lemma 2.26), hence it is 0.

Put $B = A \otimes_k k'$ and $\Phi : B \to A$. We have $B/(\mathfrak{m} \otimes 1)B = k \otimes_k k' = k'$, a field, so $\mathfrak{n} := (\mathfrak{m} \otimes 1)B = (s_1, \ldots, s_d)B$ with $s_i = f_i \otimes 1$ is a maximal ideal. Moreover $\Phi(\mathfrak{n}) \subseteq \mathfrak{m}$, so $\Phi^{-1}(\mathfrak{m}) = \mathfrak{n}$, and $B_\mathfrak{n} = \mathcal{O}_{X^{(p)},x'}$ where $x' = F_{X/k}(x)$, which is a rational point of $X^{(p)}$ (its residue field is $B/\mathfrak{n} = k$ as a k-algebra via the second factor). The local homomorphism $\Phi_x : B_\mathfrak{n} \to A$ is $F^{\#}_{X/k,x}$.

$B_\mathfrak{n}$ is regular of dimension d with coordinate system $s_1, \ldots, s_d$. Since $\mathfrak{n}$ is generated by d elements, $d' := \dim B_\mathfrak{n} \leq d$. On the other hand $A/\mathfrak{n}A = A/(f_1^p, \ldots, f_d^p)$ is of dimension 0, and Thm. 3.12

(inequality) for $\operatorname{Spec} A \to \operatorname{Spec} B_{\mathfrak{n}}$ gives $d \le d' + 0$. So $d' = d$, and $B_{\mathfrak{n}}$ is a Noetherian local ring of dimension d with maximal ideal generated by $s_1, \dots, s_d$; it is regular (this also follows from (a) and Prop. 3.30).

Applying the argument of Prop. 2.27 to $X^{(p)}$ at the rational point x' (regular by (a) and Prop. 3.30) with the coordinate system s_i, we obtain an isomorphism $k[[S_1, \dots, S_d]] \to \widehat{B_{\mathfrak{n}}}$, $S_i \mapsto s_i$. The induced map of completions $\widehat{\Phi} : \widehat{B_{\mathfrak{n}}} \to \widehat{A}$ is continuous and k-linear, and $\widehat{\Phi}(s_i) = f_i^p$. Under the identifications with power series rings it is

$$k[[S_1, \dots, S_d]] \to k[[T_1, \dots, T_d]], \qquad S_i \mapsto T_i^p.$$

Now $k[[T_1, \dots, T_d]]$ is a free module over $k[[T_1^p, \dots, T_d^p]] \simeq k[[S_1, \dots, S_d]]$ with basis the monomials T^α, $0 \le \alpha_i < p$ (write every exponent as $pq + r$ with $0 \le r < p$). So $\widehat{\Phi}$ is flat.

Descent of flatness to the local rings. $\widehat{B_{\mathfrak{n}}}$ is flat over $B_{\mathfrak{n}}$ and $\widehat{A}$ is flat over $\widehat{B_{\mathfrak{n}}}$, so $\widehat{A}$ is flat over $B_{\mathfrak{n}}$. Also $A \to \widehat{A}$ is faithfully flat. For any $B_{\mathfrak{n}}$-module N and $i \ge 1$: $\operatorname{Tor}_i^{B_{\mathfrak{n}}}(A, N) \otimes_A \widehat{A} = \operatorname{Tor}_i^{B_{\mathfrak{n}}}(\widehat{A}, N) = 0$, hence $\operatorname{Tor}_i^{B_{\mathfrak{n}}}(A, N) = 0$. So A is flat over $B_{\mathfrak{n}}$, i.e. $F_{X/k}$ is flat at x.

(c) *Flatness.* Flatness of $F_{X/k}$ at a point is local on X, and $F_{X/k}^{-1}(U^{(p)}) = U$ for $U \subseteq X$ open (since F_X is the identity on points and $W \circ F_{X/k} = F_X$, with $W : X^{(p)} \to X$ the first projection, which is a homeomorphism by Lemma 2.24); so we may replace X by a non-empty affine open subscheme U, hence assume X of finite type over k (so X, $X^{(p)}$ are quasi-compact). First let $K = \bar{k}$ and x be a closed point of X. The relative Frobenius commutes with base change (Ex. 3.2.20(a)), $F_{X_K/K} = F_{X/k} \times_k K$ (with $(X_K)^{(p)} = (X^{(p)})_K$). Let $\pi : X_K \to X$ and $\pi' : (X^{(p)})_K \to X^{(p)}$ be the projections, flat, with $\pi' \circ F_{X_K/K} = F_{X/k} \circ \pi$. Choose a point $x' \in X_K$ over x with $k(x') = K$: the fibre $\operatorname{Spec}(k(x) \otimes_k K)$ is non-empty and finite over K, and its points have residue field K (as $k(x)$ is finite over k and K is algebraically closed). Then X_K is smooth over K (base change), and x' is a rational point, so by (b) $F_{X_K/K}$ is flat at x'. Let $y = F_{X/k}(x)$, $y' = F_{X_K/K}(x')$, $A = \mathcal{O}_{X^{(p)},y}$, $A' = \mathcal{O}_{(X^{(p)})_K,y'}$, $B = \mathcal{O}_{X,x}$, $B' = \mathcal{O}_{X_K,x'}$. The maps $A \to A'$ and $B \to B'$ are flat local homomorphisms (so $B \to B'$ is faithfully flat), and $A' \to B'$ is flat. Hence B' is flat over A (composition $A \to A' \to B'$). Then for any A-module N, $\operatorname{Tor}_i^A(B, N) \otimes_B B' = \operatorname{Tor}_i^A(B', N) = 0$, and faithful flatness of $B \to B'$ gives $\operatorname{Tor}_i^A(B, N) = 0$. So $F_{X/k}$ is flat at every closed point x of X.

Next, $F_{X/k}$ is a morphism of finite type between k-varieties, and maps closed points to closed points: if x is closed, $k(x)$ is finite over k, so $k(F(x)) \subseteq k(x)$ is finite over k, so $F(x)$ is closed. Thus Ex. 3.2 (in the corrected form proved above: $X^{(p)}$ is quasi-compact and closed points of X map to closed points) shows that $F_{X/k}$ is flat everywhere.

Surjectivity. (This also follows from Lemma 2.24, which says that $F_{X/k}$ is a homeomorphism; here is a direct argument.) Let $W : X^{(p)} \to X$ be the first projection. Then $W \circ F_{X/k} = F_X$, the absolute Frobenius of X, which is the identity on the underlying topological space; so $F_{X/k}$ is injective and W is surjective on points. We claim W is injective on points. The fibre of W over $x \in X$ is $\operatorname{Spec}(k(x) \otimes_k k')$. In the ring $C = k(x) \otimes_k k'$, for any element $z = \sum a_i \otimes \mu_i$ we have $z^p = \sum a_i^p \otimes \mu_i^p = \sum a_i^p \mu_i \otimes 1 \in k(x) \otimes 1$ (because $\mu^p = F(\mu)$ is the image in k' of $\mu \in k$). So for a prime $\mathfrak{q}$ of C and $z \in \mathfrak{q}$ we get $z^p \in \mathfrak{q} \cap (k(x) \otimes 1) = 0$, as $k(x) \otimes 1$ is a field. Hence every prime of C equals the nilradical of C, and C has a single prime. So W is injective. Now for $y \in X^{(p)}$ choose x with $F_X(x) = W(y)$ (as F_X is bijective); then $W(F_{X/k}(x)) = F_X(x) = W(y)$ and injectivity of W gives $y = F_{X/k}(x)$. So $F_{X/k}$ is surjective, hence faithfully flat.

(d) Here $p = 2$, $A = k[x,y]/(y^2 + x^3)$ and $Y = \operatorname{Spec} A$. Applying F to the coefficients of the defining equation (which lie in $\mathbb{F}_2$), $A \otimes_k k' = k[x',y']/(y'^2 + x'^3) =: B$ with $x' = x \otimes 1$, $y' = y \otimes 1$, and $\Phi : B \to A$, $x' \mapsto x^2$, $y' \mapsto y^2$ (consistent: $y^4 + x^6 = (y^2 + x^3)^2 = 0$). The ring A is isomorphic to $k[\tau^2, \tau^3] \subset k[\tau]$ via $x \mapsto \tau^2$, $y \mapsto \tau^3$ (characteristic 2 so signs are irrelevant; A is a domain, the cuspidal cubic). Then the image of Φ is $k[\tau^4, \tau^6] \simeq B$ (so Φ is injective), A is a finite B-module (generated by $1, x, y, xy$; indeed x^2, y^2 lie in the image of Φ), and the fraction fields are $\operatorname{Frac} A = k(\tau)$, $\operatorname{Frac} B = k(\tau^4, \tau^6) = k(\tau^2)$, so $[\operatorname{Frac} A : \operatorname{Frac} B] = 2$.

Suppose $F_{Y/k}$ were flat at the origin o, $\mathfrak{n} = (x', y') \subset B$. The only prime of A above $\mathfrak{n}$ is $\mathfrak{m} = (x, y)$, because $A/\mathfrak{n}A = k[x,y]/(y^2+x^3, x^2, y^2) = k[x,y]/(x^2, y^2)$ is local (this uses $x^3 \in (x^2)$). So $A_\mathfrak{n} := A \otimes_B B_\mathfrak{n} = A_\mathfrak{m}$ is a finite flat module over the Noetherian local ring $B_\mathfrak{n}$, hence free, say of rank r. Computing the rank at the closed point: $r = \dim_k A_\mathfrak{n}/\mathfrak{n}A_\mathfrak{n} = \dim_k k[x,y]/(x^2, y^2) = 4$. Computing the rank at the generic point: $A_\mathfrak{n} \otimes_{B_\mathfrak{n}} \operatorname{Frac} B$ is a domain finite over the field $\operatorname{Frac} B$, hence a field containing A, and equals $\operatorname{Frac} A$; so $r = 2$. Contradiction. Therefore $F_{Y/k}$ is not flat at o. (Of course Y is not smooth at o, in accordance with (c).) □

Citations without a chapter prefix ("Thm. 3.12", "Cor. 2.12", ...) refer to Chapter 4 of Liu; results of other chapters are cited as "Thm. 1.3.16" (Chapter 1, Section 3) and so on. "Homomorphism of local rings" always means a local homomorphism.

Exercise 3.14

> **Problem.** Let us keep the notation and hypotheses of Lemma 3.16 (so $A \to B$ is a flat homomorphism of Noetherian local rings and $b \in B$ is an element whose image in $B/\mathfrak{m}_A B$ is not a zero divisor). Let $J = \operatorname{Ann}(b)$. (a) Show that bB is flat over A. (b) Using the exact sequence $0 \to J \to B \to B$, where the last map is multiplication by b, show that $J \otimes_A A/\mathfrak{m}_A = 0$. Deduce from this that $J = 0$ and that b does not divide zero in B.

Solution. Put $k = A/\mathfrak{m}_A$. The homomorphism $A \to B$ is local, so $\mathfrak{m}_A B \subseteq \mathfrak{m}_B$.

(a) Consider the exact sequence of A-modules $0 \to bB \to B \to B/bB \to 0$. By Lemma 3.16, B/bB is flat over A, and B is flat by hypothesis. For every A-module N, the long exact sequence of $\operatorname{Tor}^A(-, N)$ contains the segment

$$\operatorname{Tor}_2^A(B/bB, N) \to \operatorname{Tor}_1^A(bB, N) \to \operatorname{Tor}_1^A(B, N).$$

The outer terms vanish (a flat module has vanishing higher Tor), hence $\operatorname{Tor}_1^A(bB, N) = 0$ for all N. A module P is flat if and only if $\operatorname{Tor}_1^A(N, P) = 0$ for all A-modules N; hence bB is flat over A.

(b) Multiplication by b factors as $B \xrightarrow{u} bB \xrightarrow{\iota} B$, where u is onto, ι is the inclusion, and $\operatorname{Ker} u = J$. So we have an exact sequence $0 \to J \to B \xrightarrow{u} bB \to 0$. Tensoring with k and using $\operatorname{Tor}_1^A(bB, k) = 0$ (part (a)) gives an exact sequence

$$0 \to J \otimes_A k \to B \otimes_A k \xrightarrow{u \otimes 1} bB \otimes_A k.$$

The composite $(\iota \otimes 1) \circ (u \otimes 1)$ is multiplication by b on $B \otimes_A k = B/\mathfrak{m}_A B$, which is injective by hypothesis. Hence $u \otimes 1$ is injective, so its kernel, which is the image of the injective map $J \otimes_A k \to B \otimes_A k$, is 0. Therefore $J \otimes_A k = J/\mathfrak{m}_A J = 0$.

Thus $J = \mathfrak{m}_A J \subseteq \mathfrak{m}_B J$. As B is Noetherian, J is a finitely generated B-module, and Nakayama's lemma gives $J = 0$. That is, $\mathrm{Ann}(b) = 0$, so b is not a zero divisor of B. □

Exercise 3.15

> **Problem.** Let X, Y be integral schemes, separated and of finite type over a locally Noetherian scheme S. Let $f : X \dashrightarrow Y$ be a birational map. Let $T \to S$ be a faithfully flat morphism such that X_T and Y_T are integral. Suppose that $f_T : X_T \dashrightarrow Y_T$ is defined everywhere and that Y is affine. We want to show that f is also defined everywhere. (a) Show that we can suppose S, T are affine (and even local). (b) Let us identify the function fields $K(X)$ and $K(Y)$. Show that f is defined everywhere if and only if $\mathcal{O}_Y(Y) \subseteq \mathcal{O}_X(X)$. (c) Let $M = (\mathcal{O}_X(X) + \mathcal{O}_Y(Y))/\mathcal{O}_X(X)$. Show that $M \otimes_{\mathcal{O}_S(S)} \mathcal{O}_T(T) = 0$. Deduce from this that $M = 0$ and that f is defined everywhere.

Solution. Conventions. The birational map f is an isomorphism of S-schemes $U \to V$ between dense open subschemes of X and Y; through it we identify $K := K(X) = K(Y)$. For a nonempty open $W \subseteq X$ (resp. $W \subseteq Y$) the ring $\mathcal{O}_X(W)$ (resp. $\mathcal{O}_Y(W)$) is a subring of K, since X, Y are integral. The map f is *defined at* $x \in X$ if some morphism $W \to Y$, with $W \ni x$ open, agrees with f on a dense open subset of $W \cap U$; it is *defined everywhere* if it extends to a morphism $X \to Y$. Since Y is separated over S and X is reduced, two such local extensions agree on overlaps (they agree on a dense open and Y is separated), so they glue: f is defined everywhere if and only if it is defined at every point. The base change f_T is the isomorphism $U_T \to V_T$; it is a map of T-schemes, and "defined everywhere" for f_T means that it extends to a T-morphism $X_T \to Y_T$. Since X_T, Y_T are integral and nonempty, U_T and V_T are dense open in them.

We prove (b), then (c) in the affine case, then the reduction (a). This order is the logical one.

(b) We prove it for any integral schemes X', Y' with Y' affine and any birational map $f' : X' \dashrightarrow Y'$ (an isomorphism $U' \to V'$ of dense opens), K being the common function field.

($\Rightarrow$) Let $g : X' \to Y'$ be an extension of f' and $C = \mathcal{O}_{Y'}(Y')$. For $c \in C$ the element $g^{\#}(c) \in \mathcal{O}_{X'}(X') \subseteq K$ restricts on a dense open subset $U'' \subseteq U'$ to $(g|_{U''})^{\#}(c) = (f'|_{U''})^{\#}(c|_{V'})$, and the latter is the element $c \in K$ by the very definition of the identification $K(Y') = K(X')$. As $\mathcal{O}_{X'}(U'') \to K$ is injective, $c = g^{\#}(c) \in \mathcal{O}_{X'}(X')$. So $C \subseteq \mathcal{O}_{X'}(X')$.

($\Leftarrow$) Suppose $C \subseteq \mathcal{O}_{X'}(X')$. The inclusion is a ring homomorphism $C \to \mathcal{O}_{X'}(X')$, hence defines a morphism $g : X' \to \operatorname{Spec} C = Y'$. On U', both $g|_{U'}$ and f' are morphisms into the affine scheme Y', so they correspond to ring homomorphisms $C \to \mathcal{O}_{X'}(U') \subseteq K$. Both are $c \mapsto c$ (the inclusion $C \subseteq K$). Hence $g|_{U'} = f'$ and f' is defined everywhere.

(c), the affine case. Let $V_0 = \operatorname{Spec} R$ be an affine open subset of S, and put $X_0 = X_{V_0}$, $Y_0 = Y_{V_0}$. We assume that Y_0 is nonempty and affine. Put $A_X = \mathcal{O}(X_0)$, $A_Y = \mathcal{O}(Y_0)$ (subrings of K), $N = A_X + A_Y \subseteq K$ and $M = N/A_X$; these are R-modules. Let $T' = \operatorname{Spec} R'$ be an affine open subset of T lying over V_0 such that $X_{T'} \neq \emptyset$, and assume that $f_{T'}$ is defined everywhere. We claim that

$$M \otimes_R R' = 0.$$

$X_{T'}$ and $Y_{T'}$ are nonempty open subschemes of the integral schemes X_T, Y_T, hence integral; $K' := K(X_{T'}) = K(Y_{T'})$ via $f_{T'}$. The schemes X_0 and Y_0 are of finite type over the Noetherian

affine scheme V_0, hence quasi-compact and quasi-separated, and so are all their open subsets. Therefore flat base change for global sections gives, for every open $W \subseteq X_0$,

$$\mathcal{O}(W_{T'}) = \mathcal{O}(W) \otimes_R R', \qquad \mathcal{O}(Y_{T'}) = A_Y \otimes_R R'.$$

(For X_0 write $\mathcal{O}(W)$ as the kernel of a map between finite products of rings of affine opens, and use that R' is flat over R.)

Step 1: an injection $K \otimes_R R' \hookrightarrow K'$. We have $K = \varinjlim_W \mathcal{O}(W)$, over the nonempty opens $W \subseteq X_0$, so $K \otimes_R R' = \varinjlim_W \mathcal{O}(W_{T'})$. Each $W_{T'}$ is a nonempty open subscheme of the integral scheme $X_{T'}$. (It is nonempty because $X_{T'} \to X_0$ is flat; its image is stable under generization and so contains the generic point of X_0, which lies in W.) So the maps $\mathcal{O}(W_{T'}) \to K'$ are injective and compatible, and they induce an injection

$$j : K \otimes_R R' \hookrightarrow K'.$$

By construction $j(A_X \otimes R') = \mathcal{O}(X_{T'})$. Moreover, $j(A_Y \otimes R') = \mathcal{O}(Y_{T'})$: an element $c \in A_Y$ corresponds, under $f : U \to V$, to an element of $\mathcal{O}(U \cap X_0) \subseteq K$, and the element $c \otimes r$ of $\mathcal{O}(Y_{T'})$ restricts on $V_{T'}$ to the element of $\mathcal{O}(V_{T'})$ that corresponds to $c \otimes r \in \mathcal{O}((U \cap X_0)_{T'})$ under the isomorphism $f_{T'}$. These are the identifications used to define K'.

Step 2. Since R' is flat, $N \otimes_R R' \to K \otimes_R R'$ is injective, hence so is $\psi = j \circ (\text{this}) : N \otimes_R R' \to K'$. Because $N = A_X + A_Y$, the image of ψ is $\mathcal{O}(X_{T'}) + \mathcal{O}(Y_{T'})$. By hypothesis $f_{T'}$ is defined everywhere and $Y_{T'}$ is affine, so by (b) $\mathcal{O}(Y_{T'}) \subseteq \mathcal{O}(X_{T'})$. Hence $\psi(N \otimes R') = \mathcal{O}(X_{T'}) = \psi(A_X \otimes R')$. As ψ is injective, the injection $A_X \otimes R' \to N \otimes R'$ is onto. By flatness, $M \otimes_R R' = (N \otimes R')/(A_X \otimes R') = 0$. This proves the claim.

(a) The reduction, and the end of the proof. Here we use that the preimage in the affine scheme Y of an affine open subset of S is again affine; this is automatic if S is separated (e.g. affine, or quasi-projective over a field or $\mathbf{Z}$), which we assume for this step. Let $x \in X$ with image $s \in S$ and take an affine open neighborhood $V_0 = \operatorname{Spec} R$ of s, with X_0, Y_0, A_X, A_Y, M as above. Let $\sigma \in S$ be the image of the generic point of X; it is also the image of the generic point of Y (as f is an S-isomorphism $U \cong V$), and σ is a generization of s. Hence $\sigma \in V_0$, so X_0, Y_0 are nonempty, and f restricts to a birational map $f_0 : X_0 \dashrightarrow Y_0$. It suffices to show $A_Y \subseteq A_X$, i.e. $M = 0$: indeed, then (b) applied to f_0 shows that f_0 is defined everywhere, in particular at x, so that f is defined at x; and x is arbitrary.

Let $\mathfrak{p} = \operatorname{Ker}(R \to K)$, a prime of R (the point σ). Both A_X and A_Y are $R/\mathfrak{p}$-modules, so $\mathfrak{p}M = 0$ and $M_\mathfrak{q} = 0$ for every prime $\mathfrak{q} \not\supseteq \mathfrak{p}$. Let $\mathfrak{q} \supseteq \mathfrak{p}$ be a prime of R. By surjectivity of $T \to S$ there is $t \in T$ over $\mathfrak{q}$; let $T' = \operatorname{Spec} R'$ be an affine open neighborhood of t lying over V_0. Since $T' \to V_0$ is flat, going down gives $t' \in T'$ over $\mathfrak{p}$; the generic point ξ of X_0 lies over $\mathfrak{p}$, and $k(\xi) \otimes_{k(\mathfrak{p})} k(t') \neq 0$, so $X_{T'} \neq \emptyset$. Moreover $f_{T'}$ is defined everywhere: the T-morphism $X_T \to Y_T$ extending f_T maps $X_{T'}$ into $Y_{T'}$. By the affine case, $M \otimes_R R' = 0$, hence $M \otimes_R O_{T,t} = 0$. Now $R_\mathfrak{q} \to O_{T,t}$ is a flat local homomorphism, hence faithfully flat, and $M \otimes_R O_{T,t} = M_\mathfrak{q} \otimes_{R_\mathfrak{q}} O_{T,t}$. Therefore $M_\mathfrak{q} = 0$. So $M_\mathfrak{q} = 0$ for all primes $\mathfrak{q}$ of R, i.e. $M = 0$ (Lemma 1.2.12). □

Exercise 3.16

Problem. Let B be a semi-local ring, that is, having only a finite number of maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$. Let $\widehat{B}$ be the formal completion of B for the topology defined by $\bigcap_i \mathfrak{m}_i$. Show

that $\widehat{B} \simeq \bigoplus_i \widehat{B}_i$, where $\widehat{B}_i$ is the formal completion of $B_{\mathfrak{m}_i}$ for the topology defined by its maximal ideal.

Solution. No Noetherian hypothesis is needed. Let the $\mathfrak{m}_i$ be pairwise distinct, $J = \bigcap_i \mathfrak{m}_i$ and $P = \mathfrak{m}_1 \cdots \mathfrak{m}_n$. By definition $\widehat{B} = \varprojlim_k B/J^k$.

Step 1: the J-adic and P-adic topologies agree. Since $P \subseteq \bigcap_i \mathfrak{m}_i = J$, we have $P^k \subseteq J^k$. Conversely $J^n \subseteq P$: a product $a_1 \cdots a_n$ of n elements of J lies in $\mathfrak{m}_1 \cdots \mathfrak{m}_n$, because $a_i \in \mathfrak{m}_i$. Therefore $J^{nk} \subseteq P^k \subseteq J^k$ for all $k \geq 1$. These inclusions give mutually inverse isomorphisms between $\varprojlim_k B/J^k$ and $\varprojlim_k B/P^k$ (a map of inverse systems in each direction after reindexing, composing to the canonical transition maps). So $\widehat{B} = \varprojlim_k B/P^k$.

Step 2: Chinese remainder theorem. For $i \neq j$ we have $\mathfrak{m}_i + \mathfrak{m}_j = B$, hence $\mathfrak{m}_i^k + \mathfrak{m}_j^k = B$ for all k. For pairwise comaximal ideals the intersection equals the product, so $\bigcap_i \mathfrak{m}_i^k = \prod_i \mathfrak{m}_i^k = P^k$, and the Chinese remainder theorem gives a canonical isomorphism, compatible with k,

$$B/P^k \xrightarrow{\sim} \prod_{i=1}^{n} B/\mathfrak{m}_i^k.$$

Passing to inverse limits, $\widehat{B} \simeq \prod_i \varprojlim_k B/\mathfrak{m}_i^k$.

Step 3: $B/\mathfrak{m}_i^k = B_{\mathfrak{m}_i}/\mathfrak{m}_i^k B_{\mathfrak{m}_i}$. The ring $B/\mathfrak{m}_i^k$ is local with maximal ideal $\mathfrak{m}_i/\mathfrak{m}_i^k$: if $s \notin \mathfrak{m}_i$ there are $a \in B$ and $m \in \mathfrak{m}_i$ with $as + m = 1$, so $as \equiv 1 - m$, which is a unit modulo $\mathfrak{m}_i^k$ (its inverse is $1 + m + \cdots + m^{k-1}$). Hence every element of $B \setminus \mathfrak{m}_i$ is already invertible in $B/\mathfrak{m}_i^k$, and localization at $\mathfrak{m}_i$ does not change this ring: $B/\mathfrak{m}_i^k \to B_{\mathfrak{m}_i}/\mathfrak{m}_i^k B_{\mathfrak{m}_i}$ is an isomorphism (compatible with k). Hence $\varprojlim_k B/\mathfrak{m}_i^k = \widehat{B}_i$, and $\widehat{B} \simeq \bigoplus_i \widehat{B}_i$ (a finite product is a finite direct sum). □

Exercise 3.17

Problem. Let A be a Noetherian local ring, and let B be a finite A-algebra with maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$. Show that $B \otimes_A \widehat{A} \simeq \bigoplus_i \widehat{B_{\mathfrak{m}_i}}$, where $\widehat{\ }$ denotes the formal completion for the topology defined by the maximal ideal.

Solution. Let $\mathfrak{m} = \mathfrak{m}_A$ and $J = \bigcap_i \mathfrak{m}_i$. Since B is a finitely generated A-module and A is Noetherian, Cor. 1.3.14 gives a canonical isomorphism

$$B \otimes_A \widehat{A} \simeq \varprojlim_k B/\mathfrak{m}^k B,$$

the completion of the A-module B for the $\mathfrak{m}$-adic topology.

The topologies $(\mathfrak{m}^k B)_k$ and $(J^k)_k$ coincide. Since B is integral over A, each maximal ideal $\mathfrak{m}_i$ contracts to a maximal ideal of A (if $B/\mathfrak{m}_i$ is a field integral over the subring $A/(\mathfrak{m}_i \cap A)$, the latter is a field), hence $\mathfrak{m}_i \cap A = \mathfrak{m}$ and $\mathfrak{m}B \subseteq \mathfrak{m}_i$ for all i. So $\mathfrak{m}B \subseteq J$ and $\mathfrak{m}^k B \subseteq J^k$. Conversely, $B/\mathfrak{m}B$ is a finite-dimensional algebra over $A/\mathfrak{m}$, hence an Artinian ring, and its Jacobson radical $J/\mathfrak{m}B$ is nilpotent. So $J^N \subseteq \mathfrak{m}B$ for some N, and $J^{Nk} \subseteq \mathfrak{m}^k B$. As in Exercise 3.16, Step 1, $\varprojlim_k B/\mathfrak{m}^k B = \varprojlim_k B/J^k$, the formal completion of the semi-local ring B for the topology defined by $J = \bigcap_i \mathfrak{m}_i$.

By Exercise 3.16 this is $\bigoplus_i \widehat{B_{\mathfrak{m}_i}}$. The resulting isomorphism $B \otimes_A \widehat{A} \to \bigoplus_i \widehat{B_{\mathfrak{m}_i}}$ is canonical: it sends $b \otimes a$ to $(a \cdot \hat{b}_i)_i$, where $\hat{b}_i$ is the image of b in $\widehat{B_{\mathfrak{m}_i}}$ and $\widehat{A} \to \widehat{B_{\mathfrak{m}_i}}$ is induced by $A \to B_{\mathfrak{m}_i}$. $\square$

Exercise 3.18

> **Problem.** Let X be a quasi-projective scheme over a locally Noetherian scheme S. Let G be a finite group acting on X. Hence the quotient $Y := X/G$ exists (Exercise 3.3.23). (a) Let G act on an A-algebra B, and C be a flat A-algebra. Then G acts on $B \otimes_A C$ by the identity on the second component. Let us denote the A-linear map $g - \mathrm{Id}_B : B \to B$ by φ_g. Show that $\mathrm{Ker}(\varphi_g \otimes \mathrm{Id}_C) = \mathrm{Ker}(\varphi_g) \otimes_A C$ and that $\mathrm{Ker}(\varphi_g \otimes \mathrm{Id}_C) \cap \mathrm{Ker}(\varphi_{g'} \otimes \mathrm{Id}_C) = (\mathrm{Ker}(\varphi_g) \cap \mathrm{Ker}(\varphi_{g'})) \otimes_A C$ in $B \otimes_A C$. Deduce from this that $(B \otimes_A C)^G = B^G \otimes_A C$. Show that the quotient morphism $f : X \to Y$ commutes with flat base change. (b) Let us fix a point $x \in X$. Let us set $D := \{g \in G \mid gx = x\}$. This is the decomposition group of x. Show that D acts canonically on $\mathcal{O}_{X,x}$ as well as on the formal completion $\widehat{\mathcal{O}}_{X,x}$. Let $y = f(x)$. Show that $(\widehat{\mathcal{O}}_{X,x})^D = \widehat{\mathcal{O}}_{Y,y}$ (use Exercise 3.17 and (a)). Show that $\mathcal{O}_{Y,y} = (\mathcal{O}_{X,x})^D$ if $G = D$. (c) Show that the morphism $f : X \to Y$ factors into $X \to X/D$ and $X/D \to Y$. Let z be the image of x in X/D. Show that $X/D \to Y$ is etale at z.

Solution. Recollections (Exercises 2.3.21 and 3.3.23). We use the following facts about $f : X \to Y = X/G$. (Q1) Every G-orbit of X is contained in a G-stable affine open subset $U = \operatorname{Spec} B$. Then $f(U) = \operatorname{Spec} B^G$ is open in Y, $f^{-1}(f(U)) = U$, and such $f(U)$ cover Y. (Q2) f is finite and surjective, G acts transitively on the fibers of f, and Y is of finite type over S. In the situation of (Q1), B^G is Noetherian and B is a finite B^G-module (each $b \in B$ is a root of $\prod_g (T - gb) \in B^G[T]$; Artin–Tate). (Q3) (universal property) every G-invariant morphism $X \to Z$ factors uniquely through f. (Q4) For every subgroup $H \subseteq G$ the quotient X/H exists and has the same properties. We let G act on rings of G-stable affine opens by $g\cdot$ so that the prime of B attached to gx is g applied to the prime attached to x.

(a) Since C is A-flat, $- \otimes_A C$ preserves kernels. The sequence $0 \to \operatorname{Ker} \varphi_g \to B \xrightarrow{\varphi_g} B$ is exact, so $0 \to \operatorname{Ker} \varphi_g \otimes C \to B \otimes C \xrightarrow{\varphi_g \otimes 1} B \otimes C$ is exact, i.e. $\mathrm{Ker}(\varphi_g \otimes 1) = \mathrm{Ker}(\varphi_g) \otimes_A C$ (a submodule of $B \otimes C$ by flatness). More generally let $g_1, \dots, g_r \in G$ and consider $\Phi = (\varphi_{g_1}, \dots, \varphi_{g_r}) : B \to B^r$. Then $\operatorname{Ker} \Phi = \bigcap_i \operatorname{Ker} \varphi_{g_i}$ and $\Phi \otimes 1 : B \otimes C \to (B \otimes C)^r$ is $(\varphi_{g_i} \otimes 1)_i$, so $\mathrm{Ker}(\Phi \otimes 1) = \bigcap_i \mathrm{Ker}(\varphi_{g_i} \otimes 1)$. Exactness of $0 \to \operatorname{Ker} \Phi \to B \to B^r$ and flatness give $\bigcap_i \mathrm{Ker}(\varphi_{g_i} \otimes 1) = (\bigcap_i \operatorname{Ker} \varphi_{g_i}) \otimes_A C$. For $r = 2$ this is the second assertion. Taking for $g_1, \dots, g_r$ all the elements of G, we get $(B \otimes C)^G = \bigcap_g \mathrm{Ker}(\varphi_g \otimes 1) = (\bigcap_g \operatorname{Ker} \varphi_g) \otimes C = B^G \otimes_A C$.

Flat base change. Let $T \to S$ be a flat morphism. G acts on $X_T = X \times_S T$ through X, and $f_T : X_T \to Y_T$ is G-invariant. Let $U = \operatorname{Spec} B$ be as in (Q1) for a point $x \in X$ with image $s \in S$. Choose an affine open neighborhood $\operatorname{Spec} A_0$ of s and let $W \subseteq U$ be the preimage of $\operatorname{Spec} A_0$ in U; it is open, G-stable (G acts by S-automorphisms) and contains Gx. Let J be the ideal of B defining $U \setminus W$. By prime avoidance there is $h \in J$ not in any of the finitely many primes of Gx; then $N(h) = \prod_{g \in G} g(h) \in B^G$ and $D(N(h)) \subseteq D(h) \subseteq W$ is a G-stable principal open subset of U containing Gx. Replacing U by it, we may assume that $U = \operatorname{Spec} B$ lies over $\operatorname{Spec} A_0$ (it is still a G-stable affine open, with the properties of (Q1)). Let $\operatorname{Spec} C$ be an affine open of T over $\operatorname{Spec} A_0$. Then C is a flat A_0-algebra, $U \times_S \operatorname{Spec} C = \operatorname{Spec}(B \otimes_{A_0} C)$ is a G-stable affine open subset of X_T, and by the above its quotient is $\operatorname{Spec}((B \otimes C)^G) = \operatorname{Spec}(B^G \otimes_{A_0} C) = f(U) \times_S \operatorname{Spec} C$. These G-stable affine opens cover X_T and every point has a G-stable affine neighborhood, so by Exercise 2.3.21(c) the quotient X_T/G exists and is

obtained by gluing these quotients, i.e. $X_T/G = Y_T$ with quotient morphism f_T. This says that the formation of the quotient commutes with flat base change.

(b) *Action.* An element $g \in D$ is an automorphism of X fixing x, so it induces an automorphism of the local ring $\mathcal{O}_{X,x}$, preserving $\mathfrak{m}_x$. It is therefore continuous for the $\mathfrak{m}_x$-adic topology and extends to $\widehat{\mathcal{O}}_{X,x}$. Because G acts by S-automorphisms, D also acts on $k(x)$, trivially on $k(y)$ (the action on $k(x)$ is the one used in Exercise 3.19).

Invariants. Take $U = \operatorname{Spec} B \supseteq Gx$ as in (Q1). Let $\mathfrak{q} \subset B^G$ be the prime ideal of y, $\Sigma = B^G \setminus \mathfrak{q}$, $A' = \Sigma^{-1}B^G = \mathcal{O}_{Y,y}$ (a Noetherian local ring) and $B' = \Sigma^{-1}B$. The group G acts on B' by A'-algebra automorphisms and $(B')^G = A'$: if b/s is G-invariant with $s \in \Sigma$, then for each g there is $t_g \in \Sigma$ with $t_g(gb - b) = 0$; with $t = \prod_g t_g \in \Sigma$ we get $tb \in B^G$, and $b/s = tb/ts \in A'$. By (Q2), B' is a finite A'-algebra. Its maximal ideals are the primes of B lying over $\mathfrak{q}$, i.e. the points of $f^{-1}(y) = Gx$ (they lie over $\mathfrak{m}_{A'}$ since B' is finite over the local ring A'), and the localization of B' at the maximal ideal of x' is $\mathcal{O}_{X,x'}$. By Exercise 3.17 there is a canonical isomorphism

$$\Phi : \; B' \otimes_{A'} \widehat{A'} \;\xrightarrow{\sim}\; \bigoplus_{x' \in Gx} \widehat{\mathcal{O}}_{X,x'}, \qquad b \otimes a \mapsto \left(a\, \hat{b}_{x'}\right)_{x'}.$$

It is G-equivariant, where G acts on the left side through B' and on the right side by the isomorphisms $g : \widehat{\mathcal{O}}_{X,x'} \to \widehat{\mathcal{O}}_{X,gx'}$, permuting the summands (this is clear on pure tensors, as g is A'-linear). The ring $\widehat{A'} = \widehat{\mathcal{O}}_{Y,y}$ is flat over A' (Thm. 1.3.15), so by (a)

$$(B' \otimes_{A'} \widehat{A'})^G = (B')^G \otimes_{A'} \widehat{A'} = \widehat{A'}.$$

On the right side, a family $(u_{x'})_{x'}$ is G-invariant if and only if $u_{gx} = g\, u_x$ for all g; this forces $u_x \in (\widehat{\mathcal{O}}_{X,x})^D$, and conversely every $u \in (\widehat{\mathcal{O}}_{X,x})^D$ gives the invariant family $(g\, u)_{gx}$, which is well defined because u is D-invariant. So the projection to the x-component is a bijection from the G-invariants of $\bigoplus_{x'} \widehat{\mathcal{O}}_{X,x'}$ onto $(\widehat{\mathcal{O}}_{X,x})^D$. Composing with Φ and with the identification $\widehat{A'} = (B' \otimes \widehat{A'})^G$, the element $a \in \widehat{A'}$ goes to the image of a under the map $\rho : \widehat{\mathcal{O}}_{Y,y} \to \widehat{\mathcal{O}}_{X,x}$ induced by f. We have shown:

ρ is injective, and its image is exactly $(\widehat{\mathcal{O}}_{X,x})^D$.

This is the statement $(\widehat{\mathcal{O}}_{X,x})^D = \widehat{\mathcal{O}}_{Y,y}$ (the equality being realized by ρ).

The case $G = D$. Then $Gx = \{x\}$, so B' is local, and $B' = \mathcal{O}_{X,x}$ (a local ring is its own localization at its maximal ideal). Hence $\mathcal{O}_{Y,y} = A' = (B')^G = (\mathcal{O}_{X,x})^D$; moreover $\mathcal{O}_{X,x}$ is finite over $\mathcal{O}_{Y,y}$.

(c) D is a finite group acting on X, so X/D exists (Q4), and f is D-invariant because it is G-invariant. By (Q3) it factors uniquely as $X \xrightarrow{\pi} X/D \xrightarrow{h} Y$. Both X/D and Y are of finite type over S, so locally Noetherian, and h is of finite type (indeed finite). Let $z = \pi(x)$.

Apply (b) to the group D acting on X: the decomposition group of x in D is D itself, so $\widehat{\mathcal{O}}_{X/D,z} \xrightarrow{\rho_D} \widehat{\mathcal{O}}_{X,x}$ is injective with image $(\widehat{\mathcal{O}}_{X,x})^D$. Apply (b) to G: $\widehat{\mathcal{O}}_{Y,y} \xrightarrow{\rho} \widehat{\mathcal{O}}_{X,x}$ is injective with the same image. Since $\rho = \rho_D \circ \hat{h}^{\#}$ (with $\hat{h}^{\#} : \widehat{\mathcal{O}}_{Y,y} \to \widehat{\mathcal{O}}_{X/D,z}$ induced by h), the map $\hat{h}^{\#}$ is injective and has the same image as ρ after composing with the injection ρ_D; hence $\hat{h}^{\#}$ is surjective. So

$$\hat{h}^{\#} : \widehat{\mathcal{O}}_{Y,y} \to \widehat{\mathcal{O}}_{X/D,z} \;\text{ is an isomorphism.}$$

The residue field of a Noetherian local ring and of its completion coincide, so $k(z) = k(y)$. By Prop. 3.26 (applied to h at z), h is etale at z. □

Remark (used in Exercise 3.19). We proved a bit more: $k(z) = k(y)$, and $\widehat{\mathcal{O}}_{Y,y} \to \widehat{\mathcal{O}}_{X/D,z}$ is an isomorphism; with etaleness this gives $\mathfrak{m}_y \mathcal{O}_{X/D,z} = \mathfrak{m}_z$ and flatness of $\mathcal{O}_{Y,y} \to \mathcal{O}_{X/D,z}$.

Exercise 3.19

Problem. We keep the notation and hypotheses of the preceding exercise. Let $x \in X$. Then the decomposition group D naturally acts on $k(x)$. This induces a group homomorphism $D \to \mathrm{Aut}(k(x)/k(y))$. The inertia group I of x is by definition the kernel of this homomorphism. We want to show that $f : X \to X/G$ is etale at x if and only if I acts trivially on $\mathcal{O}_{X,x}$. (a) Show that $X \to X/G$ is etale at x if and only if $X \to X/D$ is etale at x. In what follows we will therefore reduce to the case that $G = D$, that is, $f^{-1}(y) = \{x\}$. Let us write $A = \mathcal{O}_{Y,y}$ and $B = \mathcal{O}_{X,x}$. Show that $B^D = A$ and that B is finite over A. (b) Let k' be the separable closure of $k(y)$ in $k(x)$. For any element $b \in B$, let $\bar{b}$ denote its image in $k(x)$. Let $\theta \in B$ be such that $k' = k(y)[\bar{\theta}]$. Show that $P(T) := \prod_{g \in D}(T - g(\theta)) \in A[T]$ and that its image in $k(y)[T]$ vanishes in $\bar{\theta}$. Let $\tau \in \mathrm{Aut}(k'/k(y))$. Show that there exists a $g \in D$ such that $g(\bar{\theta}) = \tau(\bar{\theta})$. Deduce from this that $\tau = \bar{g}$ and that $D \to \mathrm{Aut}(k(x)/k(y)) = \mathrm{Aut}(k'/k(y))$ is surjective. Show that $k'/k(y)$ is Galois, and that we have an exact sequence of groups $1 \to I \to D \to \mathrm{Gal}(k'/k(y)) \to 1$. (c) Let $C = A[T]/(P(T))$. This is a finite etale local A-algebra. Let D act on $C \otimes_A B$ as the identity on the first component. Show that $\mathrm{Spec}\, C = \mathrm{Spec}(C \otimes_A B)/D$, and that B is etale over A if and only if $C \otimes_A B$ is etale over C. Show that the decomposition group of every point of $\mathrm{Spec}(C \otimes_A B)_z$, where z is the closed point of $\mathrm{Spec}\, C$, is equal to I. Hence, by (c), we have reduced to the case when $D = I$, and $k(x)$ is purely inseparable over $k(y)$. (d) Using Proposition 3.26, show that $X \to Y$ is etale at x if and only if I acts trivially on $\mathcal{O}_{X,x}$. Show that in general $X/I \to Y$ is etale at the image of x in X/I.

Solution. Conventions. We use the facts (Q1)–(Q4) recalled in Exercise 3.18 and its results (a), (b), (c). Put $Y = X/G$, $y = f(x)$. For a local homomorphism $A \to B$ of Noetherian local rings (always the local rings of a morphism of finite type of locally Noetherian schemes) we say that $A \to B$ is *etale* if (F) B is flat over A, (U1) $\mathfrak{m}_A B = \mathfrak{m}_B$, and (U2) $k(B)/k(A)$ is a (finite) separable extension. By Def. 3.17, $X \to Y$ is etale at x if and only if $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ is etale in this sense; "unramified" means (U1)+(U2). The group D acts on $k(x)$ through its action on $\mathcal{O}_{X,x}$ (Exercise 3.18(b)); it acts trivially on $k(y)$.

Two lemmas.

Lemma L. Let $A \to C \to B$ be local homomorphisms of Noetherian local rings such that $A \to C$ is flat, $\mathfrak{m}_A C = \mathfrak{m}_C$ and $k(A) = k(C)$ (via $A \to C$). Then (i) B satisfies (U1), (U2) over A if and only if it does over C; (ii) B is flat over A if and only if it is flat over C. Hence $A \to B$ is etale if and only if $C \to B$ is etale.

Proof. (i) $\mathfrak{m}_A B = (\mathfrak{m}_A C)B = \mathfrak{m}_C B$ and $k(B)/k(A) = k(B)/k(C)$. (ii) If B is C-flat it is A-flat. Conversely let B be A-flat. Put $\kappa = A/\mathfrak{m}_A$; then $C/\mathfrak{m}_C = C/\mathfrak{m}_A C = C \otimes_A \kappa$. If $F_\bullet \to \kappa$ is a free resolution over A, then $C \otimes_A F_\bullet$ is a free resolution of $C \otimes_A \kappa$ over C (C is A-flat), so

$$\mathrm{Tor}_1^C(C/\mathfrak{m}_C, B) = H_1\big((C \otimes_A F_\bullet) \otimes_C B\big) = H_1(F_\bullet \otimes_A B) = \mathrm{Tor}_1^A(\kappa, B) = 0.$$

By the local flatness criterion (the one used for Lemma 3.16: for a local homomorphism $C \to B$ of Noetherian local rings, B is C-flat if and only if $\mathrm{Tor}_1^C(k(C), B) = 0$), B is flat over C. □

Lemma E. Let $A \to B$ be a local homomorphism of Noetherian local rings with B finite over A; put $k_A = k(A)$, $k_B = k(B)$. Let $k' \subseteq k_B$ be a finite Galois extension of k_A, $k' = k_A(\bar\theta)$, with minimal polynomial $\bar m$ of $\bar\theta$ over k_A. Let $Q \in A[T]$ be any monic lift of $\bar m$, and put $C = A[T]/(Q)$, $B_1 = C \otimes_A B = B[T]/(Q)$. Then: (i) C is a finite free A-module (hence faithfully flat), a local ring with maximal ideal $\mathfrak{m}_A C$ and residue field k'. (ii) The maximal ideals of B_1 are the ideals $\mathfrak{n}_\sigma = \mathrm{Ker}(B_1 \to k_B,\ b \mapsto \bar b,\ T \mapsto \sigma(\bar\theta))$, $\sigma \in \mathrm{Gal}(k'/k_A)$; they are pairwise distinct, lie over $\mathfrak{m}_B$, and have residue field k_B. (iii) $A \to B$ is etale if and only if $C \to (B_1)_{\mathfrak{n}_\sigma}$ is etale for every σ.

Proof. (i) C is free with basis $1, T, \dots, T^{r-1}$, $r = \deg Q$, and $C/\mathfrak{m}_A C = k_A[T]/(\bar m) = k'$ is a field. Every maximal ideal of the finite A-algebra C contains $\mathfrak{m}_A$, so $\mathfrak{m}_A C$ is the only maximal ideal. (ii) B_1 is finite over B, so its maximal ideals contain $\mathfrak{m}_B B_1$ and correspond to the maximal ideals of $B_1/\mathfrak{m}_B B_1 = k_B[T]/(\bar m)$. As k'/k_A is Galois, $\bar m = \prod_\sigma (T - \sigma\bar\theta)$ with distinct roots in $k' \subseteq k_B$, so $k_B[T]/(\bar m) \simeq \prod_\sigma k_B$, with the evident maximal ideals. (iii) Put $E = B/\mathfrak{m}_A B$ and note $B_1/\mathfrak{m}_A B_1 = E \otimes_{k_A} k'$ and $\mathfrak{m}_A B_1 = \mathfrak{m}_C B_1$.

Flatness. If B is A-flat, then B_1 is C-flat by base change, hence so are its localizations. Conversely suppose $(B_1)_{\mathfrak{n}_\sigma}$ is C-flat for every σ. These are all the localizations of B_1 at maximal ideals, so B_1 is C-flat by Cor. 1.2.15. Let $N' \hookrightarrow N$ be an injection of A-modules. Then $C \otimes N' \to C \otimes N$ is injective (C is A-flat), hence so is $B_1 \otimes_C (C \otimes N') \to B_1 \otimes_C (C \otimes N)$, i.e. $(B \otimes N') \otimes_A C \to (B \otimes N) \otimes_A C$. As C is faithfully flat over A, $B \otimes N' \to B \otimes N$ is injective. So B is A-flat.

Unramifiedness. Suppose B is unramified over A. Then $E = k_B$ and $B_1/\mathfrak{m}_C B_1 = k_B \otimes_{k_A} k' \simeq \prod_\sigma k_B$ (the same computation as in (ii), with the extension k'/k_A). A localization of this product of fields at a maximal ideal is the field k_B, so $\mathfrak{m}_C (B_1)_{\mathfrak{n}_\sigma} = \mathfrak{n}_\sigma (B_1)_{\mathfrak{n}_\sigma}$, and the residue extension is k_B/k', separable because k_B/k_A is. Conversely suppose all $(B_1)_{\mathfrak{n}_\sigma}$ are unramified over C. The Artinian ring $E \otimes_{k_A} k' = B_1/\mathfrak{m}_C B_1$ is the product of its localizations at its maximal ideals, namely $(B_1)_{\mathfrak{n}_\sigma}/\mathfrak{m}_C (B_1)_{\mathfrak{n}_\sigma} = k(\mathfrak{n}_\sigma) = k_B$, which are fields. So $E \otimes_{k_A} k'$ is reduced; since $E \to E \otimes_{k_A} k'$ is injective, E is reduced. A reduced local Artinian ring is a field: $E = k_B$, i.e. $\mathfrak{m}_A B = \mathfrak{m}_B$. Finally $k_B = k(\mathfrak{n}_\sigma)$ is separable over k' by hypothesis, and k'/k_A is separable, so k_B/k_A is separable. □

(a) Let z be the image of x in X/D and $h : X/D \to Y$ the induced morphism. Put $A_0 = \mathcal{O}_{Y,y}$, $C_0 = \mathcal{O}_{X/D,z}$, $B_0 = \mathcal{O}_{X,x}$ (this B_0 is called B below, once $G = D$). By Exercise 3.18(c) and its Remark, $A_0 \to C_0$ is flat, $\mathfrak{m}_{A_0} C_0 = \mathfrak{m}_{C_0}$ and $k(C_0) = k(A_0)$. By Lemma L, $A_0 \to B_0$ is etale if and only if $C_0 \to B_0$ is. In other words, $X \to X/G$ is etale at x if and only if $X \to X/D$ is etale at x. The group D acts on X, the point x has decomposition group D (and the same inertia group I) for this action, and the fiber of $X \to X/D$ over z is the D-orbit $\{x\}$ of x. So it suffices to prove the theorem when $G = D$.

Assume now $G = D$ (that is, we replace Y by X/D and y by z, which is legitimate by the above). By Exercise 3.18(b), $A := \mathcal{O}_{Y,y} = B^D$ with $B = \mathcal{O}_{X,x}$, and B is finite over A. In particular $k(x)$ is a finite extension of $k(y)$.

(b) Let k' be the separable closure of $k(y)$ in the finite extension $k(x)$ and $\theta \in B$ a lift of a primitive element $\bar\theta$ of the finite separable extension $k'/k(y)$. The coefficients of $P(T) = \prod_{g \in D} (T - g\theta) \in B[T]$ are D-invariant, hence lie in $B^D = A$. Since $g = \mathrm{id}$ is one of the factors, $P(\theta) = 0$. Reducing modulo $\mathfrak{m}_B$: the image $\bar P \in k(y)[T]$ of P satisfies $\bar P = \prod_g (T - \overline{g\theta})$ in $k(x)[T]$, and $\bar P(\bar\theta) = 0$; here $\overline{g\theta} = \bar g(\bar\theta)$, where $\bar g$ is the automorphism of $k(x)$ induced by g.

Let $\bar{m} \in k(y)[T]$ be the minimal polynomial of $\bar{\theta}$. It divides $\bar{P}$, so it splits in $k(x)[T]$ into linear factors, and each root ρ is of the form $\bar{g}(\bar{\theta})$. Since $\bar{m}$ is separable and its roots lie in $k(x)$, they lie in k'. Hence $k' = k(y)(\bar{\theta})$ contains all roots of the separable polynomial $\bar{m}$: $k'/k(y)$ is Galois.

Let $\tau \in \mathrm{Aut}(k'/k(y))$. Then $\tau(\bar{\theta})$ is a root of $\bar{m}$, so a root of $\bar{P} = \prod_g (T - \bar{g}\bar{\theta})$; hence $\tau(\bar{\theta}) = \bar{g}(\bar{\theta})$ for some $g \in D$. Every automorphism of $k(x)/k(y)$ preserves k' (the separable closure), so $\bar{g}|_{k'} \in \mathrm{Aut}(k'/k(y))$ agrees with τ on the generator $\bar{\theta}$; thus $\bar{g}|_{k'} = \tau$. Consequently $D \to \mathrm{Aut}(k'/k(y))$, $g \mapsto \bar{g}|_{k'}$, is surjective. The restriction map $\mathrm{Aut}(k(x)/k(y)) \to \mathrm{Aut}(k'/k(y))$ is injective: $k(x)/k'$ is purely inseparable, so an automorphism of $k(x)$ that is the identity on k' fixes the elements u with $u^{p^e} \in k'$ (because p^e-th power is injective), i.e. everything. Hence it is an isomorphism (the composite $D \to \mathrm{Aut}(k(x)/k(y)) \to \mathrm{Aut}(k'/k(y))$ being onto), $D \to \mathrm{Aut}(k(x)/k(y))$ is surjective, and its kernel I is the kernel of $D \to \mathrm{Gal}(k'/k(y))$. This gives the exact sequence $1 \to I \to D \to \mathrm{Gal}(k'/k(y)) \to 1$. In particular I is normal in D, and $k(x)^I \supseteq k'$.

A correction to (c) as printed. Each $\tau \in \mathrm{Gal}(k'/k(y))$ equals $\bar{g}|_{k'}$ for exactly $|I|$ elements $g \in D$, so $\bar{P} = \prod_\tau (T - \tau\bar{\theta})^{|I|} = \bar{m}^{|I|}$. Hence $A[T]/(P)$ is etale over A only if $I = 1$ (its special fiber is $k(y)[T]/(\bar{m}^{|I|})$, nonreduced for $|I| > 1$). So in (c) one has to take instead $C = A[T]/(Q)$, where $Q \in A[T]$ is a monic lift of the minimal polynomial $\bar{m}$ of $\bar{\theta}$ (no map $C \to B$ is needed, only $C \otimes_A B$). This is what we do.

(c) Let $C = A[T]/(Q)$, $B_1 = C \otimes_A B = B[T]/(Q)$ and $X_1 = \operatorname{Spec} B_1$, which is affine and of finite type over $\operatorname{Spec} A$. By Lemma E(i), C is a finite free local A-algebra, and it is etale over A since its special fiber is the separable field extension k' of $k(y)$. The group D acts on B_1 through B, by C-algebra automorphisms. Since C is A-flat, Exercise 3.18(a) gives $B_1^D = B^D \otimes_A C = C$, i.e. $\operatorname{Spec} C = X_1/D$.

By Lemma E(iii), B is etale over A if and only if $(B_1)_{\mathfrak{n}_\sigma}$ is etale over C for all $\sigma \in \mathrm{Gal}(k'/k(y))$. These are all the local rings of X_1 at closed points, and the fiber of $X_1 \to \operatorname{Spec} C$ over the closed point z of $\operatorname{Spec} C$ consists of the $\mathfrak{n}_\sigma$ (the fiber is $\operatorname{Spec}(E \otimes_{k(y)} k')$, a spectrum of an Artinian ring, so its points are the maximal ideals $\mathfrak{n}_\sigma$).

Decomposition groups. For $g \in D$ and $\sigma \in \mathrm{Gal}(k'/k(y))$, the ideal $g\mathfrak{n}_\sigma$ is the kernel of $\bar{g} \circ \mathrm{ev}_\sigma \circ g^{-1} : B_1 \to k(x)$, which is $b \mapsto \bar{b}$, $T \mapsto \bar{g}(\sigma\bar{\theta})$. So $g\mathfrak{n}_\sigma = \mathfrak{n}_{\bar{g}|_{k'} \circ \sigma}$. Since $D \to \mathrm{Gal}(k'/k(y))$ is onto, D permutes the $\mathfrak{n}_\sigma$ transitively, and the stabilizer of every $\mathfrak{n}_\sigma$ is $\{g : \bar{g}|_{k'} = \mathrm{id}\} = I$. Hence the decomposition group of every point of X_1 over z is I. (In particular, the $(B_1)_{\mathfrak{n}_\sigma}$ are all isomorphic over C, via the elements of D, so "for all σ" can be replaced by "for $\sigma = \mathrm{id}$".)

Reduction. Let $\mathfrak{n} = \mathfrak{n}_{\mathrm{id}}$ and $R = (B_1)_{\mathfrak{n}} = \mathcal{O}_{X_1,\mathfrak{n}}$. The group I (the decomposition group of $\mathfrak{n}$) acts on R, and it acts trivially on the residue field $k(\mathfrak{n}) = k(x)$, by the very definition of I. So for the action of the group I on X_1 at $\mathfrak{n}$ the decomposition group and the inertia group both equal I. By the exact sequence of (b) applied to this action, $k(x)$ is purely inseparable over the residue field of R^I (Step 3 below). Thus the question for B over A has been reduced, via B_1 over C, to the situation "$D = I$ and $k(x)$ purely inseparable over $k(y)$", as claimed. The proof is completed in (d).

(d) We prove, with $G = D$:

B is etale over A if and only if I acts trivially on $B = \mathcal{O}_{X,x}$.

Step 1. Let $\mathfrak{n}$, R be as in (c). By (c), B is etale over A iff $C \to R$ is etale.

Step 2. Apply Exercise 3.18 to D acting on X_1 at the point $\mathfrak{n}$, whose decomposition group is

I. Let $\mathfrak{n}'$ be the image of $\mathfrak{n}$ in X_1/I and $C' = \mathcal{O}_{X_1/I,\mathfrak{n}'}$. By 3.18(c) and the Remark, $C \to C'$ is flat, $\mathfrak{m}_C C' = \mathfrak{m}_{C'}$ and $k(C') = k(C) = k'$. By Exercise 3.18(b), applied to I acting on X_1 (the decomposition group of $\mathfrak{n}$ in I is I), $C' = R^I$, R is finite over C', and $\widehat{C'} \to \widehat{R}$ is injective with image $(\widehat{R})^I$. By Lemma L, $C \to R$ is etale iff $C' \to R$ is etale.

Step 3. (Here $D = I$.) Apply the sequence of (b) to the action of I on R (that is, on X_1 at $\mathfrak{n}$): the inertia group is all of I, so $\mathrm{Gal}(k''/k(C')) = 1$ where k'' is the separable closure of $k(C')$ in $k(\mathfrak{n}) = k(x)$. Hence $k'' = k(C')$, i.e. $k(x)$ is purely inseparable over $k(C')$.

Step 4: $C' \to R$ is etale iff I acts trivially on R. If I acts trivially on R, then $C' = R^I = R$ and $C' \to R$ is the identity, which is etale. Conversely suppose $C' \to R$ is etale. Then $k(\mathfrak{n})/k(C')$ is separable (U2) and purely inseparable (Step 3), so $k(\mathfrak{n}) = k(C')$. This is the hypothesis of Prop. 3.26 for the morphism $X_1 \to X_1/I$ at $\mathfrak{n}$ (a finite morphism of finite type between locally Noetherian schemes). By Prop. 3.26, $\widehat{C'} \to \widehat{R}$ is an isomorphism. By Step 2 its image is $(\widehat{R})^I$, so I acts trivially on $\widehat{R}$, hence on $R \subseteq \widehat{R}$ (Thm. 1.3.16(b); $R \to \widehat{R}$ is injective, being faithfully flat).

Step 5: I acts trivially on R iff it acts trivially on B. If I acts trivially on B, it acts trivially on $B_1 = B[T]/(Q)$ (it acts on coefficients), hence on its localization R. Conversely, $B \to B_1$ is injective (B_1 is free over B) and $B_1 \to R$ is a localization, so $B \to R$ is flat; it is local ($\mathfrak{n} \cap B = \mathfrak{m}_B$), hence faithfully flat (Cor. 1.2.20), hence injective (Exercise 1.2.19(a)). It is I-equivariant, so if I acts trivially on R it acts trivially on B.

Steps 1–5 prove the boxed statement. With (a) we get the theorem for general G: *$f : X \to X/G$ is etale at x iff I acts trivially on $\mathcal{O}_{X,x}$.*

$X/I \to Y$ is etale at the image of x. First let $G = D$. Replacing X by a G-stable affine open neighborhood $\operatorname{Spec} B_X$ of x (see (Q1)) we may assume X affine, so that $X' = X/I = \operatorname{Spec} B_X^I$ is again affine. Let x' be the image of x in X'. The group $\bar{G} = D/I$ acts on X', $X'/\bar{G} = X/D = Y$, and the fiber of $X' \to Y$ over y is $\{x'\}$, so the decomposition group of x' is all of $\bar{G}$. By Exercise 3.18(b) applied to I acting on X, $\mathcal{O}_{X',x'} = \mathcal{O}_{X,x}^I$, and by Step 3 (the same argument, for I acting on X at x) $k(x)$ is purely inseparable over $k(x')$. The elements of k' are separable over $k(y) \subseteq k(x')$ and lie in $k(x)$, hence they lie in $k(x')$ (an element which is both separable and purely inseparable over $k(x')$ belongs to $k(x')$): $k' \subseteq k(x')$. The inclusion $k(x') \subseteq k(x)$ is D-equivariant, so the composite $D/I \to \mathrm{Aut}(k(x')/k(y)) \to \mathrm{Aut}(k'/k(y))$ is the isomorphism $D/I \simeq \mathrm{Gal}(k'/k(y))$ of (b). Hence $D/I \to \mathrm{Aut}(k(x')/k(y))$ is injective: the inertia group of x' for the action of $\bar{G}$ is trivial. In particular it acts trivially on $\mathcal{O}_{X',x'}$, so by the theorem just proved (for the group $\bar{G}$ acting on X', with $G = D$-type decomposition group), $X' \to X'/\bar{G} = Y$ is etale at x'.

For general G: $X/I \to X/D$ is etale at the image of x by the case just proved, applied to the group D acting on X (the decomposition group of x in D is D, and its inertia group is I). Also $X/D \to X/G$ is etale at z by Exercise 3.18(c). A composition of etale morphisms is etale (Prop. 3.22(c)). Hence $X/I \to X/G$ is etale at the image of x. □

Exercise 3.20

Problem. Let $X = \operatorname{Spec} k[T_1, \ldots, T_n]/(F_1, \ldots, F_r)$ be an affine variety over a field k, and x a closed point of X. Let us consider $J_x = \big((\partial F_i/\partial T_j)(x)\big)_{i,j}$ as a matrix with coefficients in

$k(x)$. Show that X is smooth at x if and only if $\operatorname{rank} J_x = n - \dim \mathcal{O}_{X,x}$. Hence the Jacobian criterion is a criterion for smoothness.

Solution. Let $\bar{k}$ be an algebraic closure of k and $\pi : X_{\bar{k}} \to X$ the projection; $X_{\bar{k}} = V(F_1, \dots, F_r) \subseteq \mathbb{A}^n_{\bar{k}}$ is defined by the same polynomials. Since x is closed, $k(x)$ is a finite extension of k, so the fiber $\pi^{-1}(x) = \operatorname{Spec}(k(x) \otimes_k \bar{k})$ is the spectrum of a finite $\bar{k}$-algebra. Its points x' are closed in $X_{\bar{k}}$, with residue field $\bar{k}$, and correspond to the k-embeddings $\sigma : k(x) \to \bar{k}$ (the maximal ideal of x' is the kernel of $k(x) \otimes \bar{k} \to \bar{k}$, $a \otimes b \mapsto \sigma(a)b$). As a rational point of $\mathbb{A}^n_{\bar{k}}$, $x' = (\sigma(t_1), \dots, \sigma(t_n))$, where $t_j \in k(x)$ is the image of T_j.

Fix such an x'. Since the F_i have coefficients in k,

$$\frac{\partial F_i}{\partial T_j}(x') = \sigma\Big(\frac{\partial F_i}{\partial T_j}(x)\Big), \quad \text{i.e. } J_{x'} = \sigma(J_x),$$

and applying a field homomorphism does not change the rank: $\operatorname{rank} J_{x'} = \operatorname{rank} J_x$.

The morphism π is flat (a base change of $\operatorname{Spec} \bar{k} \to \operatorname{Spec} k$) and the local ring of the fiber $\pi^{-1}(x)$ at x' is Artinian, of dimension 0. By Thm. 3.12 (the case of equality for flat morphisms),

$$\dim \mathcal{O}_{X_{\bar{k}},x'} = \dim \mathcal{O}_{X,x} + 0 = \dim \mathcal{O}_{X,x}.$$

By the Jacobian criterion (Thm. 2.19) for the rational point x' of $X_{\bar{k}} \subseteq \mathbb{A}^n_{\bar{k}}$, $X_{\bar{k}}$ is regular at x' if and only if $\operatorname{rank} J_{x'} = n - \dim \mathcal{O}_{X_{\bar{k}},x'}$. By what we just proved, this is equivalent to $\operatorname{rank} J_x = n - \dim \mathcal{O}_{X,x}$ — a condition which does not depend on x'. By Def. 3.28, X is smooth at x if and only if all the points x' of $X_{\bar{k}}$ above x are regular, i.e. if and only if $\operatorname{rank} J_x = n - \dim \mathcal{O}_{X,x}$. □

Exercise 3.21

Problem. Let $F \in k[T_0, \dots, T_n]$ be a homogeneous polynomial of degree d over a field k. We suppose that d is prime to the characteristic of k. Show that the closed subscheme $V_+(F)$ of $\mathbb{P}^n_k$ is smooth over k if and only if the $\partial F/\partial T_i$ are without common zero in $\bar{k}$.

Solution. *Hypotheses.* We assume $F \neq 0$ and $d \geq 1$, $n \geq 1$ (for $F = 0$ the scheme is $\mathbb{P}^n_k$, which is smooth although all partial derivatives vanish; for $d = 0$, F is a nonzero constant and $V_+(F) = \emptyset$). "Common zero in $\bar{k}$" means a common zero in $\mathbb{P}^n(\bar{k})$, i.e. a nonzero vector of $\bar{k}^{n+1}$ (the partial derivatives are homogeneous, so this is a projective notion).

Lemma A (valid for any $d \geq 1$ and any characteristic). Let $F \neq 0$ be homogeneous of degree $d \geq 1$ over k, $X = V_+(F) \subseteq \mathbb{P}^n_k$, and let $x \in X(\bar{k})$. Then $X_{\bar{k}}$ is regular at x if and only if the numbers $\frac{\partial F}{\partial T_j}(x)$, $0 \leq j \leq n$, are not all zero.

Proof. Choose i with $x_i \neq 0$ and normalize $x_i = 1$. On the chart $D_+(T_i) \simeq \mathbb{A}^n_{\bar{k}}$ with coordinates $t_j = T_j/T_i$ ($j \neq i$), $X_{\bar{k}}$ is $V(f)$ with $f = F(t_0, \dots, 1, \dots, t_n) \neq 0$, and x is a rational point, with $f(x) = 0$. The local ring $\mathcal{O} = \mathcal{O}_{\mathbb{A}^n,x}$ is regular of dimension n and $f \neq 0$ is a nonunit in the domain $\mathcal{O}$, so $\dim \mathcal{O}/f\mathcal{O} = n - 1$ (Thm. 2.5.15, as in the proof of Thm. 3.12). By the Jacobian criterion (Thm. 2.19), $X_{\bar{k}}$ is regular at x if and only if the $1 \times n$ matrix $(\partial f/\partial t_j(x))_{j \neq i}$ has rank $n - (n-1) = 1$, i.e. if and only if it is nonzero. Now $\partial f/\partial t_j(x) = \frac{\partial F}{\partial T_j}(x)$ for $j \neq i$. Euler's

identity $\sum_j T_j \frac{\partial F}{\partial T_j} = dF$ at x (where $F(x) = 0$, $x_i = 1$) gives $\frac{\partial F}{\partial T_i}(x) = -\sum_{j \neq i} x_j \frac{\partial F}{\partial T_j}(x)$. So all $\frac{\partial F}{\partial T_j}(x)$ vanish if and only if those with $j \neq i$ do. □

Proof of the exercise. By Cor. 2.17, $X_{\bar{k}}$ is regular if and only if it is regular at its closed points, which are the points $x \in X(\bar{k})$ ($\bar{k}$ is algebraically closed). By Lemma A, X is smooth over k if and only if there is no $x \in X(\bar{k})$ at which all $\partial F/\partial T_j$ vanish. Now let $x \in \mathbb{P}^n(\bar{k})$ be a common zero of all the partial derivatives. Euler's identity gives $d\,F(x) = \sum_j x_j \frac{\partial F}{\partial T_j}(x) = 0$, and d is invertible in k, so $F(x) = 0$, i.e. $x \in X(\bar{k})$. Hence "there is no point of $X(\bar{k})$ where all partials vanish" is equivalent to "the partials have no common zero in $\mathbb{P}^n(\bar{k})$". □

Exercise 3.22

Problem. Let C be a conic over a field k, that is $C = V_+(F) \subset \mathbb{P}^2_k$, where $F(x, y, z)$ is a homogeneous polynomial of degree 2. (a) Show that C is reduced (resp. irreducible) if and only if F is not a square modulo k^* (resp. F is irreducible). (b) Let us suppose that C contains a rational point P. (1) Show that if C is singular at P, then (up to isomorphism) C is of the form $V_+(G)$, with $G(x, y) \in k[x, y]$ homogeneous of degree 2. (2) Let us suppose C is regular at P. Let D be a line in $\mathbb{P}^2_k$ passing through P and different from $PT_{C,P}$ (see the notation in Exercise 2.9). Show that D meets C at another rational point Q. Show that $C \simeq V_+(ax^2 + yz)$ with $a \in k$ (let R be the intersection point of the lines $PT_{C,P}$ and $PT_{C,Q}$, and reduce to $P = (1, 0, 0)$, $Q = (0, 1, 0)$ and $R = (0, 0, 1)$). Deduce from this that either C is the union of two distinct lines, or $C \simeq \mathbb{P}^1_k$. (c) Let us suppose $\mathrm{char}(k) \neq 2$. Show that $C \simeq V_+(ax^2 + by^2 + cz^2)$, and that C is smooth if and only if $abc \neq 0$. (d) Let us suppose $\mathrm{char}(k) = 2$, $C(k) = \emptyset$, and that C is not smooth. (1) Show that there exist $a, b, c \in k$ such that $C \simeq V_+(ax^2 + by^2 + cz^2)$. Show that C does not contain any smooth point. (2) Show that $\mathcal{O}_C(D_+(z))$ is equal to the integral closure of $k[x/z]$ in $K(C)$. Deduce from this that C is normal, and therefore regular.

Solution. Throughout, $F \neq 0$. "Regular at P" is meant for the scheme C over k (the ring $\mathcal{O}_{C,P}$ is regular). Recall (Exercise 2.9) that for $P \in C(k)$ the "tangent line" is $PT_{C,P} = \{t : \sum_i \frac{\partial F}{\partial T_i}(P) t_i = 0\}$.

(a) Let $S = k[x, y, z]$. *Possible shapes of F.* Since $\deg F = 2$, either F is irreducible over k, or $F = L_1 L_2$ with L_1, L_2 linear forms over k. In the second case either L_1, L_2 are not proportional, or $F = \lambda L^2$ with $\lambda \in k^*$ and L a linear form (this is the case "F is a square modulo k^*").

Case F irreducible. $S/(F)$ is a graded domain, so $C = \mathrm{Proj}\, S/(F)$ is integral (in particular reduced and irreducible).

Case $F = L_1 L_2$, L_1, L_2 not proportional. Then F is squarefree, $S/(F)$ is reduced, and so are the rings of the charts $D_+(T_i) \cap C$ (degree-0 parts of the localizations $(S/F)_{T_i}$). So C is reduced. Set-theoretically $C = V_+(L_1) \cup V_+(L_2)$, a union of two distinct lines, neither of which contains the other: C is reducible.

Case $F = \lambda L^2$. After a linear change of coordinates we may take $L = z$. In the chart $D_+(x)$ the coordinate ring of C is $k[Y, Z]/(\lambda Z^2)$ ($Y = y/x$, $Z = z/x$), in which $Z \neq 0$ but $Z^2 = 0$. So C is not reduced; but $C_{\mathrm{red}} = V_+(z) \simeq \mathbb{P}^1_k$ is irreducible, so C is irreducible.

Conclusion. C is reduced if and only if F is not a square modulo k^* (i.e. not of the form λL^2). Moreover C is integral if and only if F is irreducible. *Note on the printed statement:*

the equivalence "C irreducible $\iff$ F irreducible" is false as stated for $F = \lambda L^2$, where C is irreducible (but not reduced) although F is not irreducible. The correct statements are: C is irreducible if and only if F is irreducible or of the form λL^2; C is integral if and only if F is irreducible. This is what is proved above.

(b) Choose coordinates such that $P = (1:0:0)$. Since $F(P) = 0$, F has no x^2 term:

$$F = x\,L(y,z) + q(y,z), \qquad L = L_y y + L_z z,\ q \text{ a binary quadratic form.}$$

In the chart $x = 1$ the point P is the origin of $\mathbb{A}^2$ and $f = L + q$. Let $\mathfrak{m} = (y,z)$ in the regular local ring $\mathcal{O} = k[y,z]_{\mathfrak{m}}$. The element $f = L + q \in \mathfrak{m}$ is nonzero, and $f \notin \mathfrak{m}^2$ if and only if $L \neq 0$. By Cor. 2.12, $\mathcal{O}/f\mathcal{O} = \mathcal{O}_{C,P}$ is regular if and only if $L \neq 0$.

(1) If C is singular at P then $L = 0$ and $F = q(y,z)$ is a binary quadratic form: $C = V_+(G)$ with $G(y,z) = q$ (renaming the variables).

(2) Now $L \neq 0$. The gradient of F at P is $(0, L_y, L_z)$, so $T := PT_{C,P} = \{L_y y + L_z z = 0\}$ is a line through P. A line D through P has the form $\{\lambda y + \mu z = 0\}$, with $(\lambda,\mu) \neq 0$, and its points are $(s:\mu t:-\lambda t)$, $(s:t) \in \mathbb{P}^1(k)$. One has $D = T$ iff (λ,μ) is proportional to (L_y, L_z), iff $\ell := L(\mu,-\lambda) = L_y\mu - L_z\lambda = 0$. We have

$$F(s,\mu t,-\lambda t) = s\,t\,L(\mu,-\lambda) + t^2 q(\mu,-\lambda) = t\,(\ell s + q_0 t), \qquad q_0 = q(\mu,-\lambda).$$

Suppose $D \neq T$, i.e. $\ell \neq 0$. The zeros of this form on D are $t = 0$ (the point P) and $(s:t) = (-q_0 : \ell)$, i.e. the rational point $Q = (-q_0 : \mu\ell : -\lambda\ell)$, which differs from P since $\ell \neq 0$. So D meets C in P and in another rational point Q.

Normal form. Choose new coordinates with $P = (1:0:0)$, $Q = (0:1:0)$ (complete the vectors representing P, Q to a basis of k^3). Since $F(P) = F(Q) = 0$,

$$F = uxy + vxz + wyz + ez^2.$$

The line $D = PQ$ is $\{z = 0\}$. The gradient at P is $(0,u,v)$, so $T = \{uy + vz = 0\}$, and $D \neq T$ means $u \neq 0$. (In particular $(u,v) \neq 0$ and $(u,w) \neq 0$, so P and Q are regular points.) At Q the gradient is $(u,0,w)$, so $PT_{C,Q} = \{ux + wz = 0\}$. The lines $PT_{C,P}$ and $PT_{C,Q}$ meet at $R = (-w:-v:u)$, whose third coordinate is $u \neq 0$. Hence P, Q, R are linearly independent and we may take them as the new coordinate points: put

$$x = x' - wz', \qquad y = y' - vz', \qquad z = uz'.$$

A direct computation gives

$$\begin{aligned} F &= u(x' - wz')(y' - vz') + v(x' - wz')\,uz' + w(y' - vz')\,uz' + e\,u^2 z'^2 \\ &= u\,x'y' + u(eu - vw)\,z'^2. \end{aligned}$$

Thus $C = V_+(x'y' + e'z'^2)$ with $e' = eu - vw \in k$ (u is a unit). Renaming $(z', x', y') \to (x, y, z)$ and $a = e'$: $C \simeq V_+(ax^2 + yz)$.

Conclusion. If $a = 0$, $C = V_+(yz)$ is the union of the two distinct lines $y = 0$ and $z = 0$. If $a \neq 0$, $C \cap V_+(y,z) = \operatorname{Proj} k[x]/(ax^2) = \emptyset$, so C is covered by the opens $C \cap D_+(y)$ and $C \cap D_+(z)$. Dividing the equation by y^2, resp. z^2:

$$C \cap D_+(y) = \operatorname{Spec} k[X,Z]/(aX^2 + Z) \simeq \mathbb{A}^1_k \text{ (coordinate } X = x/y), \qquad C \cap D_+(z) \simeq \mathbb{A}^1_k \text{ (coordinate } X' = x/z).$$

On the overlap $\{y \neq 0, z \neq 0\}$, where $X \neq 0$, we have $z/y = -aX^2$ and so $X' = x/z = X/(z/y) = -1/(aX)$. So C is obtained by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{G}_{m,k}$ via $X' = -1/(aX)$; this is $\mathbb{P}^1_k$ (replace the coordinate X' by $-aX'$, which makes the gluing $X' \cdot X = 1$). Hence $C \simeq \mathbb{P}^1_k$.

(c) Let $\operatorname{char} k \neq 2$. Let $B(u,v) = \frac{1}{2}\big(F(u+v) - F(u) - F(v)\big)$, a symmetric bilinear form on k^3 with $F(v) = B(v,v)$. We show that k^3 has a B-orthogonal basis; then in the corresponding coordinates $F = ax^2 + by^2 + cz^2$ $(a,b,c \in k)$. If $B(v,v) = 0$ for all v then $B = 0$ by polarization $(B(u,v) = \frac{1}{2}(B(u+v,u+v) - B(u,u) - B(v,v)))$ and every basis is orthogonal. Otherwise take v_1 with $B(v_1,v_1) \neq 0$; then $k^3 = kv_1 \oplus v_1^\perp$ (since $v - \frac{B(v,v_1)}{B(v_1,v_1)} v_1 \in v_1^\perp$), and we apply the same argument to the restriction of B to the plane $v_1^\perp$, and then to a line. So $C \simeq V_+(ax^2 + by^2 + cz^2)$, with $(a,b,c) \neq 0$ since $F \neq 0$.

Smoothness: the degree 2 is prime to $\operatorname{char} k$. By Exercise 3.21, C is smooth if and only if $2ax, 2by, 2cz$ have no common zero in $\mathbb{P}^2(\bar{k})$. If $abc \neq 0$ this holds. If, say, $c = 0$, then $(0:0:1)$ is a common zero, and similarly for $a = 0$ or $b = 0$. So C is smooth if and only if $abc \neq 0$.

(d) Let $\operatorname{char} k = 2$. Write $F = ax^2 + by^2 + cz^2 + uxy + vxz + wyz$. Then

$$\frac{\partial F}{\partial x} = uy + vz, \qquad \frac{\partial F}{\partial y} = ux + wz, \qquad \frac{\partial F}{\partial z} = vx + wy,$$

and by Lemma A (Exercise 3.21; it holds for $d = 2$ in characteristic 2) a point $x \in C(\bar{k})$ is a singular point of $C_{\bar{k}}$ if and only if all three partials vanish at x.

(1) *Diagonal form.* Suppose $(u,v,w) \neq (0,0,0)$. The common zeros in $\mathbb{P}^2(\bar{k})$ of the three partials are given by the kernel of the matrix $N = \begin{pmatrix} 0 & u & v \\ u & 0 & w \\ v & w & 0 \end{pmatrix}$. We have $N\,(w,v,u)^t = (2uv, 2uw, 2vw)^t = 0$, and $(w,v,u) \neq 0$, so $\operatorname{rank} N \leq 2$. On the other hand $\operatorname{rank} N \geq 2$: if $u \neq 0$ the first two rows are independent; if $u = 0$ the rows are $(0,0,v), (0,0,w), (v,w,0)$ with $(v,w) \neq 0$, and the third row is not a multiple of $(0,0,1)$. So there is exactly one common zero, namely $p = (w:v:u)$, a point of $\mathbb{P}^2(k)$. As C is not smooth, $C_{\bar{k}}$ has a singular point (Cor. 2.17), which must be a common zero of the partials, i.e. p, so $p \in C(\bar{k})$ and $F(p) = 0$ with $p \in \mathbb{P}^2(k)$: $C(k) \neq \emptyset$, a contradiction. Hence $u = v = w = 0$ and

$$F = ax^2 + by^2 + cz^2.$$

Then all three partials vanish identically, so every point of $C_{\bar{k}}$ is singular (Lemma A): C has no smooth point.

(2) *The coefficients.* $C(k) = \emptyset$ gives $F(1,0,0) = a \neq 0$, $b \neq 0$, $c \neq 0$. Fix an algebraic closure $\bar{k}$ and square roots $\sqrt{a}, \sqrt{b}, \sqrt{c} \in \bar{k}$. We claim that $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are linearly independent over k. Indeed, if $\lambda\sqrt{a} + \mu\sqrt{b} + \nu\sqrt{c} = 0$ with $\lambda, \mu, \nu \in k$ not all zero, squaring (characteristic 2) gives $a\lambda^2 + b\mu^2 + c\nu^2 = 0$, i.e. $(\lambda : \mu : \nu) \in C(k)$. Put $\rho = \sqrt{a}/\sqrt{b}$ and $\delta = \sqrt{c}/\sqrt{b}$ (so $\rho^2 = a/b$, $\delta^2 = c/b$). Then $1, \rho, \delta$ are linearly independent over k (divide the relation by $\sqrt{b}$). Let $k_0 = k(\rho,\delta) = k[\rho,\delta]$, a finite (purely inseparable) extension of k.

Let $X = x/z$, $Y = y/z$. Then

$$A := \mathcal{O}_C(D_+(z)) = k[X,Y]/(aX^2 + bY^2 + c) = k[X,Y]/(Y^2 + \alpha X^2 + \gamma), \qquad \alpha = \rho^2,\ \gamma = \delta^2.$$

A is a free $k[X]$-module with basis $1, Y$. Consider the k-algebra homomorphism

$$\iota : A \to k_0[X], \qquad X \mapsto X,\ Y \mapsto \theta := \rho X + \delta;$$

it is well defined since $\theta^2 = \alpha X^2 + \gamma$. Choose a k-basis $e_1 = 1, e_2 = \rho, e_3 = \delta, e_4, \dots$ of k_0. Then $k_0[X] = \bigoplus_i e_i k[X]$, and the e_i are also linearly independent over $k(X)$ in $k_0(X)$ (indeed $k_0 \otimes_k k(X)$ is a localization of $k_0[X]$, hence a subring of $k_0(X)$). For $r, s \in k(X)$,

$$r + s\theta = r \cdot e_1 + sX \cdot e_2 + s \cdot e_3 \qquad (*)$$

has coordinates $(r, sX, s, 0, \dots)$. In particular $\iota(r + sY) = 0$ forces $r = s = 0$ for $r, s \in k[X]$: ι is injective. So A is a domain, identified with $k[X] + k[X]\theta \subseteq k_0[X]$, and $K(C) = \mathrm{Frac}(A) = k(X) + k(X)\theta \subseteq k_0(X)$.

A is the integral closure of $k[X]$ in $K(C)$. A is finite over $k[X]$, so it is contained in this integral closure A'. Conversely let $u = r + s\theta \in K(C)$ $(r, s \in k(X))$ be integral over $k[X]$. Then u is integral over the normal domain $k_0[X]$ (a PID), whose fraction field contains u, so $u \in k_0[X]$. By $(*)$, the coordinates of u with respect to the basis (e_i) over $k(X)$ are r and s (on e_1 and e_3), and they lie in $k[X]$ since $u \in k_0[X] = \bigoplus e_i k[X]$. Hence $u = r + s\theta \in A$. So $A = A'$.

Normality and regularity of C. F is irreducible (a reducible F would be a product of two linear forms, and a line has a rational point), so C is integral, and the above shows that $\mathcal{O}_C(D_+(z))$ is a normal domain. The hypotheses ($C(k) = \emptyset$, $\mathrm{char}\, k = 2$, F diagonal) are symmetric in x, y, z (permuting the variables permutes a, b, c), so the same holds for $D_+(x)$ and $D_+(y)$, which together with $D_+(z)$ cover C. Hence C is a normal scheme. A normal Noetherian local ring of dimension 1 is a discrete valuation ring (Prop. 1.12), and the generic point is regular (a field). So every local ring of C is regular: C is regular. (By (1), C is nowhere smooth: it is a regular, non-smooth conic, cf. Rem. 3.34.) □

Exercise 3.23

> **Problem.** Let k be a field and let $F(x, y, z) \in k[x, y, z]$ be a homogeneous polynomial of degree 2 defining a smooth conic $V_+(F)$ over k. (a) If $\mathrm{char}(k) \neq 2$, show that after a change of variables, one can write $F = ax^2 + by^2 + cz^2$ for some $a, b, c \in k^*$. Let us suppose moreover that k is finite with q elements. Show that the sets $\{ax^2 \mid x \in k\}$ and $\{-c - by^2 \mid y \in k\}$ both have $(q + 1)/2$ elements. Deduce from this that there exists a rational point in $V_+(F)$. (b) Let us suppose that $\mathrm{char}(k) = 2$ and that k is perfect. Show that after a suitable change of variables, one has $F = xy + z^2$. Deduce from this that $V_+(F)$ contains a rational point.

Solution. (Here $F \neq 0$, since the zero polynomial does not define a conic; a smooth conic is also nonempty over $\bar{k}$.)

(a) By Exercise 3.22(c), after a linear change of variables $F = ax^2 + by^2 + cz^2$, and since the conic is smooth, $abc \neq 0$ (Exercise 3.22(c)).

Let now $k = \mathbb{F}_q$, q odd. The squaring map $k^* \to k^*$ is a 2-to-1 homomorphism (its kernel is $\{\pm 1\}$, of order 2 as q is odd), so k^* contains $(q - 1)/2$ nonzero squares, and k contains $(q - 1)/2 + 1 = (q + 1)/2$ squares (including 0). The map $x \mapsto ax^2$ has as image $a \cdot \{\text{squares}\}$, which has $(q+1)/2$ elements since $a \neq 0$. Likewise $y \mapsto -c - by^2$ has image $-c - b \cdot \{\text{squares}\}$, the image of the set of squares under the bijection $t \mapsto -c - bt$ of k, hence of cardinality $(q + 1)/2$. The two sets are subsets of k, which has q elements, and $\frac{q+1}{2} + \frac{q+1}{2} = q + 1 > q$. So they intersect: there are $x_0, y_0 \in k$ with $ax_0^2 = -c - by_0^2$, i.e. $F(x_0, y_0, 1) = 0$. The point $(x_0 : y_0 : 1)$ is a rational point of $V_+(F)$ (for the new coordinates; hence also for the original ones, by the linear change of variables).

(b) Let $\operatorname{char} k = 2$, k perfect, $F = ax^2 + by^2 + cz^2 + uxy + vxz + wyz$. By the computation of Exercise 3.22(d), the singular points of $V_+(F)_{\bar{k}}$ are the points of $V_+(F)(\bar{k})$ at which $uy + vz$, $ux + wz$, $vx + wy$ vanish (Lemma A of Exercise 3.21).

Step 1: $(u, v, w) \neq 0$. If $u = v = w = 0$, all three partials vanish identically and every point of the nonempty $V_+(F)_{\bar{k}}$ is singular, contradicting smoothness.

Step 2: the point p. By Step 1 and the computation in Exercise 3.22(d), the partials have exactly one common zero $p = (w : v : u) \in \mathbb{P}^2(k)$ in $\mathbb{P}^2(\bar{k})$. Since $V_+(F)$ is smooth, $p \notin V_+(F)$, i.e. $F(p) \neq 0$.

Step 3: normal form. Choose a linear change of variables such that $p = (0 : 0 : 1)$. The partial derivatives of a form transform contragrediently under linear changes of variables, so the property "all partial derivatives vanish at p" is invariant under linear changes of variables; in the new coordinates F is still smooth, $F(p) \neq 0$, and all three partials vanish at p. Write the new form again as $F = a'x^2 + b'y^2 + c'z^2 + u'xy + v'xz + w'yz$. Then $\frac{\partial F}{\partial x}(p) = v'$ and $\frac{\partial F}{\partial y}(p) = w'$ vanish, so $v' = w' = 0$ and $c' = F(p) \neq 0$. Also $u' \neq 0$, because otherwise the new form would be diagonal, and by Step 1 (applied in the new coordinates) it would not be smooth. Thus

$$F = a'x^2 + u'xy + b'y^2 + c'z^2, \qquad u'c' \neq 0.$$

Since k is perfect, every element of k is a square. Replace z by $\sqrt{c'}\, z$ to get $c' = 1$. Then

$$z^2 + a'x^2 + b'y^2 = (z + \sqrt{a'}\, x + \sqrt{b'}\, y)^2 \quad \text{(characteristic 2)},$$

and the substitution $z_1 = z + \sqrt{a'}x + \sqrt{b'}y$ is invertible. In the variables x, y, z_1 we get $F = u'xy + z_1^2$, and putting $x_1 = u'x$, $F = x_1 y + z_1^2$. This is the form $xy + z^2$ in the new variables.

A rational point. In these coordinates $(1 : 0 : 0)$ satisfies $xy + z^2 = 0$. Hence $V_+(F)$ has a rational point (the corresponding point of the original coordinates). □

Exercise 3.24

> **Problem.** Let X be an algebraic variety over a field k. Show that the following properties are equivalent: (i) X is smooth; (ii) for every finite extension l of k, X_l is regular; (iii) for every purely inseparable extension l of k, X_l is regular; (iv) for every (not necessarily algebraic) extension l of k, X_l is regular.

Solution. We first prove a lemma on regular varieties over algebraically closed fields.

Lemma B. Let $k_1 \subseteq k_2$ be algebraically closed fields and Z a regular algebraic variety over k_1. Then Z_{k_2} is regular. (This is a special case of the stability of smoothness under base change, Prop. 3.38, whose proof relies on exactly this fact.)

Proof. Regularity is local, so we may assume Z affine, $Z = V(I) \subseteq \mathbb{A}^n_{k_1}$. As Z is regular, its local rings are domains (Prop. 2.11), so its irreducible components are pairwise disjoint; hence $Z = Z_1 \sqcup \cdots \sqcup Z_s$ with Z_j open and closed in Z, integral, of dimension d_j. Each Z_j is affine, $Z_j = V(I_j) \subseteq \mathbb{A}^n_{k_1}$ with I_j prime, generated by $G_{j1}, \ldots, G_{jm}$; and $Z_{k_2} = \bigsqcup (Z_j)_{k_2}$, $(Z_j)_{k_2} = V(I_j k_2[T]) \subseteq \mathbb{A}^n_{k_2}$. Fix j and write $d = d_j$, $G_i = G_{ji}$, $M = (\partial G_i / \partial T_l)$, an $m \times n$ matrix over $k_1[T]$.

For a closed point $w \in Z_j$ (a k_1-rational point) we have $\dim \mathcal{O}_{Z_j,w} = d$ (Prop. 2.5.23(a)). By Prop. 2.5, $\dim T_{Z_j,w} = n - \operatorname{rank} M(w)$, and $\dim T_{Z_j,w} \geq \dim \mathcal{O}_{Z_j,w} = d$ (Prop. 2.2(a)), so $\operatorname{rank} M(w) \leq n-d$ with equality if and only if Z_j is regular at w (Thm. 2.19). Let δ be the ideal of $k_1[T]/I_j$ generated by the minors of M of order $n-d$. Since Z_j is regular at all its closed points, δ is contained in no maximal ideal, so $\delta = (1)$: there are $c_\alpha \in k_1[T]$ and minors Δ_α of order $n-d$ with $1 = \sum_\alpha c_\alpha \Delta_\alpha$ in $k_1[T]/I_j$. This identity holds a fortiori in $k_2[T]/I_j k_2[T]$. Hence for every k_2-point w' of $(Z_j)_{k_2}$ some minor of order $n-d$ of $M(w')$ is nonzero, i.e. $\operatorname{rank} M(w') \geq n-d$, so $\dim T_{(Z_j)_{k_2},w'} = n - \operatorname{rank} M(w') \leq d$.

On the other hand Z_j is integral over the algebraically closed field k_1, hence geometrically integral (cf. Exercise 3.2.14), so $(Z_j)_{k_2}$ is integral; its function field is generated over k_2 by a transcendence basis of $K(Z_j)/k_1$ and an algebraic extension of it, so $\dim (Z_j)_{k_2} = \operatorname{trdeg}_{k_2} = d$. Hence $\dim \mathcal{O}_{(Z_j)_{k_2},w'} = d$ for every closed point w' (Prop. 2.5.23(a)). We get $d = \dim \mathcal{O} \leq \dim T_{w'} \leq d$, so $(Z_j)_{k_2}$ is regular at w'. The closed points of $(Z_j)_{k_2}$ are the k_2-points (k_2 is algebraically closed), so $(Z_j)_{k_2}$ is regular by Cor. 2.17. □

(i)⇒(iv). Let X be smooth, so $X_{\bar{k}}$ is regular, and let $l \supseteq k$ be any extension. Choose an algebraic closure $\bar{l}$ of l and a k-embedding $\bar{k} \hookrightarrow \bar{l}$. Then $X_{\bar{l}} = (X_{\bar{k}})_{\bar{l}}$ is regular by Lemma B. Thus X_l is smooth over l (Def. 3.28), hence regular by Cor. 3.32.

(iv)⇒(ii) and (iv)⇒(iii) are trivial (finite and purely inseparable extensions are extensions).

(iii)⇒(i). Let $l \subseteq \bar{k}$ be the perfect closure of k (the union of the k^{1/p^e} if $\operatorname{char} k = p > 0$, and $l = k$ if $\operatorname{char} k = 0$). It is a purely inseparable extension of k and a perfect field. By (iii), X_l is a regular algebraic variety over the perfect field l, hence smooth over l (Cor. 3.33). Since $\bar{k}$ is an algebraic closure of l, $X_{\bar{k}} = (X_l)_{\bar{k}}$ is regular, i.e. X is smooth.

(ii)⇒(i). Assume (ii). Since smoothness means regularity of $X_{\bar{k}}$, which is local on X, we may assume $X = V(F_1, \ldots, F_r) \subseteq \mathbb{A}^n_k$ affine. By Cor. 2.17 it suffices to show that $X_{\bar{k}}$ is regular at its closed points $x' = (t_1, \ldots, t_n) \in \bar{k}^n$. Let $l = k(t_1, \ldots, t_n)$, a finite extension of k, and w the l-rational point $(t_1, \ldots, t_n)$ of X_l, so x' lies over w. By (ii), X_l is regular at w, so by the Jacobian criterion (Thm. 2.19) $\operatorname{rank} J_w = n - \dim \mathcal{O}_{X_l,w}$. The morphism $X_{\bar{k}} \to X_l$ is flat and its fiber over w is $\operatorname{Spec}(l \otimes_l \bar{k}) = \operatorname{Spec} \bar{k}$, of dimension 0, so $\dim \mathcal{O}_{X_{\bar{k}},x'} = \dim \mathcal{O}_{X_l,w}$ by Thm. 3.12. The Jacobian matrices of $X_{\bar{k}}$ at x' and of X_l at w are the same matrix (entries in $l \subseteq \bar{k}$). So $\operatorname{rank} J_{x'} = n - \dim \mathcal{O}_{X_{\bar{k}},x'}$, and $X_{\bar{k}}$ is regular at x' by Thm. 2.19. □

Exercise 3.25

> **Problem.** Let $f : X \to Y$ be a morphism of schemes, let $x \in X$, and $y = f(x)$. (a) Show that we have an exact sequence of $k(x)$-vector spaces: $0 \to T_{X_y,x} \to T_{X,x} \to T_{Y,y} \otimes_{k(y)} k(x)$. (b) Let us suppose that f admits a section $g : Y \to X$, and that $x = g(y)$. Show that $k(x) = k(y)$, and that we have a decomposition $T_{X,x} \simeq T_{X_y,x} \oplus T_{Y,y}$. (c) Let us suppose that X, Y are locally Noetherian and regular, and that f is of finite type. Show that under the hypothesis of (b), X_y is smooth at x. (d) Let us keep the hypothesis above. Let $d = \dim_x X_y$. Show that we have an isomorphism $\widehat{\mathcal{O}}_{X,x} \simeq \widehat{\mathcal{O}}_{Y,y}[[T_1, \ldots, T_d]]$, and that f is smooth at x.

Solution. Put $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$, with maximal ideals $\mathfrak{m}_y, \mathfrak{m}_x$ and residue fields $k(y), k(x)$, and let $\mathfrak{n} = \mathfrak{m}_x/\mathfrak{m}_y B$ be the maximal ideal of $\mathcal{O}_{X_y,x} = B/\mathfrak{m}_y B$.

(a) We have $\mathfrak{n} = \mathfrak{m}_x/\mathfrak{m}_y B$ and $\mathfrak{n}^2 = (\mathfrak{m}_x^2 + \mathfrak{m}_y B)/\mathfrak{m}_y B$, so $\mathfrak{n}/\mathfrak{n}^2 = \mathfrak{m}_x/(\mathfrak{m}_x^2 + \mathfrak{m}_y B)$. Hence the sequence of $k(x)$-vector spaces

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \otimes_{k(y)} k(x) \xrightarrow{\alpha} \mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \mathfrak{n}/\mathfrak{n}^2 \longrightarrow 0$$

(where $\alpha(a \otimes \lambda) = \lambda f^\#(a)$) is exact: the image of α is $(\mathfrak{m}_y B + \mathfrak{m}_x^2)/\mathfrak{m}_x^2$ and the cokernel is $\mathfrak{m}_x/(\mathfrak{m}_x^2 + \mathfrak{m}_y B) = \mathfrak{n}/\mathfrak{n}^2$. The functor $\mathrm{Hom}_{k(x)}(-, k(x))$ is exact, so the dual sequence

$$0 \to T_{X_y,x} \to T_{X,x} \xrightarrow{\alpha^\vee} \mathrm{Hom}_{k(x)}\big(\mathfrak{m}_y/\mathfrak{m}_y^2 \otimes_{k(y)} k(x), k(x)\big) = \mathrm{Hom}_{k(y)}(\mathfrak{m}_y/\mathfrak{m}_y^2, k(x))$$

is exact. The right-hand side contains $T_{Y,y} \otimes_{k(y)} k(x)$ (equality if $\dim_{k(y)} \mathfrak{m}_y/\mathfrak{m}_y^2 < \infty$, e.g. if Y is locally Noetherian), and $\alpha^\vee$ is the tangent map $T_{f,x}$ of Def. 2.1. (In general the target of $T_{f,x}$ in Def. 2.1 is to be read as $\mathrm{Hom}_{k(y)}(\mathfrak{m}_y/\mathfrak{m}_y^2, k(x))$; the exercise is stated with $T_{Y,y} \otimes k(x)$, which is the same space in the Noetherian situation of (b)–(d).) This is the required exact sequence.

(b) Let $g^\# : k(x) = k(g(y)) \to k(y)$ and $f^\# : k(y) \to k(x)$ be the maps induced by g and f on residue fields. Since $f \circ g = \mathrm{id}_Y$, we have $g^\# \circ f^\# = \mathrm{id}_{k(y)}$. The homomorphism $g^\#$ of fields is injective, and surjective by this relation, so it is an isomorphism; hence $f^\# = (g^\#)^{-1}$ and $k(x) = k(y)$ (via $f^\#$). We identify the two fields; then $T_{Y,y} \otimes_{k(y)} k(x) = T_{Y,y}$.

By (a), $0 \to T_{X_y,x} \to T_{X,x} \xrightarrow{T_{f,x}} T_{Y,y}$ is exact. Applying Prop. 2.2(b) to $f \circ g = \mathrm{id}_Y$ gives $T_{f,x} \circ T_{g,y} = \mathrm{id}_{T_{Y,y}}$ (the identification $k(x) = k(y)$ being used for the base change of scalars). So $T_{f,x}$ is surjective and has the section $T_{g,y}$, and the exact sequence $0 \to T_{X_y,x} \to T_{X,x} \to T_{Y,y} \to 0$ splits: $T_{X,x} \simeq T_{X_y,x} \oplus T_{Y,y}$.

(c) Now X, Y are locally Noetherian and regular, f is of finite type, so X_y is an algebraic variety over $k(y)$. Since $k(x) = k(y)$, the point x is a rational point of X_y, hence closed, and $k(x)$ is separable over $k(y)$. By Prop. 3.30, X_y is smooth at x if and only if it is regular at x. By (b), $\dim T_{X_y,x} = \dim T_{X,x} - \dim T_{Y,y}$, and since X, Y are regular at x, y this equals $\dim B - \dim A$. By Thm. 3.12, $\dim \mathcal{O}_{X_y,x} \geq \dim B - \dim A$, and by Prop. 2.2(a), $\dim \mathcal{O}_{X_y,x} \leq \dim T_{X_y,x}$. Hence

$$\dim B - \dim A \leq \dim \mathcal{O}_{X_y,x} \leq \dim T_{X_y,x} = \dim B - \dim A,$$

so $\dim \mathcal{O}_{X_y,x} = \dim T_{X_y,x}$: X_y is regular at x, hence smooth at x. (Here f need not be flat.)

(d) Let $d = \dim \mathcal{O}_{X_y,x} = \dim B - \dim A$ (by (c) the fiber ring is regular, of dimension $\dim T_{X_y,x}$), $e = \dim A$. Choose $a_1, \ldots, a_e$ generating $\mathfrak{m}_y$. Since $\mathcal{O}_{X_y,x}$ is regular of dimension d, $\mathfrak{n}$ is generated by d elements $\bar{t}_1, \ldots, \bar{t}_d$; lift them to $t_1, \ldots, t_d \in \mathfrak{m}_x$. Then $\mathfrak{m}_x = \mathfrak{m}_y B + (t_1, \ldots, t_d) = (f^\# a_1, \ldots, f^\# a_e, t_1, \ldots, t_d)$, with $e + d = \dim B$ generators; so $(f^\# a_i, t_j)$ is a coordinate system of B.

The local homomorphism $f^\# : A \to B$ induces $\widehat{A} \to \widehat{B}$ (continuity), and $\widehat{B}$ is complete, so there is a well defined homomorphism

$$\varphi : \ P := \widehat{A}[[T_1, \ldots, T_d]] \longrightarrow \widehat{B}, \qquad T_j \mapsto t_j,$$

(the series $\sum a_\alpha t^\alpha$ converge since $t_j \in \mathfrak{m}_x \widehat{B}$). P is a Noetherian complete local ring (Prop. 1.3.7) with maximal ideal $(\mathfrak{m}_y \widehat{A}, T_1, \ldots, T_d)$, generated by $e + d$ elements.

φ is surjective. Since $k(x) = k(y)$ we have $B = f^\#(A) + \mathfrak{m}_x$, and since $\mathfrak{m}_x = (f^\# a_i, t_j)$ we have $\mathfrak{m}_x^n = \varphi_0((\mathfrak{m}_y, T)^n) + \mathfrak{m}_x^{n+1}$ for all n, where $\varphi_0 : A[T_1, \ldots, T_d] \to B$ is the polynomial version

of φ. Hence $A[T]/(\mathfrak{m}_y, T)^n \to B/\mathfrak{m}_x^n$ is surjective for all n. Passing to the limit and using Lemma 1.3.3 exactly as in the proof of Prop. 2.27 (note $\varprojlim_n A[T]/(\mathfrak{m}_y, T)^n = \widehat{A}[[T]]$), the map $\varphi : \widehat{A}[[T]] \to \widehat{B}$ is surjective.

φ is injective. By Lemma 2.26, $\widehat{B}$ is regular of dimension $e+d$, hence a domain (Prop. 2.11); so $q = \operatorname{Ker}\varphi$ is a prime ideal of P and $P/q \simeq \widehat{B}$ has dimension $e+d$. On the other hand $\dim P \leq e+d$, because the maximal ideal of P is generated by $e+d$ elements (Cor. 2.5.14(b)). As $\dim P \geq \dim P/q = e+d$, P is a regular local ring of dimension $e+d$, hence a domain, and the equality $\dim P/q = \dim P$ forces $q = 0$ (a nonzero prime in a Noetherian local domain strictly lowers the dimension). Thus φ is an isomorphism:

$$\widehat{\mathcal{O}}_{X,x} \simeq \widehat{\mathcal{O}}_{Y,y}[[T_1, \ldots, T_d]].$$

f is smooth at x. By Def. 3.35, we need (1) f flat at x and (2) X_y smooth at x over $k(y)$. (2) is (c). For (1): $\widehat{A}$ is flat over A (Thm. 1.3.15), and $\widehat{A}[[T]]$ is flat over $\widehat{A}$ (it is the (T)-adic completion of the Noetherian ring $\widehat{A}[T]$, which is free over $\widehat{A}$, and completion is flat by Thm. 1.3.15). Hence $\widehat{B} \simeq \widehat{A}[[T]]$ is flat over A. Let $N' \to N$ be an injection of A-modules, L the kernel of $B \otimes_A N' \to B \otimes_A N$. Tensoring with $\widehat{B}$ over B (flat, Thm. 1.3.15), $\widehat{B} \otimes_B L$ is the kernel of $\widehat{B} \otimes_A N' \to \widehat{B} \otimes_A N$, which is 0. Since $\widehat{B}$ is faithfully flat over B (Thm. 1.3.16(b)), $L = 0$. So B is flat over A, i.e. f is flat at x, and f is smooth at x. □

Exercise 3.26

> **Problem.** Let X be a scheme of finite type over a locally Noetherian regular scheme S. Let $x \in X$. (a) Show that in a neighborhood of x, there exists a closed immersion $i : X \to Z$ into a scheme Z that is of finite type over S and regular at x (locally embed X into an affine space over S and use Theorem 3.36). (b) Using exact sequence (2.1) in the proof of Proposition 2.5, show that one can successively reduce the dimension of $T_{Z,i(x)}$ in such a way that the canonical map $T_{X,x} \to T_{Z,i(x)}$ is an isomorphism.

Solution. (a) The statement is local on X. Let $s \in S$ be the image of x, $V = \operatorname{Spec} R$ an affine open neighborhood of s in S (R is Noetherian and regular), and $U = \operatorname{Spec} R[T_1, \ldots, T_n]/I$ an affine open neighborhood of x in X lying over V (possible since X is of finite type over S). Let $Z = \mathbb{A}^n_V = \operatorname{Spec} R[T_1, \ldots, T_n]$ and $i : U \to Z$ the closed immersion defined by I. The scheme Z is of finite type over V, hence over S. The morphism $Z \to V$ is flat (the algebra $R[T]$ is free over R), of finite type, and its fibers $\mathbb{A}^n_{k(v)}$ are smooth over $k(v)$ (Def. 3.28 and the examples after it). So $Z \to V$ is smooth (Def. 3.35), and since V is regular, Z is regular by Thm. 3.36. In particular Z is regular at $i(x)$.

(b) Let $z = i(x)$, $\mathfrak{m}_z$ the maximal ideal of $\mathcal{O}_{Z,z}$ and $J \subseteq \mathfrak{m}_z$ the ideal of i at z, so $\mathcal{O}_{X,x} = \mathcal{O}_{Z,z}/J$ and $\mathfrak{m}_x = \mathfrak{m}_z/J$. As in the proof of Prop. 2.5 (exact sequence (2.1), which is valid in general), we have an exact sequence

$$0 \to (J + \mathfrak{m}_z^2)/\mathfrak{m}_z^2 \to \mathfrak{m}_z/\mathfrak{m}_z^2 \to \mathfrak{m}_x/\mathfrak{m}_x^2 \to 0.$$

Dualizing, $T_{X,x} \to T_{Z,z}$ is always injective, and it is an isomorphism if and only if $J \subseteq \mathfrak{m}_z^2$.

Suppose $J \not\subseteq \mathfrak{m}_z^2$ and choose $g \in J \setminus \mathfrak{m}_z^2$ (so $g \in \mathfrak{m}_z$, $g \neq 0$). Let $\mathcal{I} \subseteq \mathcal{O}_Z$ be the (quasi-coherent) ideal sheaf of the closed immersion i, so that $\mathcal{I}_z = J$. Choose an affine open neighborhood W

of z and a section $\tilde{g} \in \mathcal{I}(W)$ whose germ at z is g. Let $Z_1 = V(\tilde{g}) \subseteq W$, a closed subscheme of W, of finite type over S. Since $\tilde{g}$ lies in the ideal of i, the closed immersion i restricted to $i^{-1}(W)$ factors through a closed immersion $i_1 : i^{-1}(W) \to Z_1$; we replace X by the open neighborhood $i^{-1}(W)$ of x. We have $\mathcal{O}_{Z_1,z} = \mathcal{O}_{Z,z}/g\mathcal{O}_{Z,z}$. As $\mathcal{O}_{Z,z}$ is regular and $g \in \mathfrak{m}_z \setminus \mathfrak{m}_z^2$, $g \neq 0$, Cor. 2.12 shows that $\mathcal{O}_{Z_1,z}$ is regular, of dimension $\dim \mathcal{O}_{Z,z} - 1$. So Z_1 is of finite type over S and regular at z. Moreover, with $\mathfrak{m}_1 = \mathfrak{m}_z/(g)$, the exact sequence (2.1) for (g) gives $\dim \mathfrak{m}_1/\mathfrak{m}_1^2 = \dim \mathfrak{m}_z/\mathfrak{m}_z^2 - \dim((g) + \mathfrak{m}_z^2)/\mathfrak{m}_z^2 = \dim \mathfrak{m}_z/\mathfrak{m}_z^2 - 1$, i.e. $\dim T_{Z_1,z} = \dim T_{Z,z} - 1$, since $g \notin \mathfrak{m}_z^2$.

We repeat this with (Z_1, z) in place of (Z, z). Each step lowers $\dim T_{Z,z}$ by exactly 1, and this dimension is a nonnegative integer, so the process stops after at most $\dim T_{Z,z}$ steps. It stops only when $J \subseteq \mathfrak{m}_z^2$, i.e. when $T_{X,x} \to T_{Z,z}$ is an isomorphism. At the end we have a closed immersion of a neighborhood of x in X into a scheme Z of finite type over S, regular at x, inducing an isomorphism on tangent spaces at x. □

4.4 Zariski Main Theorem and applications

Results of the text are cited by their numbers in Liu. Those of earlier chapters carry the chapter in front: "Prop. 3.3.18" is Proposition 18 of Section 3.3, "Prop. 2.5.19" is Proposition 19 of Section 2.5. Exercise numbers in the text of a solution ("Ex. 4.2") refer to exercises of Section 4.4. An algebraic variety is a separated scheme of finite type over a field (not necessarily reduced), and "quasi-finite" means "of finite type with finite fibres" (Ex. 3.2.8).

Honest notes. Exercise 4.5 is false as printed (a cuspidal curve and a non-reduced scheme are counterexamples in characteristic 0); it is proved under the additional hypotheses that Y is normal and X is reduced with all components dominating Y, and the solution says exactly what is and is not proved. In Exercise 4.4 the algebra B must be nonzero. In Exercise 4.6 the induction does not close with the global fibre dimensions of the hint, so it is run with the local fibre dimension $\dim_x X_{f(x)}$ at a point; this is a slightly stronger statement and contains the one asked for. Exercises 4.1, 4.3, 4.4 and 4.6 use the finiteness theorem for projective morphisms (Thm. 5.3.2, in the form of Rem. 3.3.20: $\mathcal{O}_X(X)$ is finite over a Noetherian ring A when X is proper over A), through Thm. 4.4, Cor. 4.8 or Lemma 4.13; the book itself uses this theorem "in anticipation" at these places. Exercise 4.2 does not use it.

Fact (J). *Let S be a scheme of finite type over a field κ. Every nonempty locally closed subset $L \subseteq S$ contains a point which is closed in S.* Indeed, L with its reduced structure is a scheme of finite type over κ; let $\operatorname{Spec} C \subseteq L$ be a nonempty affine open subset of L and $\mathfrak{m}$ a maximal ideal of C. By the Nullstellensatz $C/\mathfrak{m}$ is finite over κ, so the corresponding point u has $k(u)$ algebraic over κ, hence is closed in S (Ex. 2.5.9). Two consequences: a closed subset of S containing all closed points of S is S; and if the underlying set of S is finite, then every point of S is closed (a point of a finite T_0 space is locally closed, so Fact (J) applies to it).

Exercise 4.1

> **Problem.** Let $f : X \to Y$ be a projective morphism of algebraic varieties over an algebraically closed field k. Let us suppose that f is injective and $T_{f,x}$ is injective for every

closed point $x \in X$. Show that f is a finite morphism, and, using Nakayama's lemma, show that f is a closed immersion.

Solution. Only the injectivity of f on closed points is used. For a closed point x of X we have $k(x) = k$ (Ex. 2.5.9 and $k = \overline{k}$). Put $y = f(x)$; then $k \subseteq k(y) \subseteq k(x) = k$, so $k(y) = k$ and y is closed (Ex. 2.5.9). The tangent map $T_{f,x} : T_{X,x} \to T_{Y,y} \otimes_{k(y)} k(x) = T_{Y,y}$ is the k-dual of the map $\mathfrak{m}_y/\mathfrak{m}_y^2 \to \mathfrak{m}_x/\mathfrak{m}_x^2$ induced by $f^{\#}$; these are finite-dimensional vector spaces, so $T_{f,x}$ is injective if and only if $\mathfrak{m}_y/\mathfrak{m}_y^2 \to \mathfrak{m}_x/\mathfrak{m}_x^2$ is surjective.

Step 1: the fibre over a closed point y is empty or a single point. The fibre $X_y = f^{-1}(y)$ is closed in X. If it is nonempty it contains a point closed in X by Fact (J). Two distinct closed points of X cannot both lie over y, since f is injective on closed points; so X_y contains exactly one closed point x_0 of X. If $X_y \neq \{x_0\}$, then $X_y \setminus \{x_0\}$ is locally closed in X and nonempty, hence contains a point closed in X by Fact (J), which is a second closed point over y, a contradiction. So $X_y = \{x_0\}$.

Step 2: f is finite. Let $y \in Y$ be a closed point. By Step 1, X_y is finite, so by Ex. 4.2 below (whose proof does not use the present exercise) there is an affine open neighbourhood V_y of y such that $f^{-1}(V_y) \to V_y$ is finite. Let $W = \bigcup_y V_y$ over all closed points y. Then $Y \setminus W$ is closed and contains no closed point of Y, hence is empty by Fact (J). Thus the affine open sets V_y cover Y and $f^{-1}(V_y)$ is affine and finite over V_y; this implies that f is finite (Ex. 3.3.15).

Step 3: f is a closed immersion. Let $V = \operatorname{Spec} A$ be an affine open subset of Y. Since f is finite, $f^{-1}(V) = \operatorname{Spec} B$ with B a finite A-module via $\varphi : A \to B$. We show that φ is surjective. The A-module $M = B/\varphi(A)$ is finite. If $M \neq 0$ there is a maximal ideal $\mathfrak{m}$ of A with $M_{\mathfrak{m}} \neq 0$. The corresponding point y is closed in V, hence has $k(y) = A/\mathfrak{m}$ finite over k, so $k(y) = k$ and y is closed in Y. We show $M_{\mathfrak{m}} = 0$, a contradiction.

The ring $B_{\mathfrak{m}} = B \otimes_A A_{\mathfrak{m}}$ is a finite $A_{\mathfrak{m}}$-module with $B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}} = B/\mathfrak{m}B = \mathcal{O}(X_y)$. If $X_y = \emptyset$ then $B_{\mathfrak{m}} = 0$ by Nakayama's lemma and $M_{\mathfrak{m}} = 0$. Otherwise $X_y = \{x\}$ by Step 1. The maximal ideals of the finite $A_{\mathfrak{m}}$-algebra $B_{\mathfrak{m}}$ are the primes lying over $\mathfrak{m}A_{\mathfrak{m}}$, i.e. the points of X_y; so $B_{\mathfrak{m}}$ is local with maximal ideal corresponding to x, and $B_{\mathfrak{m}} = \mathcal{O}_{X,x}$. Since $T_{f,x}$ is injective, $\mathfrak{m}_x = \mathfrak{m}_y\mathcal{O}_{X,x} + \mathfrak{m}_x^2$. Let $Q = \mathfrak{m}_x/\mathfrak{m}_y\mathcal{O}_{X,x}$, a finitely generated module over the Noetherian local ring $\mathcal{O}_{X,x}$ (varieties are Noetherian). The equality gives $Q = \mathfrak{m}_x Q$, hence $Q = 0$ by Nakayama, i.e. $\mathfrak{m}_x = \mathfrak{m}_y\mathcal{O}_{X,x}$. Therefore

$$B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}} = \mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x} = \mathcal{O}_{X,x}/\mathfrak{m}_x = k(x) = k,$$

so $A_{\mathfrak{m}} \to B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}}$ is surjective, i.e. $B_{\mathfrak{m}} = \varphi(A_{\mathfrak{m}}) + \mathfrak{m}B_{\mathfrak{m}}$. Thus $M_{\mathfrak{m}} = \mathfrak{m}M_{\mathfrak{m}}$, and $M_{\mathfrak{m}} = 0$ by Nakayama's lemma applied to the finite $A_{\mathfrak{m}}$-module $M_{\mathfrak{m}}$.

Hence $A \to B$ is surjective for every affine open $V \subseteq Y$, which says that f is a closed immersion. □

Exercise 4.2

Problem. Let $f : X \to Y$ be a projective morphism. Let $y \in Y$ be such that X_y is finite. We want to show that there exists a neighborhood V of y such that $f^{-1}(V) \to V$ is finite, without using Zariski's "Main Theorem". We may suppose Y is affine. (a) Let us fix an

embedding $X \to \mathbb{P}^n_Y$. Show that there exists a non-empty principal closed subset $V_+(P)$ of $\mathbb{P}^n_Y$ such that $V_+(P) \cap X_y = \emptyset$ (use Prop. 3.3.36). (b) Show that there exists an affine neighborhood V of y such that $X \cap V_+(P) \cap \mathbb{P}^n_V = \emptyset$. (c) Show that $f^{-1}(V) = X \cap D_+(P) \cap \mathbb{P}^n_V$ and that $f^{-1}(V)$ is affine. Conclude using Prop. 3.3.18.

Solution. Let $Y = \operatorname{Spec} A$, with A any ring, and let $X \subseteq \mathbb{P}^n_Y = \operatorname{Proj} A[T_0, \dots, T_n]$ be a closed subscheme, f being the restriction of the projection $\pi : \mathbb{P}^n_Y \to Y$. Let $\mathfrak{p} \subset A$ be the prime of y and $\kappa = k(y) = \operatorname{Frac}(A/\mathfrak{p})$. The fibre of $\mathbb{P}^n_Y$ over y is $\mathbb{P}^n_\kappa$, and $F := X_y$ is a finite subset of it.

(a) The points of F correspond to finitely many homogeneous primes $\mathfrak{q}_1, \dots, \mathfrak{q}_r$ of $\kappa[T_0, \dots, T_n]$ not containing the irrelevant ideal $I_+ = (T_0, \dots, T_n)$. By the argument of Prop. 3.3.36(a) with $Z = \emptyset$ (homogeneous prime avoidance, Ex. 2.5.5) there is a homogeneous element $g \in I_+$ lying in no $\mathfrak{q}_i$ (if $r = 0$ take $g = T_0$). Then $e := \deg g \geq 1$ and $V_+(g) \cap F = \emptyset$. Multiplying g by a nonzero element of κ we may assume that its coefficients lie in the image of $A/\mathfrak{p}$, and we lift it to a homogeneous polynomial $P \in A[T_0, \dots, T_n]$ of degree e. The image of P in $\kappa[T_0, \dots, T_n]$ is a nonzero multiple of g, so $V_+(P) \cap \pi^{-1}(y) = V_+(g)$ and therefore $V_+(P) \cap X_y = \emptyset$. (If $n \geq 1$ the set $V_+(P)$ is nonempty because $\deg P \geq 1$; nonemptiness plays no role below.)

(b) The set $C = X \cap V_+(P)$ is closed in $\mathbb{P}^n_Y$, because X is closed. The morphism π is proper (Thm. 3.3.30, in particular closed), so $\pi(C)$ is closed in Y. By (a), $y \notin \pi(C)$. Choose $a \in A$ with $y \in D(a) \subseteq Y \setminus \pi(C)$ and put $V = D(a)$, an affine neighbourhood of y. Then $C \cap \pi^{-1}(V) = \emptyset$, i.e. $X \cap V_+(P) \cap \mathbb{P}^n_V = \emptyset$.

(c) We have $f^{-1}(V) = X \cap \mathbb{P}^n_V$, and $\mathbb{P}^n_V = (\mathbb{P}^n_V \cap D_+(P)) \cup (\mathbb{P}^n_V \cap V_+(P))$. By (b) the second part does not meet X, so $f^{-1}(V) = X \cap D_+(P) \cap \mathbb{P}^n_V$. Since $\deg P \geq 1$, $D_+(P) = \operatorname{Spec}\big(A[T_0, \dots, T_n]_P\big)_0$ is affine, and $D_+(P) \cap \mathbb{P}^n_V$ is the principal open subset of $D_+(P)$ defined by the image of a, hence affine. The open subscheme $f^{-1}(V)$ of X is a closed subscheme of $\mathbb{P}^n_V$ contained in the open subscheme $D_+(P) \cap \mathbb{P}^n_V$, hence a closed subscheme of this affine scheme; so $f^{-1}(V) = \operatorname{Spec} B'$ is affine.

The morphism $f^{-1}(V) \to V$ is proper (restriction of the proper morphism f over the open V, Thm. 3.3.30 and Prop. 3.3.16), and of finite type (Cor. 3.3.10). Since $f^{-1}(V)$ is proper over $V = \operatorname{Spec} A_a$, the ring $B' = \mathcal{O}(f^{-1}(V))$ is integral over A_a by Prop. 3.3.18. An algebra which is of finite type and integral is finite. Hence $f^{-1}(V) \to V$ is finite. □

Exercise 4.3

Problem. Let Y be a locally Noetherian scheme, and let $f : X \to Y$ be a quasi-finite and quasi-projective morphism. Show that f factors into an open immersion $X \to Z$ followed by a finite morphism $Z \to Y$.

Solution. By Def. 3.3.35 there is a projective morphism $p : \overline{X} \to Y$ and an open immersion of finite type $X \to \overline{X}$ over Y; we regard X as an open subscheme of $\overline{X}$. The following proposition does not use quasi-finiteness and is also used in Ex. 4.4.

Proposition R. *There exist a finite Y-scheme $\pi : Z \to Y$, a Y-morphism $g : \overline{X} \to Z$ and an open subset $\Omega \subseteq \overline{X}$ such that $g|_\Omega : \Omega \to Z$ is an open immersion and Ω contains every point $x \in \overline{X}$ which is isolated in its fibre $\overline{X}_{p(x)}$. If $Y = \operatorname{Spec} A$ is affine (and Noetherian), $Z = \operatorname{Spec} \mathcal{O}_{\overline{X}}(\overline{X})$ and g is the canonical morphism.*

Proof. For an affine open $W \subseteq Y$ let $\mathcal{A}(W) = \mathcal{O}_{\overline{X}}(p^{-1}W)$. It is a finite $\mathcal{O}_Y(W)$-algebra (Rem. 3.3.20 and Thm. 5.3.2: $p^{-1}W$ is proper over the Noetherian ring $\mathcal{O}_Y(W)$ by Thm. 3.3.30). For $a \in \mathcal{O}_Y(W)$ we have $\mathcal{A}(D(a)) = \mathcal{A}(W)_a$ (flat base change, Prop. 3.1.24). Hence $\mathcal{A} = p_*\mathcal{O}_{\overline{X}}$ is a coherent sheaf of $\mathcal{O}_Y$-algebras, and we put $Z = \operatorname{Spec}_Y \mathcal{A}$, glued from the affine schemes $\operatorname{Spec} \mathcal{A}(W)$; it is finite over Y. The canonical morphisms $p^{-1}W \to \operatorname{Spec} \mathcal{O}(p^{-1}W)$ (Prop. 2.3.25) glue to $g : \overline{X} \to Z$. If Y is affine we simply have $Z = \operatorname{Spec} \mathcal{O}_{\overline{X}}(\overline{X})$.

Let $W \subseteq Y$ be an affine open subset; it is a Noetherian scheme. The morphism $p^{-1}W \to W$ is separated of finite type (Cor. 3.3.10) and $\mathcal{O}(p^{-1}W)$ is a finite $\mathcal{O}_Y(W)$-module. By Thm. 4.4, the set Ω_W of points of $p^{-1}W$ which are isolated in their fibres is open in $p^{-1}W$, and g induces an open immersion $\Omega_W \to \operatorname{Spec} \mathcal{A}(W) = \pi^{-1}(W)$. Let $\Omega = \bigcup_W \Omega_W$; since fibres of $p^{-1}W$ and of $\overline{X}$ coincide, $\Omega \cap p^{-1}W = \Omega_W$, and Ω is exactly the set of points isolated in their fibres. Being an open immersion is local on the target $Z = \bigcup \pi^{-1}(W)$, so $g|_\Omega$ is an open immersion. □

Quasi-finiteness gives $X \subseteq \Omega$. Let $x \in X$ and $y = f(x)$. The fibre X_y is a scheme of finite type over $k(y)$ (as f is of finite type) with finitely many points, so every point of X_y is closed in X_y (Fact (J)), i.e. X_y is a finite discrete space and $\{x\}$ is open in X_y. Now $X_y = X \cap \overline{X}_y$ is open in $\overline{X}_y$ because X is open in $\overline{X}$. So $\{x\}$ is open in $\overline{X}_y$, i.e. x is isolated in its fibre in $\overline{X}$. Hence $X \subseteq \Omega$.

Consequently $g|_X : X \to Z$ is an open immersion (the restriction of the open immersion $g|_\Omega$ to the open subscheme $X \subseteq \Omega$), and $\pi : Z \to Y$ is finite. Since $\pi \circ g = p$ and $X \to \overline{X}$ is over Y, the morphism f is the composition $X \to Z \to Y$. □

Exercise 4.4

> **Problem.** Let A be a discrete valuation ring, and let B be a finitely generated flat A-algebra. Show that there exists a subalgebra C of B with a finite injective homomorphism $A[T_1, \dots, T_d] \to C$, and such that the morphism $\operatorname{Spec} B \to \operatorname{Spec} C$ is an open immersion. This is a sort of generalization of Noether's normalization lemma. Show that in general we cannot have $C = B$.

Solution. We assume $B \neq 0$ (for $B = 0$ there is no injection $A[T_1, \dots, T_d] \to C \subseteq B$; the zero ring is flat and finitely generated, so the statement needs $B \neq 0$). Let t be a uniformizer, $K = \operatorname{Frac}(A) = A[1/t]$, $k = A/tA$, and $B_K = B[1/t] = B \otimes_A K$. Since B is flat, multiplication by t is injective on B, so B is torsion free and $B \to B_K$ is injective; in particular $B_K \neq 0$.

Lemma A. *There exist $s \geq 0$ and $z_1, \dots, z_s \in B$ such that (i) the K-algebra map $K[T_1, \dots, T_s] \to B_K$, $T_i \mapsto z_i$, is injective and makes B_K a finite module; (ii) B/tB is a finite module over $k[T_1, \dots, T_s]$ via $T_i \mapsto \overline{z_i}$.*

Proof. Induction on the number m of generators of the A-algebra $B = A[x_1, \dots, x_m]$ (all such $B \neq 0$ are flat, hence torsion free). Let J be the kernel of $A[X_1, \dots, X_m] \to B$. If $J = 0$, then B is a polynomial ring and $z_i = x_i$ ($s = m$) work. In particular this covers $m = 0$: then $B = A/J$ is torsion free and nonzero, so $J = 0$. Suppose $J \neq 0$, so $m \geq 1$. Take $0 \neq g \in J$ and write $g = t^e g_1$ with $g_1 \in A[X]$ having a coefficient which is a unit. Since t is a nonzerodivisor on B, $g_1 \in J$, and the reduction $\overline{g_1} \in k[X]$ is nonzero.

Choose an integer r larger than every exponent of every variable occurring in g_1, and put $y_i = x_i - x_m^{r^i}$ for $1 \leq i \leq m-1$. Let $B' = A[y_1, \dots, y_{m-1}] \subseteq B$, a subalgebra with $m-1$

generators, torsion free hence flat and nonzero. Consider

$$G(Y_1,\dots,Y_{m-1},X) = g_1(Y_1+X^r,\dots,Y_{m-1}+X^{r^{m-1}},X) \in A[Y][X].$$

For a monomial X^α the substitution gives $X^{w(\alpha)}+$(terms of lower degree in X with coefficients in $\mathbf{Z}[Y]$), where $w(\alpha)=\alpha_m+\alpha_1 r+\cdots+\alpha_{m-1}r^{m-1}$; the numbers $w(\alpha)$ are pairwise distinct for exponent vectors with entries $<r$ (base-r digits). Hence $G = cX^N +$ lower terms in X with coefficients in $A[Y]$, where $c\in A$ is the coefficient in g_1 of the monomial of maximal weight, $c\neq 0$. Likewise, the reduction $\overline{G}=\overline{g_1}(Y_i+X^{r^i},X)\in k[Y][X]$ equals $\bar{c}'X^{N'}+$ lower terms, with $\bar{c}'\in k^\times$ the coefficient of the monomial of maximal weight in $\overline{g_1}$.

Now $G(y,x_m)=g_1(x)=0$ in B, so in B_K the element x_m satisfies an equation with leading coefficient $c\in K^\times$ over $K[y]$: it is integral over $B'_K=B'\otimes K$. As $x_i=y_i+x_m^{r^i}$, $B_K=B'_K[x_m]$ is a finite B'_K-module. In B/tB we have $\overline{G}(\bar{y},\bar{x}_m)=0$ with leading coefficient $\bar{c}'\neq 0$, so $\bar{x}_m$ is integral over the subalgebra $\overline{B'}$ of B/tB generated by the $\bar{y}_i$, and $B/tB=\overline{B'}[\bar{x}_m]$ is a finite $\overline{B'}$-module.

By induction applied to B' there are $z_1,\dots,z_s\in B'\subseteq B$ with $K[T]\to B'_K$ injective and B'_K finite over $K[T]$, and B'/tB' finite over $k[T]$. Since $B'_K\subseteq B_K$, (i) holds for B. The ring $\overline{B'}$ is a quotient of B'/tB', so it is finite over the subalgebra generated by the $\bar{z}_i$; as B/tB is finite over $\overline{B'}$, (ii) holds. □

The map to affine space. Take $z_1,\dots,z_s$ as in Lemma A, put $d=s$, $R=A[T_1,\dots,T_d]$ and let $\phi: R\to B$ send T_i to z_i. It is injective: if $\phi(h)=0$, then $h=0$ in $K[T]$ by Lemma A(i) and $B\subseteq B_K$, so $h=0$. The morphism $\operatorname{Spec} B\to\operatorname{Spec} R$ is quasi-finite: for $q\in\operatorname{Spec} R$ with residue field $\kappa(q)$, if $t\in q$ then $B\otimes_R\kappa(q)=(B/tB)\otimes_{k[T]}\kappa(q)$ is finite-dimensional by Lemma A(ii); if $t\notin q$ then $B\otimes_R\kappa(q)=B_K\otimes_{K[T]}\kappa(q)$ is finite-dimensional by Lemma A(i). Thus all fibres are finite. Also R is Noetherian, and $\operatorname{Spec} B\to\operatorname{Spec} R$ is of finite type with affine source and target, hence quasi-projective (Ex. 3.3.20(b)).

The algebra C. By Ex. 4.3 (more precisely, by Proposition R and the argument there), there is a projective R-scheme $\overline{X}$ containing $X=\operatorname{Spec} B$ as an open subscheme, and, with $D=\mathcal{O}_{\overline{X}}(\overline{X})$ (a finite R-algebra), the canonical morphism $g: X\to\operatorname{Spec} D$ is an open immersion. It corresponds to the restriction homomorphism $\psi: D\to\mathcal{O}_X(X)=B$. Let $C=\psi(D)\subseteq B$. It is an R-subalgebra of B containing $\phi(R)$ and is finite over R as a quotient of D; and $R\to C$ is injective because $R\to C\subseteq B$ is ϕ. So $R=A[T_1,\dots,T_d]\to C$ is finite and injective.

It remains to see that $q:\operatorname{Spec} B\to\operatorname{Spec} C$ is an open immersion. The morphism g factors as $X\xrightarrow{q}\operatorname{Spec} C\xrightarrow{\iota}\operatorname{Spec} D$ with ι a closed immersion. Since g is an open immersion, $g(X)$ is open in $\operatorname{Spec} D$ and contained in the closed subset $\operatorname{Spec} C$, so q is a homeomorphism onto the open subset $g(X)$ of $\operatorname{Spec} C$. For $x\in X$ the stalk map $\mathcal{O}_{\operatorname{Spec} D,g(x)}\to\mathcal{O}_{X,x}$ is the composite of the surjection $\mathcal{O}_{\operatorname{Spec} D,g(x)}\to\mathcal{O}_{\operatorname{Spec} C,q(x)}$ and of $q_x^\#$; as it is an isomorphism, the surjection is also injective, hence bijective, and so $q_x^\#$ is an isomorphism. Therefore q is an open immersion.

We cannot take $C=B$ in general. Let $B=A[X]/(tX-1)\simeq K$. It is finitely generated over A, and torsion free, hence flat. Suppose $A[T_1,\dots,T_d]\to B=K$ were finite and injective. Every element of K is algebraic over A (it is a quotient a/t^n, a root of t^nT-a), so a homomorphism $A[T_1,\dots,T_d]\to K$ with $d\geq 1$ has a nonzero kernel; hence $d=0$. But $A\to K$ is not finite: if $K=a_1A+\cdots+a_rA$, write $a_i=u_i/t^n$ with $u_i\in A$; then $K\subseteq t^{-n}A$, which is false since $t^{-n-1}\notin t^{-n}A$. So $C=B$ is impossible. For this B the proof gives $d=0$ and $C=A$, and indeed $\operatorname{Spec} K=D(t)\subseteq\operatorname{Spec} A$ is open. □

Exercise 4.5

> **Problem.** Let Y be a locally Noetherian integral scheme, and let $f : X \to Y$ be a projective morphism with geometrically connected fibres. Show that for any affine open subset V of Y, there exists an $n \geq 1$ such that $(f_*\mathcal{O}_X(V))^{p^n}$ is contained in the image of $\mathcal{O}_Y(V) \to f_*\mathcal{O}_X(V)$, where p is the characteristic of $K(Y)$ (by convention, we set $p^n = 1$ if $p = 0$). In particular, if $\operatorname{char} K(Y) = 0$, then $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is an isomorphism.

Solution. The statement as printed is false, so we first give counterexamples and then prove a corrected version. Here "geometrically connected fibres" means that every fibre is nonempty and stays connected after extension of the base field to an algebraic closure.

Counterexample 1 (Y not normal). Let k be a field of characteristic 0, $A = k[t^2, t^3] \subset B = k[t]$, $Y = \operatorname{Spec} A$ (integral, Noetherian) and $X = \operatorname{Spec} B$. The morphism $f : X \to Y$ is finite, hence projective (Ex. 3.3.22). Its fibre over the singular point is $\operatorname{Spec} k[t]/(t^2)$, a single point, and the other fibres are single reduced points; so all fibres are geometrically connected. But $f_*\mathcal{O}_X(Y) = B \neq A$ and $p^n = 1$, so $B^{p^n} = B \not\subseteq A$, and $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is not an isomorphism, although X is integral.

Counterexample 2 (X not reduced). Let k be a field of characteristic 0, $Y = \mathbb{A}^1_k$ (normal) and $X = Y \times_k \operatorname{Spec} k[\varepsilon]$, $\varepsilon^2 = 0$. Then $X \to Y$ is finite (free of rank 2), hence projective, with fibres $\operatorname{Spec} \kappa[\varepsilon]$, geometrically connected. But $f_*\mathcal{O}_X(Y) = k[T][\varepsilon] \neq k[T]$.

So we prove the following.

Theorem. *Let Y be a normal locally Noetherian integral scheme with generic point η, $K = K(Y)$, $p = \operatorname{char} K$. Let $f : X \to Y$ be a projective morphism with X reduced, every irreducible component of X dominating Y (e.g. X integral and f dominant), and with X_η nonempty and geometrically connected over K (this follows if all fibres are geometrically connected, since then $f(X)$, which is closed by Thm. 3.3.30, contains all points, i.e. f is surjective). Then for every affine open V of Y there is $n \geq 1$ with $(f_*\mathcal{O}_X(V))^{p^n} \subseteq \operatorname{Im}(\mathcal{O}_Y(V) \to f_*\mathcal{O}_X(V))$ (with $p^n = 1$ if $p = 0$). If $p = 0$, then $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is an isomorphism.*

Proof. Let $V = \operatorname{Spec} A$, A an integrally closed Noetherian domain with fraction field K, $B = \mathcal{O}_X(f^{-1}V) = f_*\mathcal{O}_X(V)$ and $L = \mathcal{O}(X_\eta)$.

(1) $L = B \otimes_A K$ and $B \to L$ is injective. Since $A \to K$ is flat and $f^{-1}(V) \to V$ is quasi-compact and separated, $\mathcal{O}(X_\eta) = B \otimes_A K$ (Prop. 3.1.24). Let $b \in B$ with $ab = 0$, $0 \neq a \in A$. At the generic point ξ of an irreducible component of X, $f(\xi) = \eta$ by hypothesis, and $\mathcal{O}_{X,\xi}$ is a field (as X is reduced) which is a $K = \mathcal{O}_{Y,\eta}$-algebra; so a is invertible there and $b_\xi = 0$. Since X is reduced and locally Noetherian, a section vanishing at all generic points vanishes (an affine open $\operatorname{Spec} R$ with R reduced has $R \hookrightarrow \prod R_{\mathfrak{p}_i}$ over its minimal primes). So $b = 0$, i.e. B is torsion free and $B \hookrightarrow B \otimes K = L$.

(2) L is a finite purely inseparable extension of K. The K-scheme X_η is projective, hence proper (Thm. 3.3.30), reduced (it is obtained from the reduced affine pieces of X by inverting $A \setminus \{0\}$), nonempty, and geometrically connected, hence connected. By Cor. 3.3.21, $L = \mathcal{O}(X_\eta)$ is a field, finite and purely inseparable over K. If $p = 0$ this means $L = K$. If $p > 0$, write $L = K(\ell_1, \ldots, \ell_r)$ with $\ell_i^{p^{n_i}} \in K$ and let $n \geq \max n_i$, $n \geq 1$. Then $\ell^{p^n} \in K$ for every $\ell \in L$, since ℓ is a polynomial in the ℓ_i with coefficients in K and $x \mapsto x^{p^n}$ is additive and multiplicative.

(3) Conclusion. Let $b \in B \subseteq L$. By Prop. 3.3.18, b is integral over A ($f^{-1}(V)$ is proper over V). Then b^{p^n} is integral over A and lies in K (in K itself, if $p = 0$, with $p^n = 1$), hence lies in A because A is integrally closed. The element $b^{p^n} \in B \subseteq L$ is therefore the image of an element of A (the map $A \to B \to L$ is the inclusion $A \subseteq K \subseteq L$), i.e. $B^{p^n} \subseteq \mathrm{Im}(A \to B)$.

If $p = 0$ this gives $B \subseteq \mathrm{Im}(A)$, so $A \to B$ is surjective. It is injective as well, because $A \to B \to L$ is the composite $A \subseteq K \to L$, injective since K is a field and $L \neq 0$. As this holds for every affine open V, $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is an isomorphism. □

What is not proved. If Y is not normal, the same argument only gives $b^{p^n} \in K \cap B$, i.e. b^{p^n} lies in the integral closure of A in K; Counterexample 1 shows that in characteristic 0 one cannot do better. For Y not normal in characteristic $p > 0$, or for X non-reduced in characteristic $p > 0$ (where torsion in B has to be controlled using the special fibres), I do not prove the printed statement. Only the geometric connectedness of the generic fibre was used. □

Exercise 4.6

> **Problem.** Let Y be a locally Noetherian irreducible scheme, and let $f : X \to Y$ be a dominant morphism of finite type with X irreducible. We are going to generalize Lemma 4.13 and show that $\dim X_\eta \leq \dim X_y$ if $X_y \neq \emptyset$. By this lemma, we may suppose that $\dim X_y \geq 1$. We will use induction on $\dim X_y + \dim \mathcal{O}_{Y,y}$. (a) Let $x \in X_y$ be such that $\dim \mathcal{O}_{X_y,x} = \dim X_y$. Show that there exists a $b \in \mathfrak{m}_x\mathcal{O}_{X,x}$ which belongs to none of the minimal prime ideals containing $\mathfrak{m}_y\mathcal{O}_{X,x}$. Restricting X if necessary, we may suppose X is affine and $b \in \mathcal{O}_X(X)$. (b) Let b be as above. Show that there exists an irreducible component Z of $V(b)$ passing through x and not contained in X_y. Let z' be the generic point of Z, and $y' = f(z')$ the generic point of $f(Z)$ (Ex. 2.4.11). Show that $\dim \mathcal{O}_{Y,y'} < \dim \mathcal{O}_{Y,y}$ and $\dim X_{y'} - \dim X_y \leq \dim Z_{y'} - \dim Z_y$. By applying the induction hypothesis to $Z \to f(Z)$ and $X \times_Y \mathrm{Spec}\, \mathcal{O}_{Y,y'} \to \mathrm{Spec}\, \mathcal{O}_{Y,y'}$, show that $\dim X_\eta \leq \dim X_y$.

Solution. Notation and the form in which the statement is proved. Let η be the generic point of Y and ξ that of X; since f is dominant, $f(\xi) = \eta$. For $y \in Y$ the fibre X_y is a scheme of finite type over $k(y)$ whose underlying space is $f^{-1}(y)$. For a scheme S of finite type over a field and $s \in S$ put $\dim_s S = \max\{\dim T : T \text{ an irreducible component of } S \text{ containing } s\}$, and for $x \in X$ put $e(x) = \dim_x X_{f(x)}$. The inequality of (b) is a comparison of fibre dimensions near x and z' rather than of whole fibres (the whole fibre $X_{y'}$ may have components far from z', and controlling them would need an upper semicontinuity theorem which is not available at this point). So we prove the local statement

Theorem. *Let Y be a locally Noetherian irreducible scheme, $f : X \to Y$ a dominant morphism of finite type, X irreducible, $x \in X$. Then* $\dim X_\eta \leq e(x)$.

This contains the exercise: if $X_y \neq \emptyset$, take x the generic point of an irreducible component of X_y of maximal dimension; the only component through x is that one (a component through x contains $\overline{\{x\}}$, hence equals it), so $e(x) = \dim X_y$.

Facts used. (D1) For an irreducible scheme S of finite type over a field, a nonempty open subset has the same dimension (Prop. 2.5.19), and an irreducible closed subset $T \subsetneq S$ has $\dim T < \dim S$. (D2) If S is an irreducible variety and $b \in \mathcal{O}(S)$ is non-nilpotent, every irreducible component of $V(b)$ has dimension $\dim S - 1$ (Cor. 2.5.26). (D3) Krull's principal ideal theorem and prime avoidance. (D4) Lemma 4.13.

Moreover, X_η is irreducible: it contains ξ, and a subset of an irreducible space containing the generic point is irreducible. By (D1), for every nonempty open $W \subseteq X$ (which contains ξ), $W_\eta = W \cap X_\eta$ is a nonempty open subset of X_η, so $\dim W_\eta = \dim X_\eta$. (5)

Proof of the Theorem. We use strong induction on $N = e(x) + \dim \mathcal{O}_{Y,y}$, $y = f(x)$, simultaneously for all data (Y, X, f, x) as in the Theorem. Put $d = e(x)$.

Case $y = \eta$. Then $X_y = X_\eta$ is irreducible, so $e(x) = \dim X_\eta$.

Case $y \neq \eta$ and $d = 0$ (the case "$\dim X_y = 0$" treated by Lemma 4.13). Then $\{x\}$ is a component of X_y and no other component contains x, so $\{x\}$ is open in X_y. Hence there is an open $W \subseteq X$ with $W \cap X_y = \{x\}$. Then W is irreducible, $W \to Y$ is of finite type, and $W_y = \{x\}$ is nonempty and finite. By Lemma 4.13, W_η has at most one point; it contains ξ, so $W_\eta = \{\xi\}$ and $\dim X_\eta = \dim W_\eta = 0$ by (5).

Case $y \neq \eta$ and $d \geq 1$. Note $\dim \mathcal{O}_{Y,y} \geq 1$.

Reduction to a good point. Let $C_1, \ldots, C_r$ be the irreducible components of X_y containing x, and F the union of the others. Then $F = X_y \cap F''$ for a closed subset F'' of X, and $x \notin F''$. Replacing X by $X \setminus F''$ we may assume that all irreducible components of X_y pass through x; by (D1) their dimensions are unchanged, and (5) holds. If x is the generic point of a component of X_y, then this component is the only one; we then replace x by a closed point of X_y (it still lies on the unique component, so $e(x) = d$ is unchanged). In every case we may therefore assume: (G1) every irreducible component of X_y contains x; (G2) x is not the generic point of a component of X_y (if x were the generic point of one component, all others would contain x, hence equal it); (G3) $e(x) = d = \dim X_y \geq 1$. Shrinking X further to an affine open neighbourhood $W = \operatorname{Spec} B_0$ of x with $f(W)$ inside an affine open $\operatorname{Spec} A_0 \subseteq Y$ preserves (G1)–(G3) and (5). So assume $X = \operatorname{Spec} B_0$ is affine.

(a) The element b. Let $R = \mathcal{O}_{X,x}$ and $\bar{R} = R/\mathfrak{m}_y R = \mathcal{O}_{X_y,x}$. The minimal primes $\bar{q}_1, \ldots, \bar{q}_r$ of $\bar{R}$ correspond to the components of X_y through x. By (G2), $\bar{q}_j \neq \bar{\mathfrak{m}}$ (the maximal ideal of $\bar{R}$), i.e. $\bar{\mathfrak{m}} \not\subseteq \bar{q}_j$ for all j. By prime avoidance there is $\bar{b} \in \bar{\mathfrak{m}} \setminus \bigcup_j \bar{q}_j$; lift it to $b \in \mathfrak{m}_x R$. Then b lies in no minimal prime q_j of R containing $\mathfrak{m}_y R$. Shrinking X to a smaller affine neighbourhood of x (this preserves (G1)–(G3) and (5)) we may assume $b \in \mathcal{O}_X(X)$.

For a component C of X_y the generic point c of C is a generization of x and $b \notin \bar{q}_C$, so b is a unit of $\mathcal{O}_{X_y,c} = \bar{R}_{\bar{q}_C}$; thus b does not vanish at c, and $V(b) \cap C$ is a proper closed subset of C. By (D1) every irreducible closed subset of $V(b) \cap X_y$ has dimension $\leq d - 1$. (6)

(b) The component Z. Irreducible components of $V(b)$ through x correspond to the minimal primes P of bR; each has $\operatorname{ht} P \leq 1$ by Krull. Suppose every such P contained $\mathfrak{m}_y R$. Take one, P. It contains a minimal prime q of $\mathfrak{m}_y R$, and $b \in P \setminus q$, so $q \subsetneq P$. The ring R has a unique minimal prime p_0 (that of ξ: every generization of x is a specialization of ξ), and $p_0 \subseteq q \subsetneq P$ with $\operatorname{ht} P \leq 1$ forces $q = p_0$. Then $\mathfrak{m}_y R \subseteq p_0$ means $f(\xi) = y$, i.e. $y = \eta$, contrary to assumption. So some irreducible component Z of $V(b)$ through x has generic point z' with $f(z') \neq y$, i.e. $Z \not\subseteq X_y$. Put $y' = f(z')$. We give Z the reduced structure and $Y' = \overline{\{y'\}}$ the reduced structure; then f induces a dominant morphism of finite type $Z \to Y'$ (as Z is reduced), with Z irreducible, Y' locally Noetherian irreducible with generic point y', and $x \in Z$, $f(x) = y \in Y'$ (as $f(x) \in f(\overline{\{z'\}}) \subseteq \overline{\{y'\}}$).

(b1) $\dim \mathcal{O}_{Y,y'} < \dim \mathcal{O}_{Y,y}$. Indeed $y' \neq y$ is a generization of y, so $\mathcal{O}_{Y,y'}$ is the localization of

$\mathcal{O}_{Y,y}$ at a prime $\mathfrak{p} \subsetneq \mathfrak{m}_y$, and $\dim \mathcal{O}_{Y,y'} = \operatorname{ht} \mathfrak{p} < \operatorname{ht} \mathfrak{m}_y = \dim \mathcal{O}_{Y,y}$.

(b2) $e_Z(x) := \dim_x Z_y \leq d-1$, where Z_y is the fibre of $Z \to Y'$ over y. Indeed $Z_y \subseteq V(b) \cap X_y$ is closed in X_y and (6) applies to each of its components. Also $\dim \mathcal{O}_{Y',y} \leq \dim \mathcal{O}_{Y,y}$, since $\mathcal{O}_{Y',y}$ is a quotient of $\mathcal{O}_{Y,y}$.

(b3) $\dim_{z'} X_{y'} \leq \dim Z_{y'} + 1$, where $Z_{y'}$ is the generic fibre of $Z \to Y'$. Let C' be an irreducible component of $X_{y'}$ containing z' (a $k(y')$-scheme of finite type; give C' the reduced structure). The set $Z_{y'} = Z \cap X_{y'}$ is the closure of $\{z'\}$ in $X_{y'}$, so $Z_{y'} \subseteq C'$. Consider two cases. If $C' \subseteq V(b)$: C' is irreducible, contains z', and lies in $V(b)$, so it lies in an irreducible component of $V(b)$ containing z', which must be Z (any such component contains $\overline{\{z'\}} = Z$ and Z is a component); then $C' \subseteq Z \cap X_{y'} = Z_{y'}$ and $C' = Z_{y'}$. If $C' \not\subseteq V(b)$: b is non-nilpotent on the integral variety C', and $Z_{y'}$ is an irreducible component of $V(b) \cap C'$: if $T \supseteq Z_{y'}$ is an irreducible closed subset of $V(b) \cap C'$, its closure $\overline{T}$ in X is irreducible, contained in $V(b)$ and contains z', hence $\overline{T} \subseteq Z$, so $T \subseteq Z \cap X_{y'} = Z_{y'}$. By (D2), $\dim Z_{y'} = \dim C' - 1$. In both cases $\dim C' \leq \dim Z_{y'} + 1$.

By (b2) and (b3), $\dim_{z'} X_{y'} - \dim_x X_y \leq \dim Z_{y'} + 1 - d \leq \dim Z_{y'} - \dim_x Z_y$; this is the local form of the inequality asked for in (b).

(c) Induction. First apply the induction hypothesis to $Z \to Y'$ and the point x: its invariant is $e_Z(x) + \dim \mathcal{O}_{Y',y} \leq d - 1 + \dim \mathcal{O}_{Y,y} < N$. It gives $\dim Z_{y'} \leq e_Z(x) \leq d-1$, because y' is the generic point of Y'. By (b3), $e(z') = \dim_{z'} X_{y'} \leq d$.

Next consider $X'' = X \times_Y \operatorname{Spec} \mathcal{O}_{Y,y'} \to Y'' = \operatorname{Spec} \mathcal{O}_{Y,y'}$. Here Y'' is a Noetherian irreducible scheme with generic point η and closed point y'. The morphism $X'' \to X$ is a homeomorphism onto the set of points of X lying over generizations of y' (localization on the affine scheme $X = \operatorname{Spec} B_0$ over $\operatorname{Spec} A_0$); this set contains ξ, so X'' is irreducible, $X'' \to Y''$ is dominant and of finite type, $X''_{y'} = X_{y'}$ and $X''_\eta = X_\eta$. The point $z' \in X''$ lies over y', and its invariant is $e(z') + \dim \mathcal{O}_{Y,y'} \leq d + \dim \mathcal{O}_{Y,y} - 1 < N$ by (b1). The induction hypothesis gives $\dim X_\eta = \dim X''_\eta \leq e(z') \leq d = e(x)$. □

What was proved. The statement of the exercise follows from the Theorem as explained above. Items (a) and (b) are proved as stated, except that the inequality $\dim X_{y'} - \dim X_y \leq \dim Z_{y'} - \dim Z_y$ is proved in the local form $\dim_{z'} X_{y'} - \dim_x X_y \leq \dim Z_{y'} - \dim_x Z_y$ (all that the induction needs), and x is chosen as in the reduction step. The proof relies on Lemma 4.13, hence on Thm. 5.3.2. □